

가려진 물체의 탐색을 위한 심층강화학습 기반 탐색 지능화 기술 연구

A Study on Deep Reinforcement Learning-Based Exploration Intelligence for Occluded Object Search

소속: 숭실대학교 기계공학과 지능로봇시스템연구실(Intelligent Robotic system Laboratory, IROL)

발표자: 박사과정 전하늘

발표일: 2025년 05월 27일 (화)

Comments

- 제안 연구의 활용 분야(application) 
- 2024년도 이전 연구 추가 
 - 연구 동향
- 이전연구 분석 2장으로 분리(논문)
- Human이 동일 조건에서 실험했을 때 결과 비교 
- 물체별 step count 결과 그래프에 box plot 추가
- X-ray의 물체별 step count의 fluctuation 이유 설명 추가 
- High-level task planner 와 Low-level task planner의 computation cost 비교 
- High-level task planner를 LLM을 통한 end-to-end로 대체 가능한지? 
 - 가능하다면, 제안 연구 방법(DRL)과 비교하였을 때 장단점

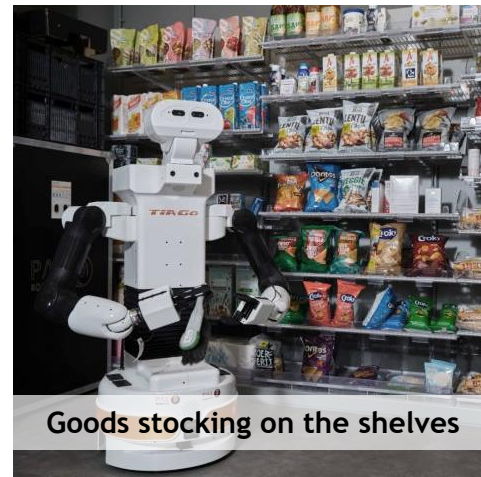
Research motivations

Non-observable target object searching

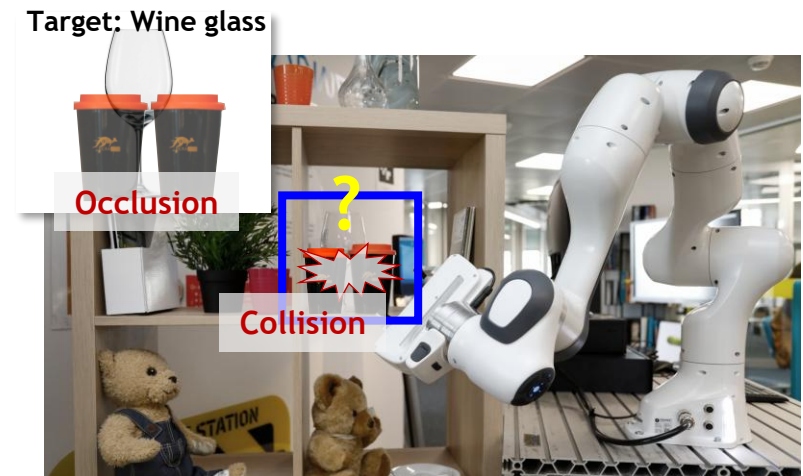
- 최근 home service⁽¹⁾, logistics⁽²⁾, retail shop⁽³⁾ 등과 같은 “복잡한 제약공간 내의 관측 불가 물체의 추론·탐색 방법”에 관한 연구가 활발히 진행되고 있음.
- **비정형 주변 물체에 의한 폐색(Occlusion) 및 제약공간(Constrained Space)** 등과 같은 다양한 환경적 불확실성에 의해 탐색 방향 추론이 어려움
- Keywords: #Hidden object, #Occluded object search, #Constrained space, #Deep Reinforcement Learning(DRL)



출처: Toyota Research Institute



출처: TIAGo(Mobile manipulator Robot) for Retail shop



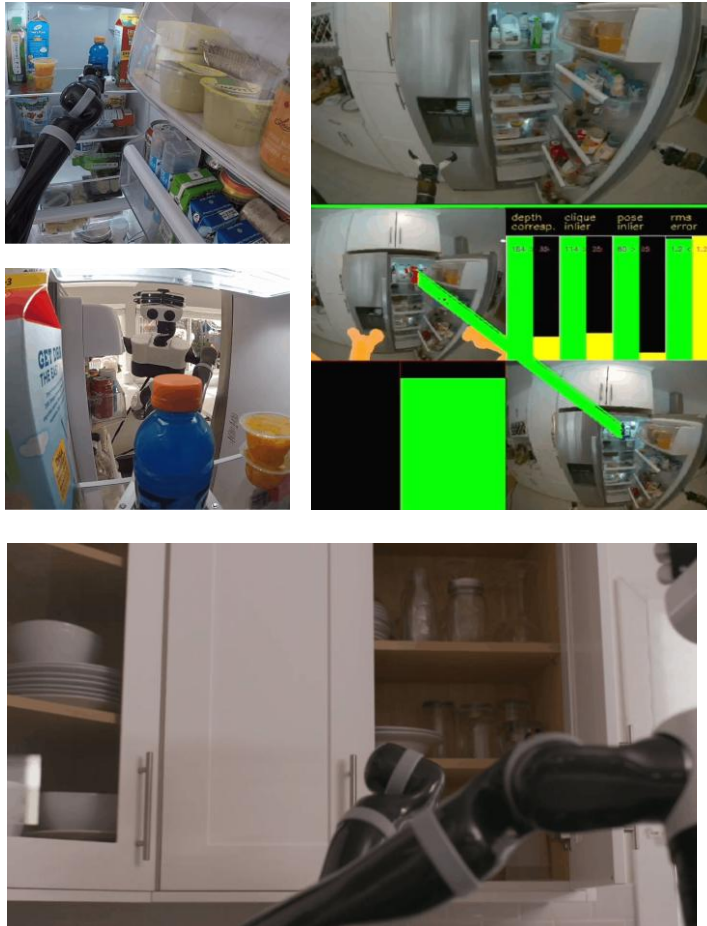
출처: Robotic Vision Australia Annual report 2019

(1) Mobile Manipulation for the HSR Intelligent Home Service Robot; 2019
 (2) Knowledge-Enabled Robotic Agents for Shelf Replenishment in Cluttered Retail Environments; 2016
 (3) Rearrangement planning of multiple movable objects by using real-time search methodology; 2002

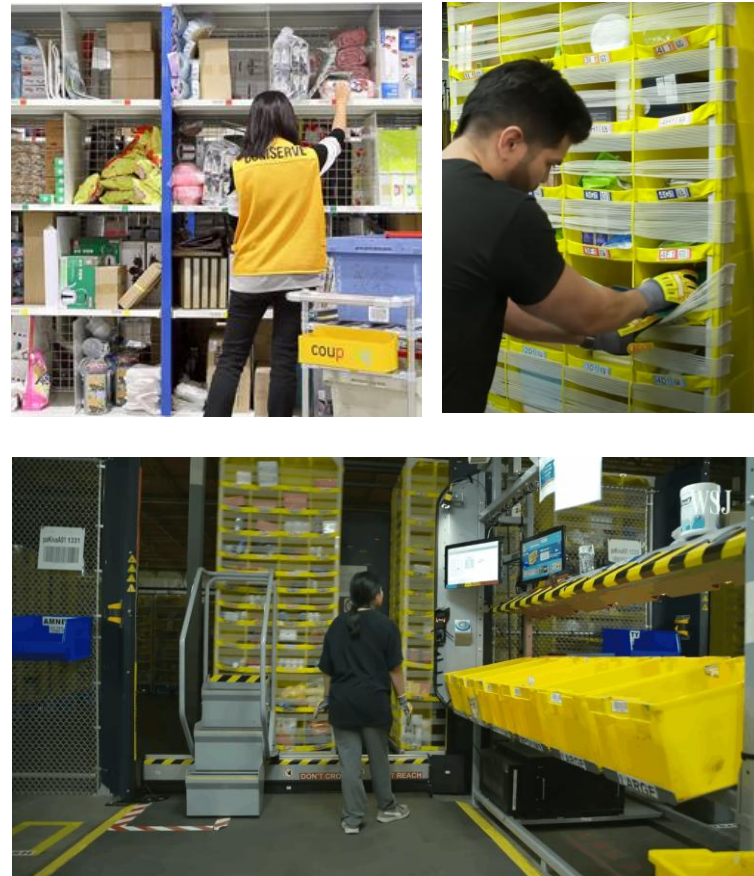
Research motivations

Applications of the proposed research

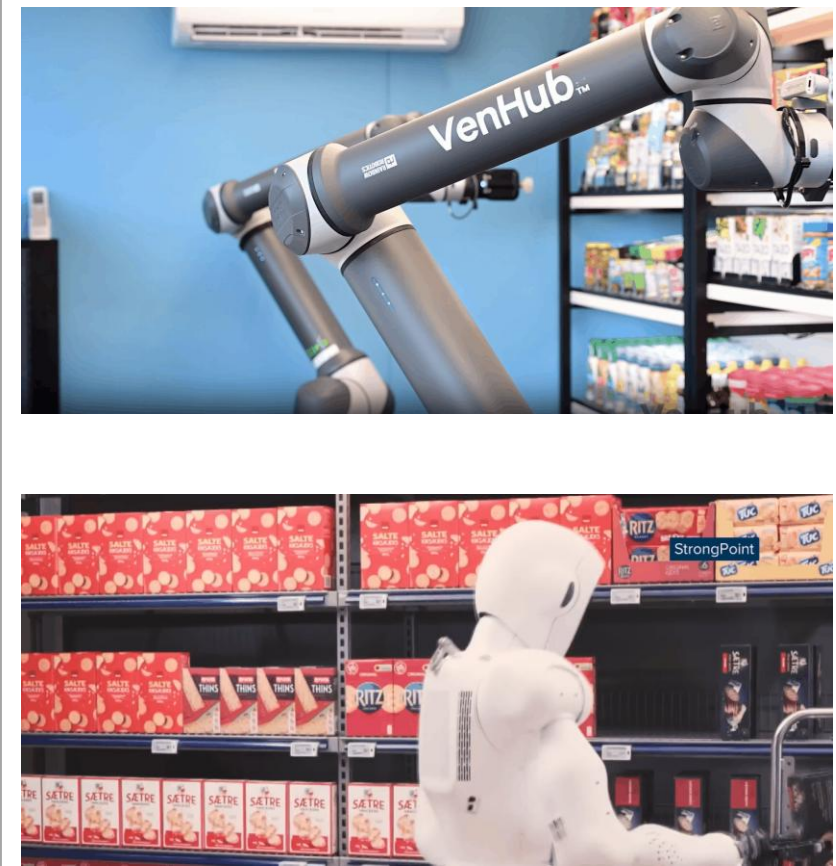
• Home service



• Logistics



• Retail shop



Research motivations

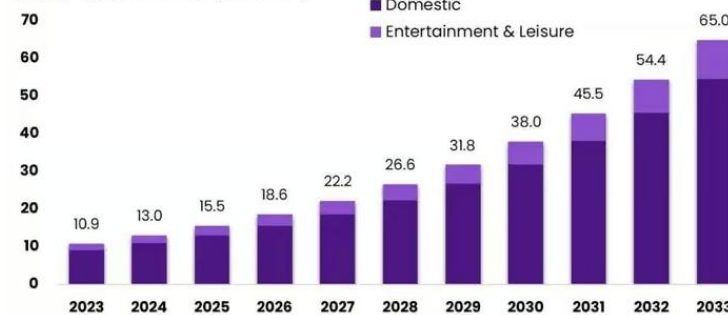
Market prospects for the proposed research

• Home service

- 2023년 시장규모: 약 101억 5천만 달러
- 2032년 시장전망: 약 288억 5천만 달러
- 연평균 성장률(CAGR): 19.1%
 - 특히, **정밀한 탐색과 물체 인식, 추론** 기반의 **지능형 상호작용이 가능한 로봇**의 수요가 급증

Global Household Robots Market

Size, by Type, 2024-2033 (USD Billion)



MARKET STATISTICS

MARKET SIZE 2023

\$
10.15 BN

CAGR 2024-2032

19.10%

MARKET SIZE 2032

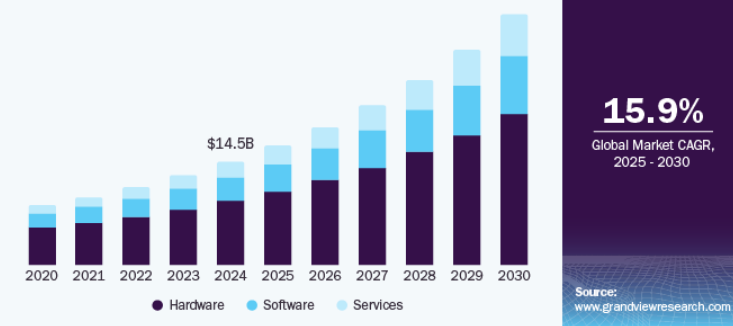
\$
48.85 BN

• Logistics

- 2024년 시장규모: 약 145억 달러
- 2030년 시장전망: 약 350억 5천만 달러
- 연평균 성장률(CAGR): 15.9%
 - 분류 및 피킹 작업, 라스트마일 배송 과정에서 로봇이 광범위 하게 도입되어 **인력 비용 절감과 작업자 안전 확보**

Logistics Robot Market

Size, by Component, 2020 - 2030 (USD Billion)



출처: Grand View Resarch

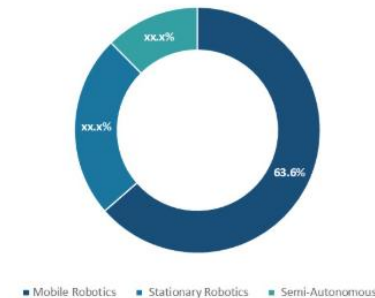
• Retail shop

- 2025년 시장규모: 약 166억 7천만 달러
- 2032년 시장전망: 1,046억 4천만 달러
- 연평균 성장률(CAGR): 30.0%
 - **재고관리, 진열 보조** 등을 수행하는 로봇의 수요 증가
 - 아시아 태평양 지역은 특히 연평균 성장률 **37.2%**로 소매 로봇 분야에서 빠르게 성장할 것으로 예상

Retail Robots Market, By Type, 2025

Total Market Size:
USD 16.67 Bn

% Market share By Type



COHERENT
MARKET INSIGHTS

63.6%

Mobile Robotics Type -
Estimated Market
Revenue Share, 2025

Retail Robots
Market

출처: Coherent Market Insights



Previous works

Timeline for Non-observable object searching

~ 2021

- ① Tree Search-based and Motion Planning with Prehensile and Non-prehensile Manipulation for Obstacle Rearrangement in Clutter
- ② Robotic Weight-based Object Relocation in Clutter via Tree-based Q-learning Approach using Breadth and Depth Search Techniques
- ③ Occlusion-Aware Search for Object Retrieval in Clutter

2022

- ① Semi-Autonomous Teleoperation via Learning Non-Prehensile Manipulation Skills
- ② Grasp Planning for Occluded Objects in a Confined Space with Lateral View Using Monte Carlo Tree Search
- ③ Efficient and High-quality Prehensile Rearrangement in Cluttered and Confined Spaces

2023

- ① Viewpoint Push Planning for Mapping of Unknown Confined Spaces
- ② Object Rearrangement Planning for Target Retrieval in a Confined Space with Lateral View

2024

- ① Language-guided Active Sensing of Confined, Cluttered Environments via Object Rearrangement Planning
- ② Object-aware interactive perception for tabletop scene exploration

Environment	가관측성 (Observability)	모든 환경 및 target object를 포함한 object들의 정보가 관측 가능. [Fully Observable (F.O) / Partially Observable (P.O) / Unobservable(U.O)]
	물체 형상 (Object configuration)	물체의 형상의 종류가 한가지 이상 [Structured / Unstructured]
Agent	자율성 (Autonomy)	Target object를 탐색하는데 있어서 중간에 사람이 관여하는지 여부 [Fully Autonomous / Manual]
	상황 인지성 (Context-awareness)	환경 및 작업의 상황(context)을 인지하여 의사결정에 활용여부 [Context-aware / Context-free]
	강인성 (Robustness)	새로운 환경이나 다양한 상황 변화에 대해 일반화 여부 [Generalizable / Specific]
	제어 (Controllability)	Target searching-grasping을 위한 동작 종류
System	방법 (Method)	숨겨진 물체를 탐색·수거를 위한 접근 방식 [Heuristic / DRL]

Previous works

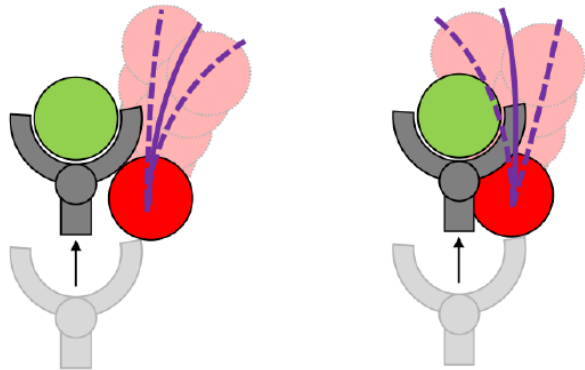
Timeline for Occluded object searching

~ 2021



- ① Tree Search-based and Motion Planning with Prehensile and Non-prehensile Manipulation for Obstacle Rearrangement in Clutter
- ② Robotic Weight-based Object Relocation in Clutter via Tree-based Q-learning Approach using Breadth and Depth Search Techniques
- ③ Occlusion-Aware Search for Object Retrieval in Clutter

①

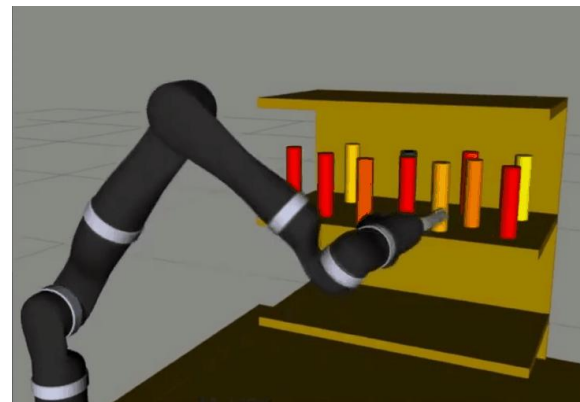


- Tree search 기반 최소의 물건을 재배치하는 경로 계획
- 장애물의 배치를 고려한 push 혹은 grasp
- Fully observable, Structured

2022

- ① Semi-Autonomous Teleoperation via Learning Non-Prehensile Manipulation Skills
- ② Grasp Planning for Occluded Objects in a Confined Space with Lateral View Using Monte Carlo Tree Search
- ③ Efficient and High-quality Prehensile Rearrangement in Cluttered and Confined Spaces

②



- Robotic weight을 고려한 물체 제거 순서 계획
- Q-Learning 기반 강화학습
- Fully observable, Structured

2023

- ① Viewpoint Push Planning for Mapping of Unknown Confined Spaces
- ② Object Rearrangement Planning for Target Retrieval in a Confined Space with Lateral View

③



- Stochastic method 기반 Unobservable Target 추정
- DRL(PPO)를 활용한 goal position 도달 경로 계획
- 카메라를 통해 안보이는 object 뒷면의 형상도 알아야 함
- Partially Observable, Structured

2024

- ① Language-guided Active Sensing of Confined, Cluttered Environments via Object Rearrangement Planning
- ② Object-aware interactive perception for tabletop scene exploration

Previous works

Timeline for Occluded object searching

~ 2021

- ① Tree Search-based and Motion Planning with Prehensile and Non-prehensile Manipulation for Obstacle Rearrangement in Clutter
- ② Robotic Weight-based Object Relocation in Clutter via Tree-based Q-learning Approach using Breadth and Depth Search Techniques
- ③ Occlusion-Aware Search for Object Retrieval in Clutter

①

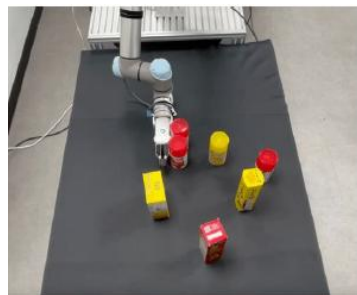


- 강화학습을 활용하여 다수의 task 수행 경로를 추출하고 사용자의 기호에 맞는 경로를 선택
- 사용자의 개입이 필수
- Fully observable, Structured

2022

- ① Semi-Autonomous Teleoperation via Learning Non-Prehensile Manipulation Skills
- ② Grasp Planning for Occluded Objects in a Confined Space with Lateral View Using Monte Carlo Tree Search
- ③ Efficient and High-quality Prehensile Rearrangement in Cluttered and Confined Spaces

②

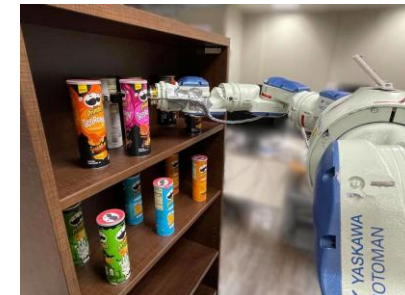
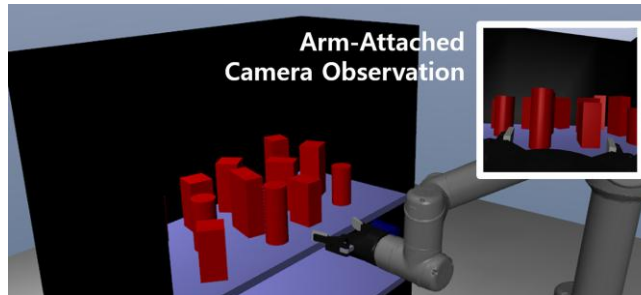


- PCD 기반 MTCS를 통한 target object 탐색 방법
- Fully observable, Structured

2023

- ① Viewpoint Push Planning for Mapping of Unknown Confined Spaces
- ② Object Rearrangement Planning for Target Retrieval in a Confined Space with Lateral View

③



- 제한된 공간에서 target object 탐색을 위해 다수의 원형 물체를 목표 위치로 재배치하는 방법 제안
- Fully observable, Structured

2024

- ① Language-guided Active Sensing of Confined, Cluttered Environments via Object Rearrangement Planning
- ② Object-aware interactive perception for tabletop scene exploration

Previous works

Timeline for Occluded object searching

~ 2021

2022

2023

2024

① Tree Search-based and Motion Planning with Prehensile and Non-prehensile Manipulation for Obstacle Rearrangement in Clutter

② Robotic Weight-based Object Relocation in Clutter via Tree-based Q-learning Approach using Breadth and Depth Search Techniques

③ Occlusion-Aware Search for Object Retrieval in Clutter

① Semi-Autonomous Teleoperation via Learning Non-Prehensile Manipulation Skills

② Grasp Planning for Occluded Objects in a Confined Space with Lateral View Using Monte Carlo Tree Search

③ Efficient and High-quality Prehensile Rearrangement in Cluttered and Confined Spaces

① Viewpoint Push Planning for Mapping of Unknown Confined Spaces

② Object Rearrangement Planning for Target Retrieval in a Confined Space with Lateral View

① Language-guided Active Sensing of Confined, Cluttered Environments via Object Rearrangement Planning

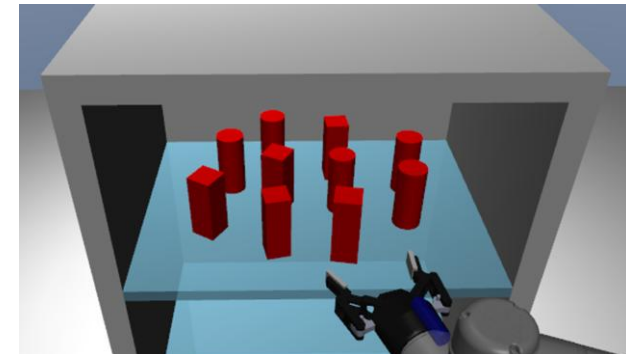
② Object-aware interactive perception for tabletop scene exploration

①



- 제한된 공간에서 가려진 영역을 탐색하기 위하여 DRL 기반 최적의 Viewpoint planning(VPP) 방법 제안
- 공간 탐색을 위해 기존 전문가들이 사전 정의한 push 네트워크 필요
- Fully observable

②



- 선반과 같이 제한된 공간에서 목표 물체를 탐색하기 위하여 sub-MCTS 기반 물체 재배치 순서 planning 수행
- Fully observable, Structured

Previous works

Timeline for Occluded object searching

~ 2021

2022

2023

2024

① Tree Search-based and Motion Planning with Prehensile and Non-prehensile Manipulation for Obstacle Rearrangement in Clutter

② Robotic Weight-based Object Relocation in Clutter via Tree-based Q-learning Approach using Breadth and Depth Search Techniques

③ Occlusion-Aware Search for Object Retrieval in Clutter

① Semi-Autonomous Teleoperation via Learning Non-Prehensile Manipulation Skills

② Grasp Planning for Occluded Objects in a Confined Space with Lateral View Using Monte Carlo Tree Search

③ Efficient and High-quality Prehensile Rearrangement in Cluttered and Confined Spaces

① Viewpoint Push Planning for Mapping of Unknown Confined Spaces

② Object Rearrangement Planning for Target Retrieval in a Confined Space with Lateral View

① Language-guided Active Sensing of Confined, Cluttered Environments via Object Rearrangement Planning

② Object-aware interactive perception for tabletop scene exploration

①



- 복잡하고 제한된 환경에서 가려진 물체 탐색을 위해 언어 기반 ROI를 선택하고 해당 viewpoint에서 occlusion을 발생시키는 물체를 제거
- 공간 탐색을 위해 사용자의 개입이 필수적
- **Partially observable, Manual**

②



- 관측 불가능한 목표 물체 탐색을 위해 voxel map 기반의 PCD 정보량이 적은 영역의 물체를 제거하면서 탐색
- **Partially observable, Structured**

Related research trends

- 기간: 2021~2024
- 주제:
 - 로봇을 활용한 물체 탐색
 - Vision 기반 물체 탐색
- 키워드:
 - Object search, occluded object, robotic exploration, DRL, LLM for robotics 등
- 저널:
 - ICRA / IROS / RSS / CoRL / RA-L / T-RO
 - CVPR / ECCV / ICCV (로봇 비전 관련)
 - arXiv, IEEE T-ETCI, Springer Robotics 등

• 2021: 휴리스틱에서 DRL로의 전환

- 휴리스틱의 기술적 한계
 - 미리 설정한 규칙에 의해 고정된 경로로 탐색하는 경향이 존재
→ 환경이 바뀌면 탐색 성능이 급격히 저하
 - 반면, 다양한 환경에서 센서의 관측 값으로 최적의 policy를 학습하므로 복잡한 환경에서 효율적으로 목표물체를 탐색 가능

• 2022: 심층 강화학습의 주류 및 성능 향상

- 주요 논문의 80% 이상이 심층 강화학습에 기반한 접근법 제안
 - DRL 방법의 성능 개선을 다룬 연구들도 대거 등장

• 2023~2024: 대형 언어 모델(LLM)의 등장과 융합 시도

- 2023년: ChatGPT 공개와 더불어 로봇 탐색 분야에 LLM이 처음 도입
- 2024년: LLM의 추론 모듈과 대규모 vision 모델 및 DRL과의 결합으로 하이브리드 시스템의 형태를 이룸

연도	출판 논문 수	휴리스틱 기반	심층 강화학습 기반	LLM 융합 기반
2021	약 10	4	6	-
2022	약 15	3	12	-
2023	약 20	3	14	3
2024	약 24	3	14	7



Previous works

Contributions

Years	No.	Environment		Agent				System
		Observability	Object configuration	Autonomy	Context awareness	Robustness	Controllability	Method
~ 2021	1	F.O	Structured	Fully Autonomous	X	Specific	Push - Grasp (Obstacle & target)	Heuristic
	2	F.O	Structured	Fully Autonomous	X	Specific	Grasp (Obstacle & target)	Q-learning (DRL)
	3	P.O	Structured	Fully Autonomous	O	Specific	Push - Grasp (Target)	PPO (DRL)
2022	1	F.O	Structured	Manual	X	Generalizable	Push - Grasp (Target)	DLPG (DRL)
	2	F.O	Structured	Fully Autonomous	X	Specific	Grasp	Heuristic
	3	F.O	Structured	Manual	X	Specific	Grasp	Heuristic
2023	1	F.O	Unstructured	Fully Autonomous	O	Generalizable	Push - Grasp (Target)	TQC (DRL)
	2	F.O	Structured	Fully Autonomous	X	Specific	Grasp	Heuristic
2024	1	P.O	Structured	Manual	X	Specific	Grasp	Heuristic
	2	P.O	Structured	Fully Autonomous	X	Specific	Grasp	Heuristic
Our		U.O	Unstructured	Fully Autonomous	O	Generalizable	Sweeping - Grasp (Obstacle & target)	PPO (DRL)

Previous works

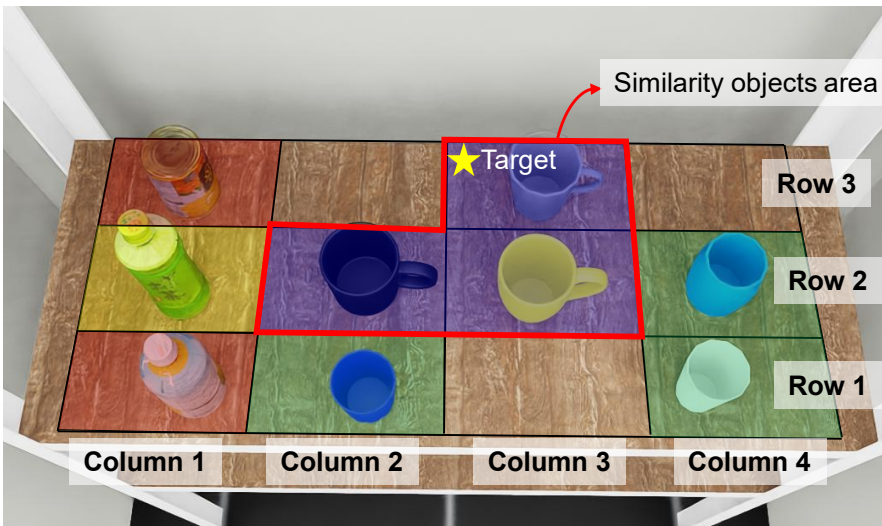
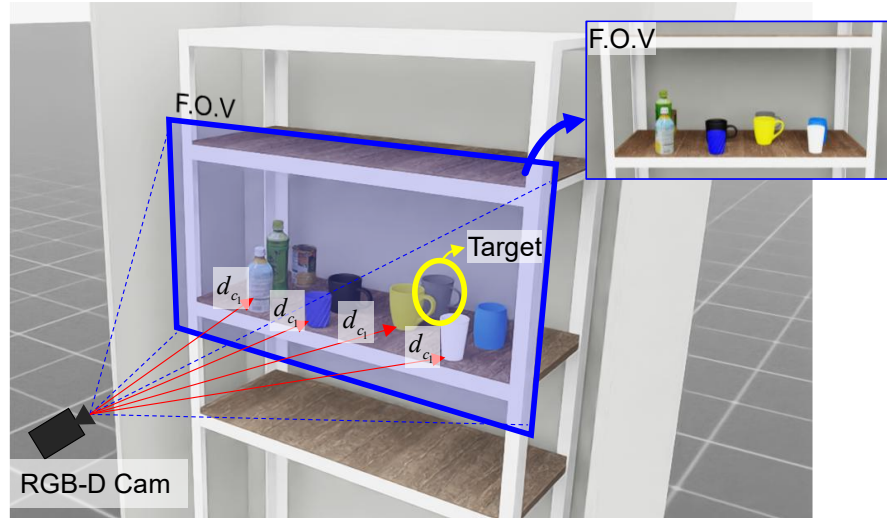
Contributions

Years	No.	Environment		Agent				System
		Observability	Object configuration	Autonomy	Context awareness	Robustness	Controllability	Method
~ 2021	1	F.O	Structured	Fully Autonomous	X	Specific	Push - Grasp (Obstacle & target)	Heuristic
	2	F.O	Structured	Fully Autonomous	X	Specific	Grasp (Obstacle & target)	Q-learning (DRL)
	3	F.O	Structured	Fully Autonomous	X	Specific	Push - Grasp (Target)	PPO (DRL)
2022	1	F.O	Structured	Manual	X	Generalizable	Push - Grasp (Target)	DLPG (DRL)
	2	F.O	Structured	Fully Autonomous	X	Specific	Grasp	Heuristic
	3	F.O	Structured	Manual	X	Specific	Grasp	Heuristic
2023	1	F.O	Structured	Fully Autonomous	X	Generalizable	Push - Grasp (Target)	TQC (DRL)
	2	F.O	Structured	Fully Autonomous	X	Specific	Grasp	Heuristic
2024	1	P.O	Structured	Manual	X	Specific	Grasp	Heuristic
	2	P.O	Structured	Fully Autonomous	X	Specific	Grasp	Heuristic
Our		U.O	Unstructured	Fully Autonomous	O	Generalizable	Sweeping - Grasp (Obstacle & target)	PPO (DRL)

- 앞서 언급한 환경에 대응하는 본 연구에서의 contribution은,
 - **Target observability:** Target이 완전히 가려져 있는 상황
 - **Object configuration:** object 종류와 형상이 다양함
 - **Autonomy:** 사람의 개입 없이 자동으로 target object를 탐색
 - **Context awareness:** 환경 및 작업의 상황(context)을 인지하여 target object를 추론 및 탐색
 - **Robustness:** 다양한 실제환경에 유동적으로 대응 가능

Method

Problem definition



■ Can
 ■ Cup
 ■ Mug
 ■ Bottle

- **Main Task** : 복잡한 선반환경에서 관측 불가능한 target object 추론·탐색

- **Assumptions**

- 비슷한 종류의 물체는 서로 근처에 위치함
- 선반의 높이 때문에 위에서 물건을 관측 및 파지할 수 없음
- 선반 속 물체들은 매번 랜덤하게 배치됨
- 매 step마다 하나의 물체에 대해 한개의 action을 수행 함

- **Shelf configuration**

- 3 x 4 배열의 grid cell에 object들이 배치 됨
- Object들은 매번 랜덤하게 $\pm 1cm$ pose uncertainty를 부여하여 배치

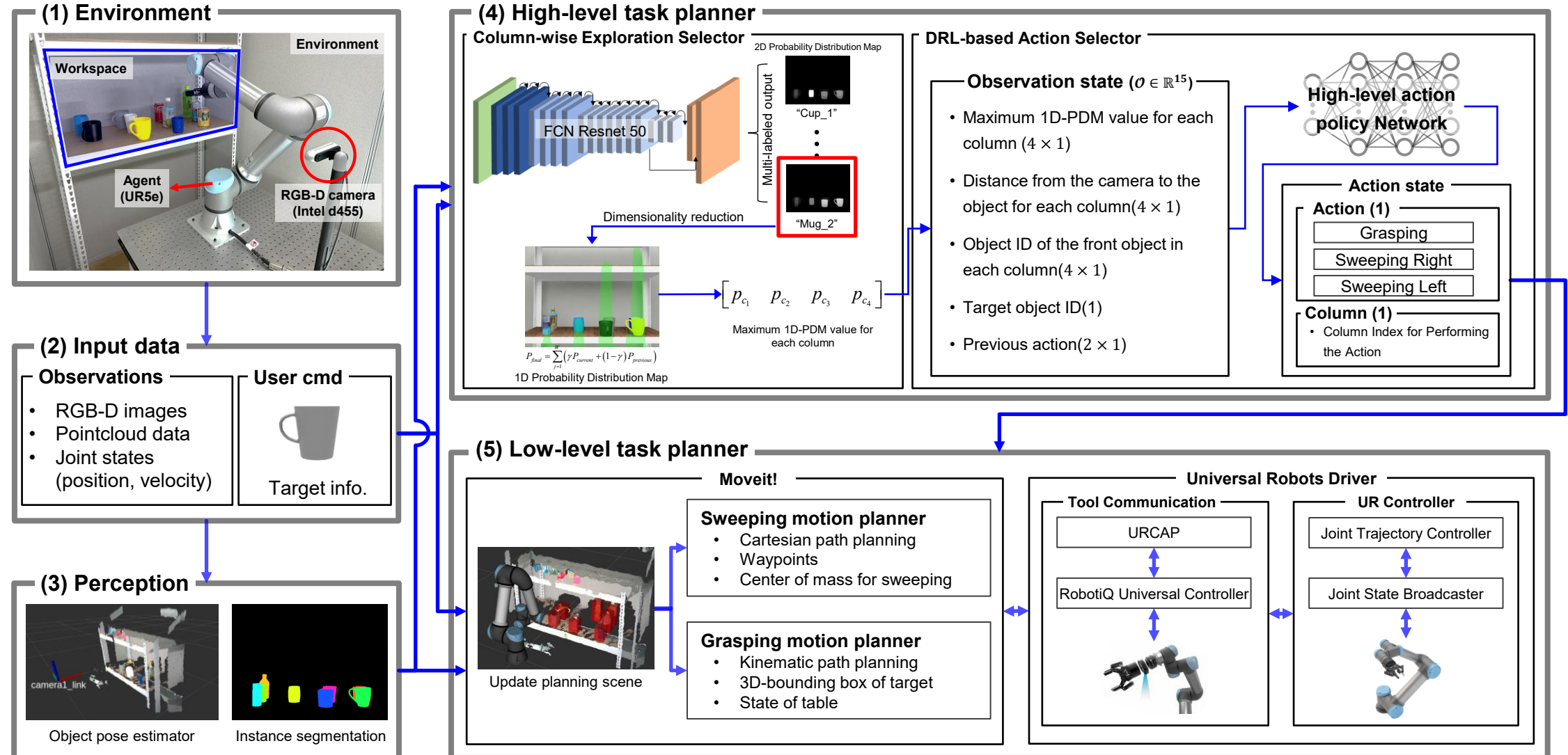
- **Object list**

- Object는 서로 다른 4개의 category로 구성되어 있음
 - Cup, Mug, Can, Bottle
- 각 category 별로 3개의 서로 다른 object들로 구성됨
 - Ex. Cup : Cup_1 / Cup_2 / Cup_3

(모든 오브젝트는 sim2real을 고려하여 3D 스캐너로 스캔하여 학습 및 simulation 수행)

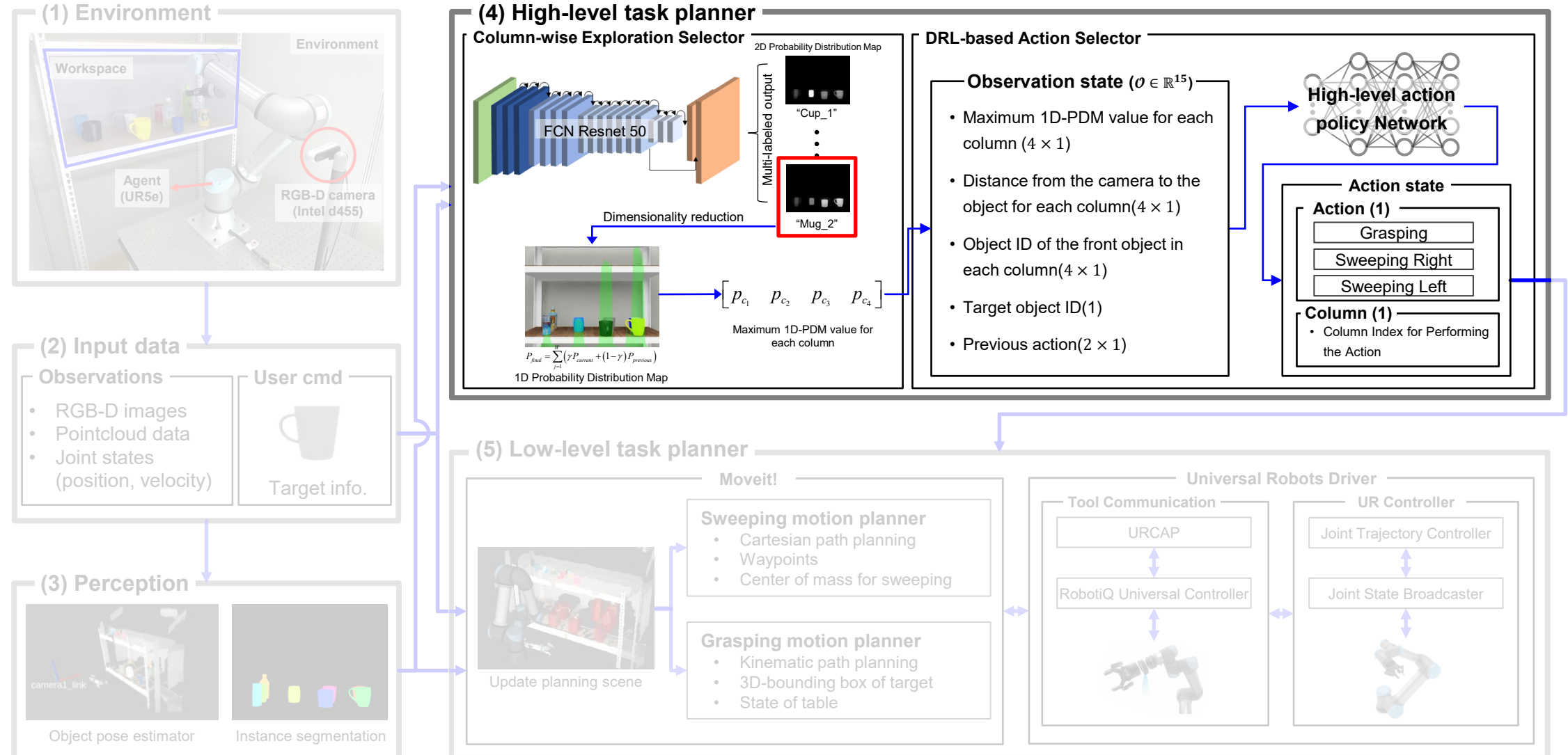
Method

Overall framework



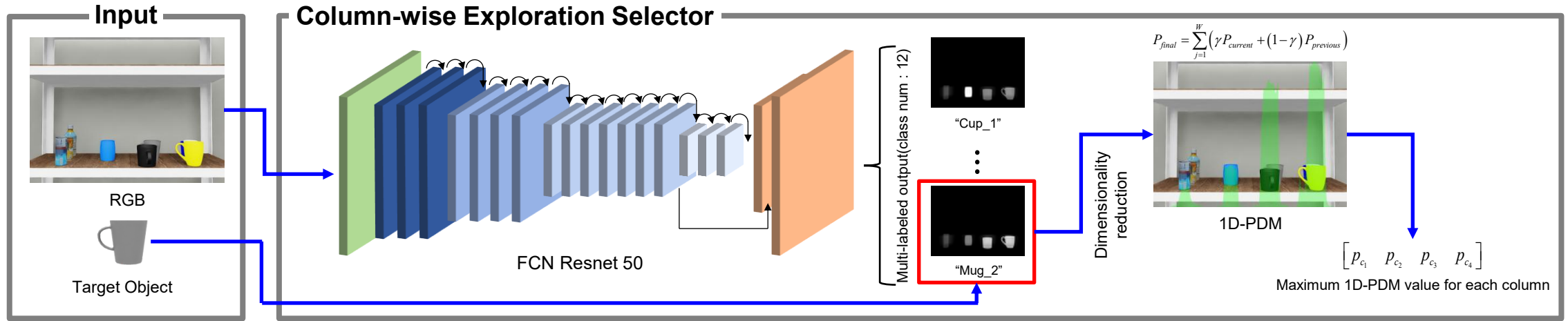
Method

Overall framework

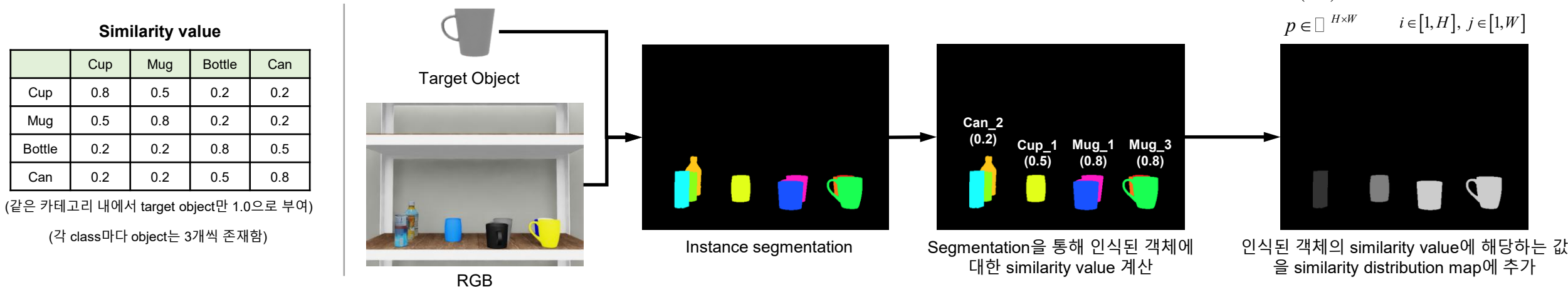


Column-wise Exploration Selector

Data generation for training the Column-wise Exploration Selector

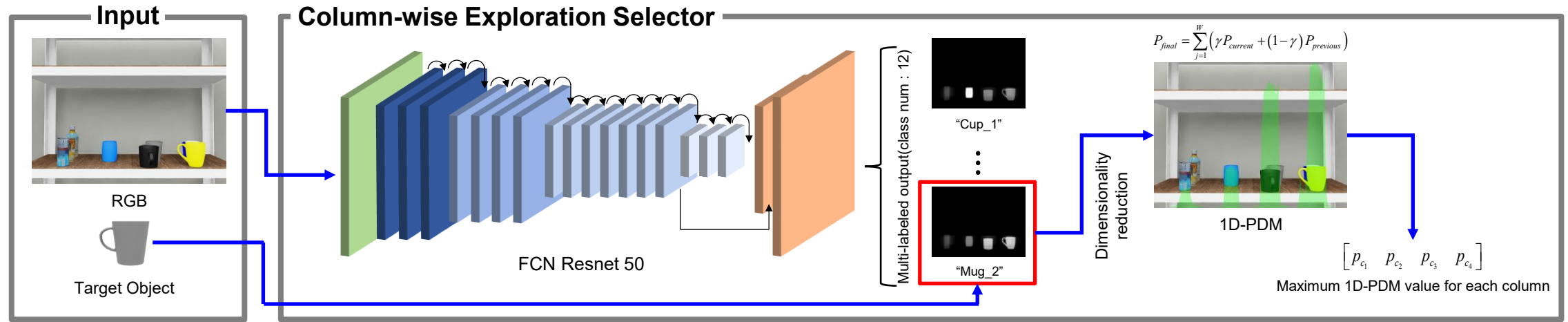


- **(Similarity distribution map) 선반내에 관측 가능한 object들 사이의 유사도 평가**
 - Instance segmentation 통해 관측 가능한 object class인식 후 사전 정의 된 similarity value를 해당 object 픽셀 값에 부여하여 similarity distribution map에 추가 → 유사도가 높은 object 일수록 흰색에 가까워 짐



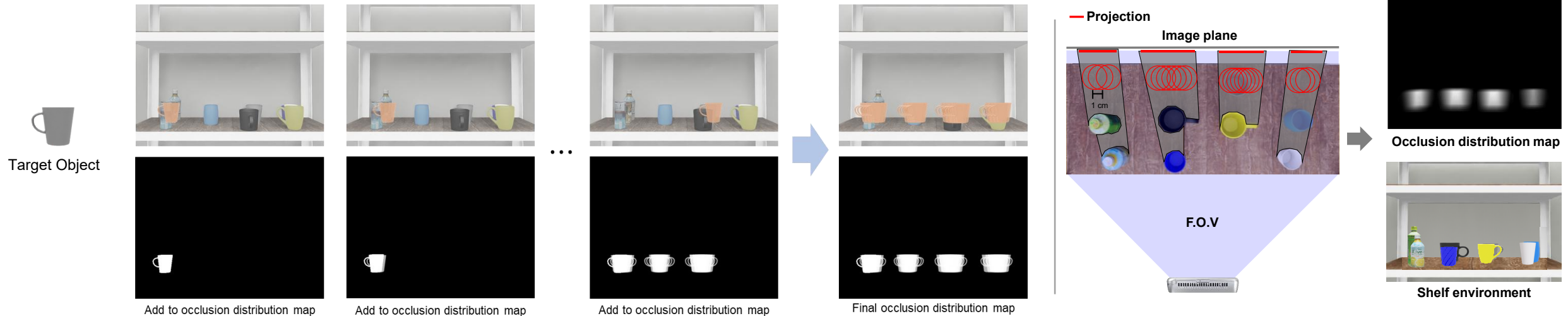
Column-wise Exploration Selector

Data generation for training the Column-wise Exploration Selector



- **(Occlusion distribution map)** Obstacle dummy에 의해 target object가 가려질 수 있는 영역에 대한 평가

- Target object를 각 raw마다 일정 간격(1cm)으로 translation 및 rotation(45°) 시켜서 obstacle dummy에 의해 가려지면 occlusion distribution map에 추가
- obstacle dummy에 의해 target object가 많이 가려질수록 해당 지역이 흰색에 가까워 짐

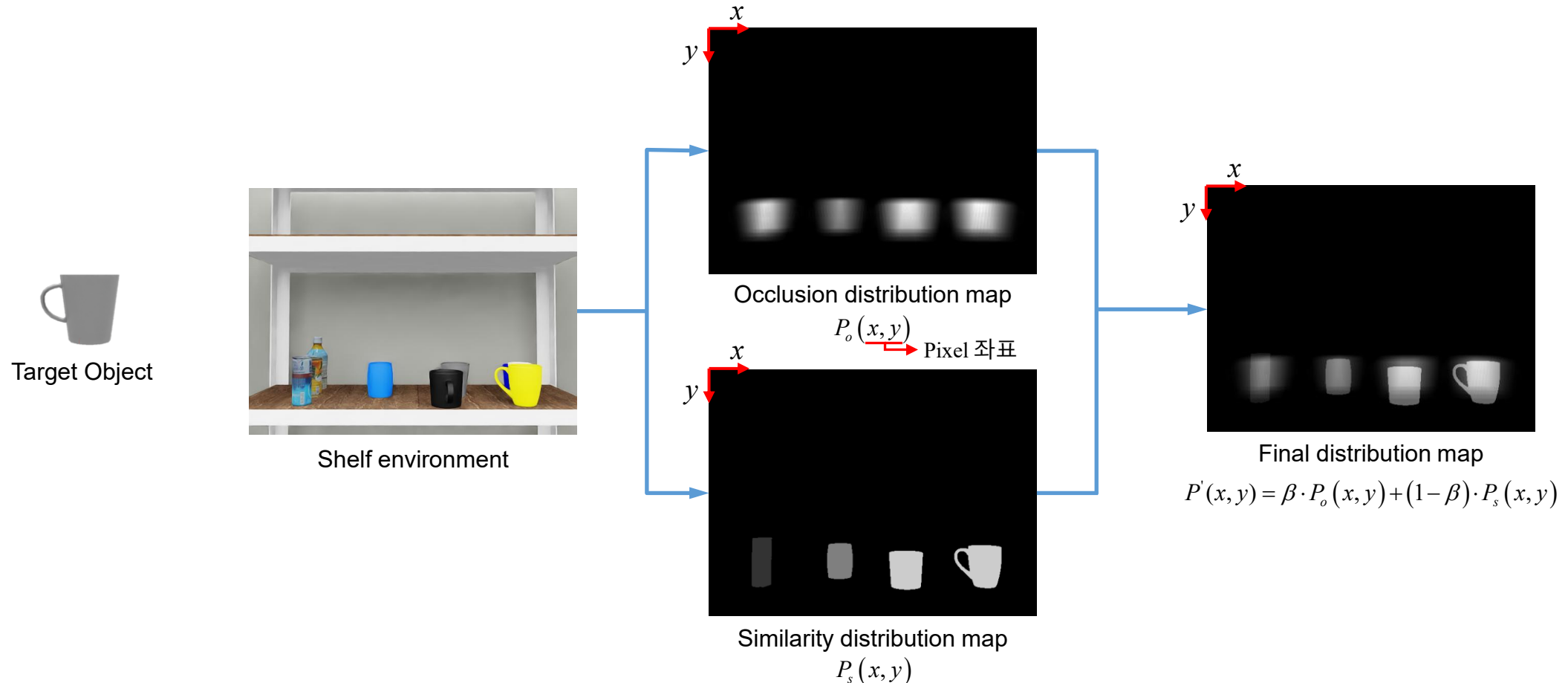


Column-wise Exploration Selector

Data generation for training the Column-wise Exploration Selector

- 최종 similarity distribution map 생성

- Obstacle dummy에 의해 생성된 occlusion distribution map과 관측 가능한 object들의 similarity distribution map을 β 의 비율로 합쳐서 최종 학습을 위한 정답 distribution map 생성



Column-wise Exploration Selector

1D Probability Distribution Map

- 2D 형태의 distribution map을 확률 계산을 위해 1D 형태의 distribution map으로 변환

- 정규화

$$p_{norm}(i, j) = \frac{p(i, j)}{\max(p)} \quad p \in \mathbb{R}^{H \times W} \quad i \in [1, H], j \in [1, W]$$

- 각 픽셀별 가중치 부여

$$p_{weighted}(i, j) = p_{norm}(i, j) \cdot e^{-g(1-p_{norm}(i, j))} \quad g: gain$$

- 1D-Probability Distribution Map(1D-PDM) 생성

$$P_{current} = \sum_{i=1}^H p_{weighted}(i, j)$$



RGB



2D-PDM



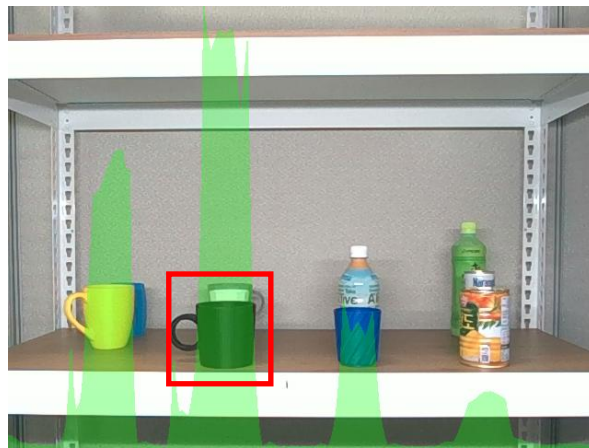
1D-PDM

Column-wise Exploration Selector

1D Probability Distribution Map

- N-1 번째 probability distribution map과 N번째 probability distribution map을 결합하여 최종 distribution을 생성

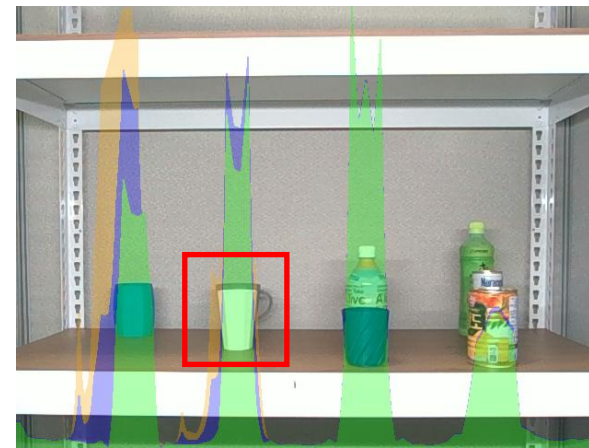
$$P_{final} = \sum_{j=1}^W (\gamma P_{current} + (1-\gamma) P_{previous}) \quad \gamma : \text{learning rate}$$



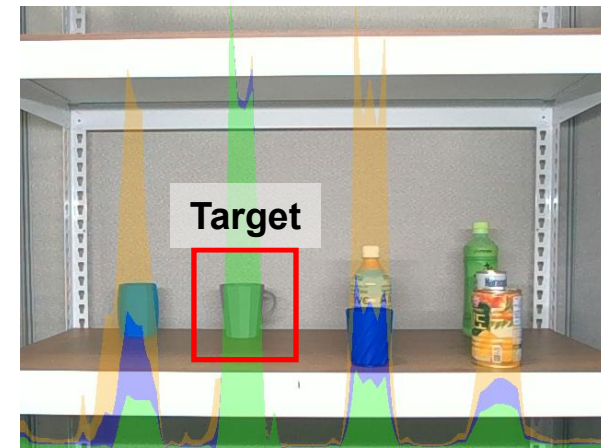
Step 1



Step 2



Step 3



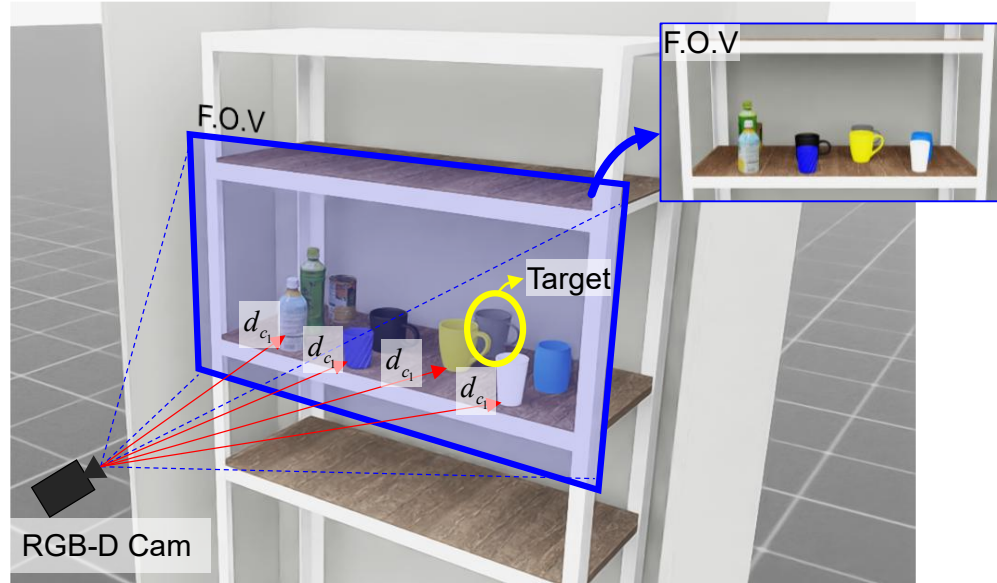
Step 4



- 1D-PDM at the N-1th step
- 1D-PDM at the nth step
- Final 1D-PDM at the nth step
- The object selected for removal at each step.

DRL-based Action Selector

MDP formulation



- **Observation** ($o \in O$) : Agent가 현재 관측하는 환경 정보

$$P = \begin{bmatrix} P_{c_1} & P_{c_2} & P_{c_3} & P_{c_4} \end{bmatrix} \quad \text{Maximum 1D-PDM value for each column}$$

$$D = \begin{bmatrix} d_{c_1} & d_{c_2} & d_{c_3} & d_{c_4} \end{bmatrix} \quad \text{Distance from the camera to the object for each column}$$

$$O = \begin{bmatrix} O_{id_{c_1}} & O_{id_{c_2}} & O_{id_{c_3}} & O_{id_{c_4}} \end{bmatrix} \quad \text{Object ID of the front object in each column}$$

$$T = \begin{bmatrix} T_{id} \end{bmatrix} \quad \text{Object ID of the front object in each column}$$

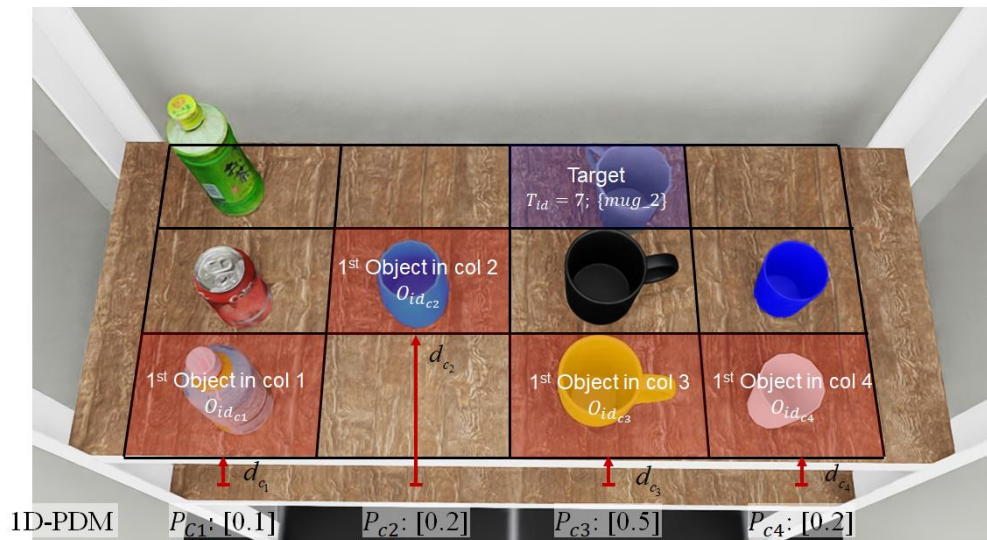
$$A_{t-1} = \begin{bmatrix} a_{t-1} & c_{t-1} \end{bmatrix} \quad \text{Previous (t-1) step action}$$

- **Action** ($a \in A$) : Agent의 행동 집합

$$A = \begin{bmatrix} a & c \end{bmatrix}$$

Target column for the action

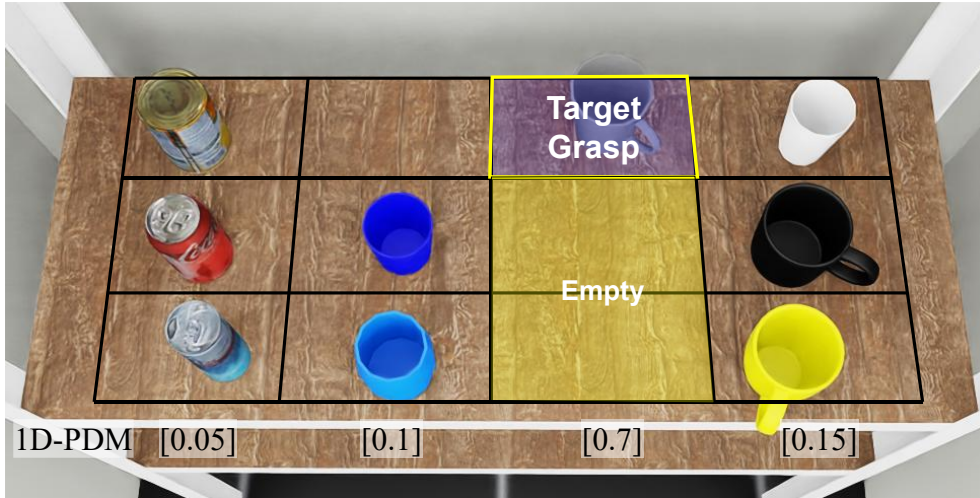
Action(Grasping: 0 / Sweeping right: 1 / Sweeping left: 2)



DRL-based Action Selector

MDP formulation

- **Reward (r):** Agent 행동에 따른 환경의 변화에 대한 보상



λ	Scales of each reward and penalty
P_t	1D-PDM at step t
d_t^{col}	Distance to the selected column at step t
d_t^{left}	Distance to the left column at step t
d_t^{right}	Distance to the right column at step t
o_t^{col}	Object ID of the selected column at step t
o_{t-1}	Object ID of the selected column at step t-1
a_t^{type}	Action type at step t
$Top - 2(P_t)$	The set of two columns with the highest 1D-PDM values at step t

- **r_{tg} : Target Grasping**

$$r_{tg} = \lambda_{tg} \cdot 1 \{ a_t^{type} = 0 \wedge o_t^{col} = T \wedge o_t^{col} \neq -1 \}$$

- **r_{swr} : Sweeping Right Reward**

$$r_{swr} = \lambda_{swr} \cdot 1 \{ a_t^{type} = 1 \wedge c_t \neq 3 \wedge o_t^{col} \neq -1 \wedge d_t^{right} > d_t^{col} \wedge c_t \in \text{Top-2}(P_t) \}$$

- **r_{swl} : Sweeping Left Reward**

$$r_{swl} = \lambda_{swl} \cdot 1 \{ a_t^{type} = 2 \wedge c_t \neq 0 \wedge o_t^{col} \neq -1 \wedge d_t^{left} > d_t^{col} \wedge c_t \in \text{Top-2}(P_t) \}$$

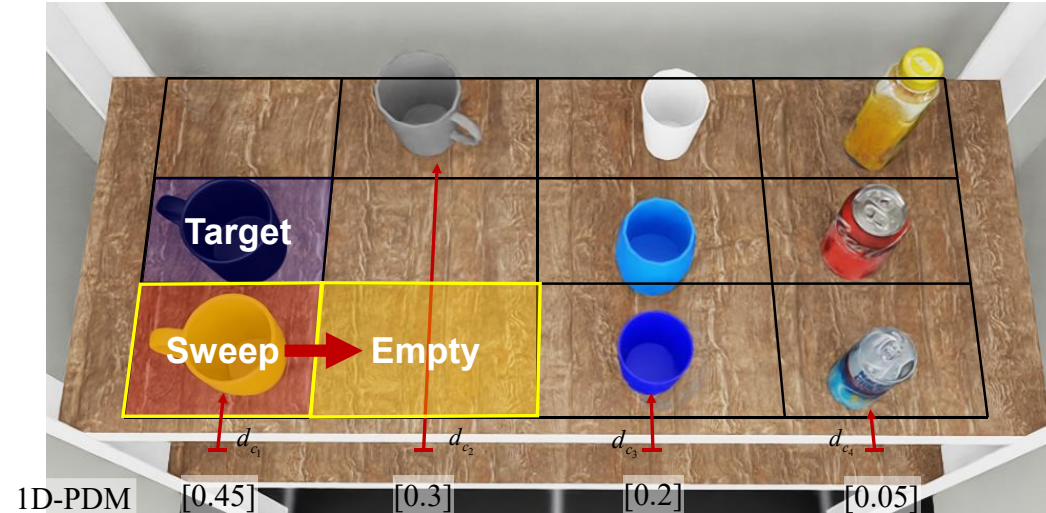
- **r_{gr} : Non-target Grasping Reward**

$$r_{gr} = \lambda_{gr} \cdot 1 \{ a_t^{type} = 0 \wedge o_t^{col} \neq T \wedge o_t^{col} \neq -1 \wedge d_t^{right} \leq d_t^{col} \wedge d_t^{left} \leq d_t^{col} \wedge c_t \in \text{Top-2}(P_t) \}$$

DRL-based Action Selector

MDP formulation

- **Reward (r):** Agent 행동에 따른 환경의 변화에 대한 보상



λ	Scales of each reward and penalty
P_t	1D-PDM at step t
d_t^{col}	Distance to the selected column at step t
d_t^{left}	Distance to the left column at step t
d_t^{right}	Distance to the right column at step t
o_t^{col}	Object ID of the selected column at step t
o_{t-1}	Object ID of the selected column at step t-1
a_t^{type}	Action type at step t
$Top - 2(P_t)$	The set of two columns with the highest 1D-PDM values at step t

- **r_{tg} : Target Grasping**

$$r_{tg} = \lambda_{tg} \cdot 1 \{ a_t^{type} = 0 \wedge o_t^{col} = T \wedge o_t^{col} \neq -1 \}$$

- **r_{swr} : Sweeping Right Reward**

$$r_{swr} = \lambda_{swr} \cdot 1 \{ a_t^{type} = 1 \wedge c_t \neq 3 \wedge o_t^{col} \neq -1 \wedge d_t^{right} > d_t^{col} \wedge c_t \in \text{Top-2}(P_t) \}$$

- **r_{swl} : Sweeping Left Reward**

$$r_{swl} = \lambda_{swl} \cdot 1 \{ a_t^{type} = 2 \wedge c_t \neq 0 \wedge o_t^{col} \neq -1 \wedge d_t^{left} > d_t^{col} \wedge c_t \in \text{Top-2}(P_t) \}$$

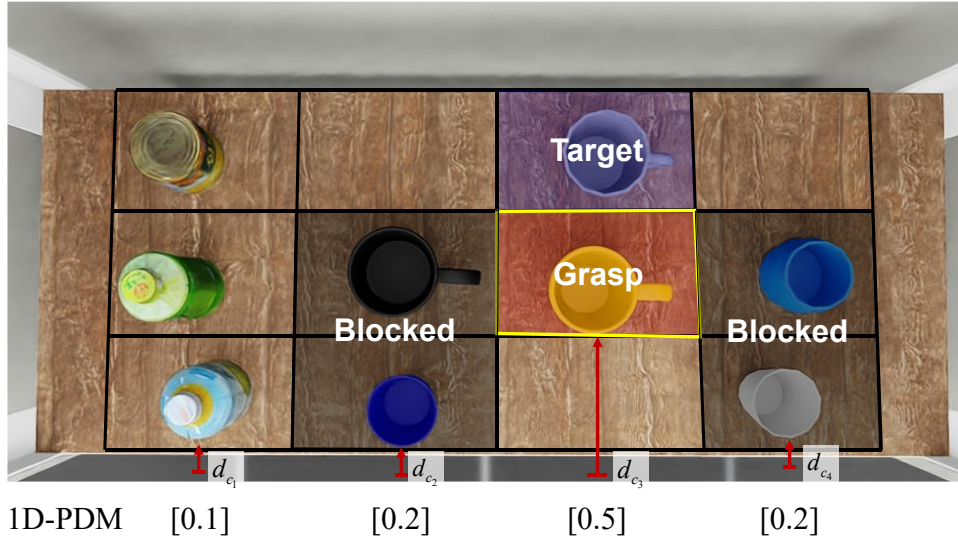
- **r_{gr} : Non-target Grasping Reward**

$$r_{gr} = \lambda_{gr} \cdot 1 \{ a_t^{type} = 0 \wedge o_t^{col} \neq T \wedge o_t^{col} \neq -1 \wedge d_t^{right} \leq d_t^{col} \wedge d_t^{left} \leq d_t^{col} \wedge c_t \in \text{Top-2}(P_t) \}$$

DRL-based Action Selector

MDP formulation

- **Reward (r):** Agent 행동에 따른 환경의 변화에 대한 보상



λ	Scales of each reward and penalty
P_t	1D-PDM at step t
d_t^{col}	Distance to the selected column at step t
d_t^{left}	Distance to the left column at step t
d_t^{right}	Distance to the right column at step t
o_t^{col}	Object ID of the selected column at step t
o_{t-1}	Object ID of the selected column at step t-1
a_t^{type}	Action type at step t
$Top - 2(P_t)$	The set of two columns with the highest 1D-PDM values at step t

- r_{tg} : Target Grasping

$$r_{tg} = \lambda_{tg} \cdot 1 \{ a_t^{type} = 0 \wedge o_t^{col} = T \wedge o_t^{col} \neq -1 \}$$

- r_{swr} : Sweeping Right Reward

$$r_{swr} = \lambda_{swr} \cdot 1 \{ a_t^{type} = 1 \wedge c_t \neq 3 \wedge o_t^{col} \neq -1 \wedge d_t^{right} > d_t^{col} \wedge c_t \in \text{Top-2}(P_t) \}$$

- r_{swl} : Sweeping Left Reward

$$r_{swl} = \lambda_{swl} \cdot 1 \{ a_t^{type} = 2 \wedge c_t \neq 0 \wedge o_t^{col} \neq -1 \wedge d_t^{left} > d_t^{col} \wedge c_t \in \text{Top-2}(P_t) \}$$

- r_{gr} : Non-target Grasping Reward

$$r_{gr} = \lambda_{gr} \cdot 1 \{ a_t^{type} = 0 \wedge o_t^{col} \neq T \wedge o_t^{col} \neq -1 \wedge d_t^{right} \leq d_t^{col} \wedge d_t^{left} \leq d_t^{col} \wedge c_t \in \text{Top-2}(P_t) \}$$

DRL-based Action Selector

MDP formulation

- **Penalty (p):** Agent의 잘못된 선택이나 목표와 상충하는 행동에 대해 부여되는 벌점



λ	Scales of each reward and penalty
P_t	1D-PDM at step t
d_t^{col}	Distance to the selected column at step t
d_t^{left}	Distance to the left column at step t
d_t^{right}	Distance to the right column at step t
o_t^{col}	Object ID of the selected column at step t
o_{t-1}	Object ID of the selected column at step t-1
a_t^{type}	Action type at step t
$Top - 2(P_t)$	The set of two columns with the highest 1D-PDM values at step t

- p_{lpg} : **Low Probability Grasping Penalty**

$$p_{lpg} = \lambda_{lpg} \cdot 1 \{a_t^{type} = 0 \wedge c_t \notin Top-2(P_t)\}$$

- p_{lpswr} : **Low Probability Sweeping Right Penalty**

$$p_{lpswr} = \lambda_{lpswr} \cdot 1 \{a_t^{type} = 1 \wedge c_t \notin Top-2(P_t)\}$$

- p_{lpswl} : **Low Probability Sweeping Left Penalty**

$$p_{lpswl} = \lambda_{lpswl} \cdot 1 \{a_t^{type} = 2 \wedge c_t \notin Top-2(P_t)\}$$

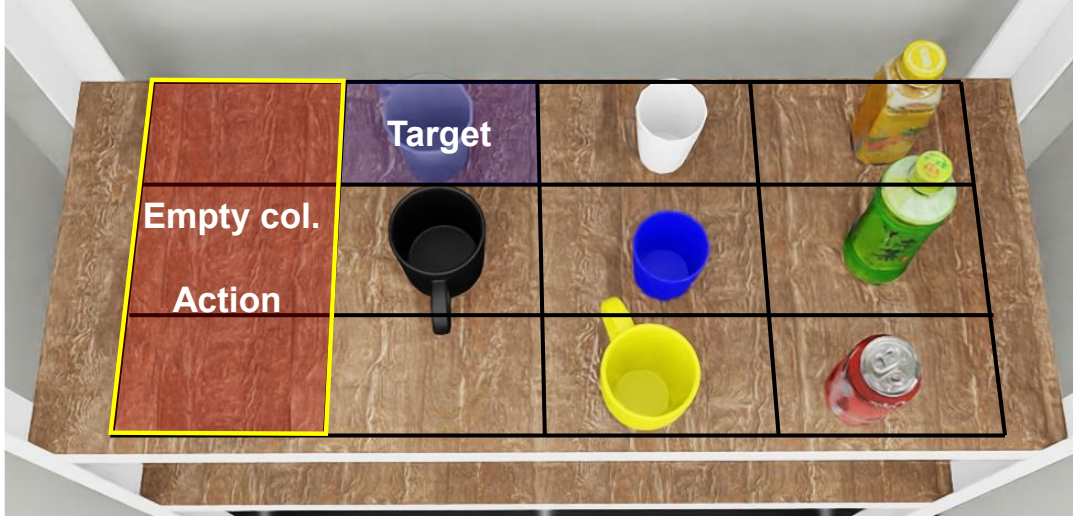
- p_{ec} : **Empty Column Action Penalty**

$$p_{ec} = \lambda_{ec} \cdot 1 \{o_t^{col} \neq -1\}$$

DRL-based Action Selector

MDP formulation

- **Penalty (p):** Agent의 잘못된 선택이나 목표와 상충하는 행동에 대해 부여되는 벌점



λ	Scales of each reward and penalty
P_t	1D-PDM at step t
d_t^{col}	Distance to the selected column at step t
d_t^{left}	Distance to the left column at step t
d_t^{right}	Distance to the right column at step t
o_t^{col}	Object ID of the selected column at step t
o_{t-1}	Object ID of the selected column at step t-1
a_t^{type}	Action type at step t
$Top - 2(P_t)$	The set of two columns with the highest 1D-PDM values at step t

- **p_{lpg} : Low Probability Grasping Penalty**

$$p_{lpg} = \lambda_{lpg} \cdot 1\{a_t^{type} = 0 \wedge c_t \notin Top-2(P_t)\}$$

- **p_{lpswr} : Low Probability Sweeping Right Penalty**

$$p_{lpswr} = \lambda_{lpswr} \cdot 1\{a_t^{type} = 1 \wedge c_t \notin Top-2(P_t)\}$$

- **p_{lpswl} : Low Probability Sweeping Left Penalty**

$$p_{lpswl} = \lambda_{lpswl} \cdot 1\{a_t^{type} = 2 \wedge c_t \notin Top-2(P_t)\}$$

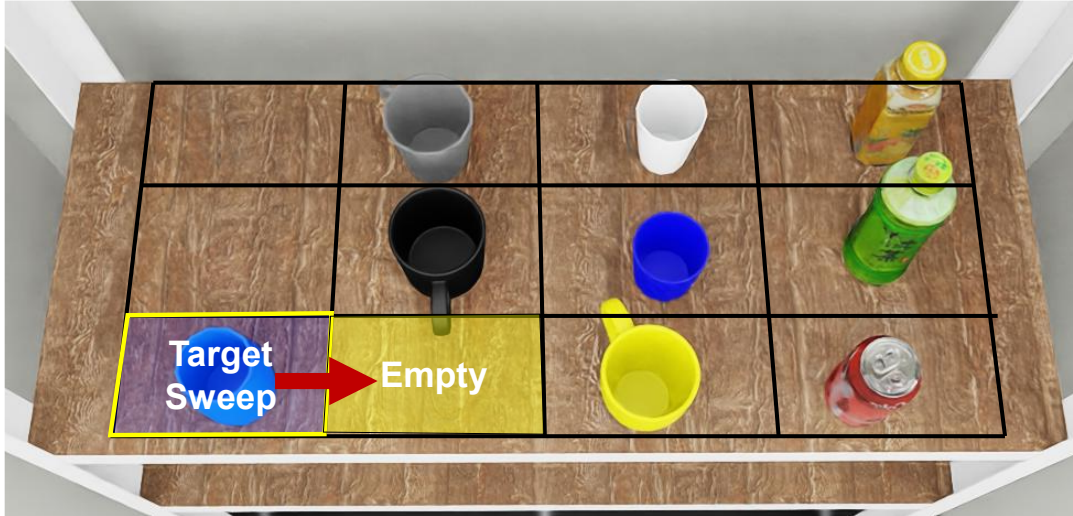
- **p_{ec} : Empty Column Action Penalty**

$$p_{ec} = \lambda_{ec} \cdot 1\{o_t^{col} \neq -1\}$$

DRL-based Action Selector

MDP formulation

- **Penalty (p):** Agent의 잘못된 선택이나 목표와 상충하는 행동에 대해 부여되는 벌점



λ	Scales of each reward and penalty
P_t	1D-PDM at step t
d_t^{col}	Distance to the selected column at step t
d_t^{left}	Distance to the left column at step t
d_t^{right}	Distance to the right column at step t
o_t^{col}	Object ID of the selected column at step t
o_{t-1}	Object ID of the selected column at step $t-1$
a_t^{type}	Action type at step t
$Top - 2(P_t)$	The set of two columns with the highest 1D-PDM values at step t

- **p_{ts} : Target Sweeping Penalty**

$$p_{ts} = \lambda_{ts} \cdot 1 \left\{ \left(a_t^{type} = 1 \vee a_t^{type} = 2 \right) \wedge o_t^{col} = T \right\}$$

- **p_{lpg} : Grasping When Sweeping Possible Penalty**

$$p_{gns} = \lambda_{gns} \cdot 1 \left\{ a_t^{type} = 0 \wedge \text{Sweeping right or left was possible at } t \right\}$$

- **p_{rsw} : Sweeping Same Object Again Penalty**

$$p_{rsw} = \lambda_{rsw} \cdot 1 \left\{ \left(a_{t-1}^{type} = 1 (or\ 2) \right) \wedge \left(a_t^{type} = 2 (or\ 1) \right) \wedge (o_{t-1} = o_t) \right\}$$

- **p_{rsg} : Grasping Previously Swept Object Penalty**

$$p_{rsg} = \lambda_{rsg} \cdot 1 \left\{ \left(a_{t-1}^{type} = 1 \vee 2 \right) \wedge (o_{t-1} = o_t) \wedge a_t = 0 \right\}$$

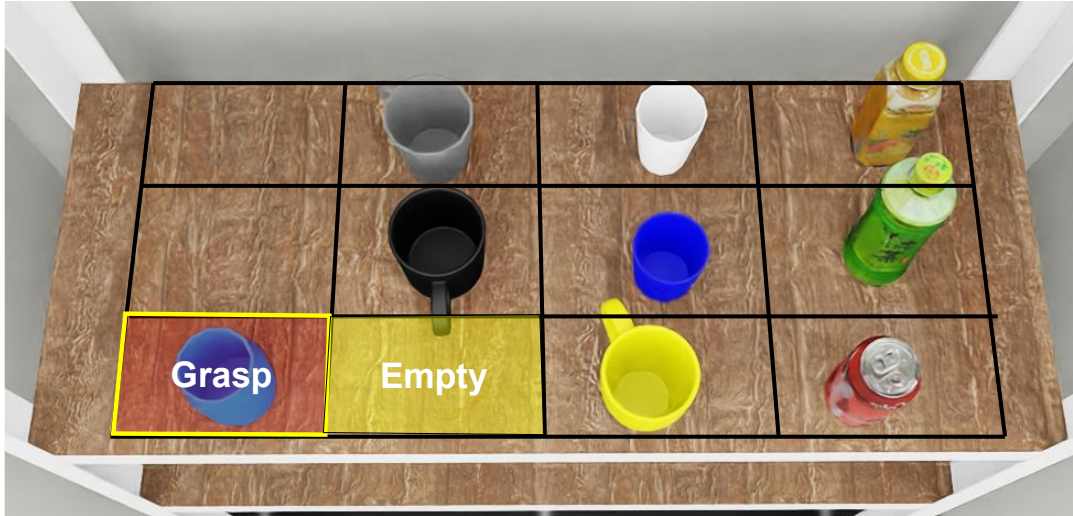
- **p_{ts} : Termination Penalty**

$$p_{term} = \lambda_{term} \cdot 1 \left\{ Out\ of\ workspace\ or\ collision \right\}$$

DRL-based Action Selector

MDP formulation

- **Penalty (p):** Agent의 잘못된 선택이나 목표와 상충하는 행동에 대해 부여되는 벌점



λ	Scales of each reward and penalty
P_t	1D-PDM at step t
d_t^{col}	Distance to the selected column at step t
d_t^{left}	Distance to the left column at step t
d_t^{right}	Distance to the right column at step t
o_t^{col}	Object ID of the selected column at step t
o_{t-1}	Object ID of the selected column at step t-1
a_t^{type}	Action type at step t
$Top - 2(P_t)$	The set of two columns with the highest 1D-PDM values at step t

- **p_{ts} : Target Sweeping Penalty**

$$p_{ts} = \lambda_{ts} \cdot 1 \left\{ \left(a_t^{type} = 1 \vee a_t^{type} = 2 \right) \wedge o_t^{col} = T \right\}$$

- **p_{lpg} : Grasping When Sweeping Possible Penalty**

$$p_{gns} = \lambda_{gns} \cdot 1 \left\{ a_t^{type} = 0 \wedge \text{Sweeping right or left was possible at } t \right\}$$

- **p_{rsw} : Sweeping Same Object Again Penalty**

$$p_{rsw} = \lambda_{rsw} \cdot 1 \left\{ \left(a_{t-1}^{type} = 1 (or\ 2) \right) \wedge \left(a_t^{type} = 2 (or\ 1) \right) \wedge (o_{t-1} = o_t) \right\}$$

- **p_{rsg} : Grasping Previously Swept Object Penalty**

$$p_{rsg} = \lambda_{rsg} \cdot 1 \left\{ \left(a_{t-1}^{type} = 1 \vee 2 \right) \wedge (o_{t-1} = o_t) \wedge a_t = 0 \right\}$$

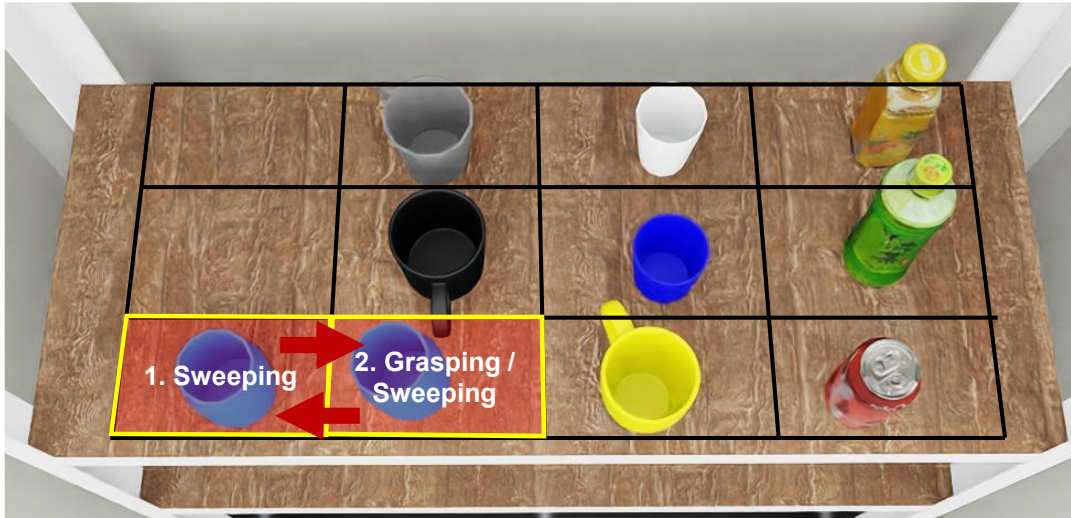
- **p_{ts} : Termination Penalty**

$$p_{term} = \lambda_{term} \cdot 1 \left\{ Out\ of\ workspace\ or\ collision \right\}$$

DRL-based Action Selector

MDP formulation

- **Penalty (p):** Agent의 잘못된 선택이나 목표와 상충하는 행동에 대해 부여되는 벌점



λ	Scales of each reward and penalty
P_t	1D-PDM at step t
d_t^{col}	Distance to the selected column at step t
d_t^{left}	Distance to the left column at step t
d_t^{right}	Distance to the right column at step t
o_t^{col}	Object ID of the selected column at step t
o_{t-1}	Object ID of the selected column at step $t-1$
a_t^{type}	Action type at step t
$Top - 2(P_t)$	The set of two columns with the highest 1D-PDM values at step t

- **p_{ts} : Target Sweeping Penalty**

$$p_{ts} = \lambda_{ts} \cdot 1 \left\{ \left(a_t^{type} = 1 \vee a_t^{type} = 2 \right) \wedge o_t^{col} = T \right\}$$

- **p_{lpg} : Grasping When Sweeping Possible Penalty**

$$p_{gns} = \lambda_{gns} \cdot 1 \left\{ a_t^{type} = 0 \wedge \text{Sweeping right or left was possible at } t \right\}$$

- **p_{rsw} : Sweeping Same Object Again Penalty**

$$p_{rsw} = \lambda_{rsw} \cdot 1 \left\{ \left(a_{t-1}^{type} = 1 (or\ 2) \right) \wedge \left(a_t^{type} = 2 (or\ 1) \right) \wedge (o_{t-1} = o_t) \right\}$$

- **p_{rsg} : Grasping Previously Swept Object Penalty**

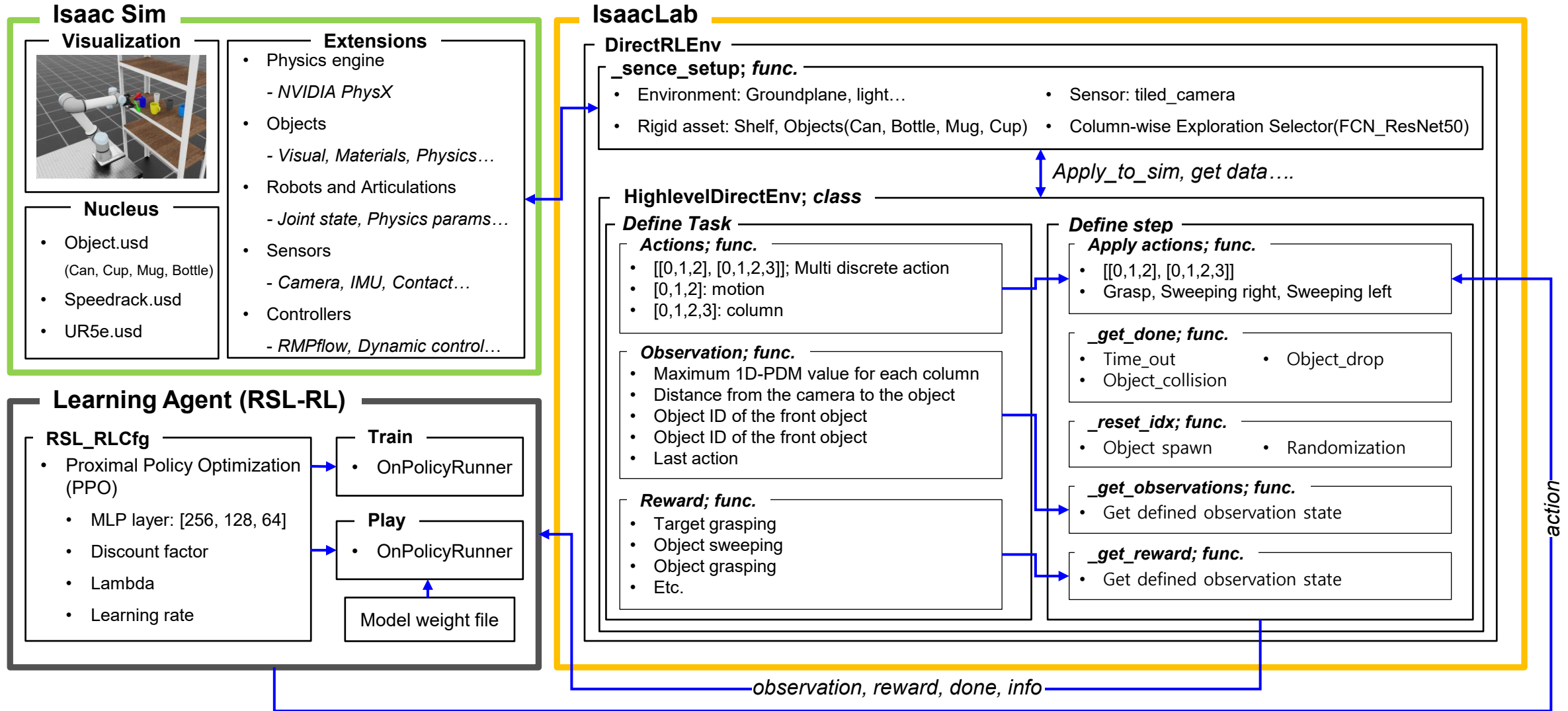
$$p_{rsg} = \lambda_{rsg} \cdot 1 \left\{ \left(a_{t-1}^{type} = 1 \vee 2 \right) \wedge (o_{t-1} = o_t) \wedge a_t = 0 \right\}$$

- **p_{ts} : Termination Penalty**

$$p_{term} = \lambda_{term} \cdot 1 \left\{ Out\ of\ workspace\ or\ collision \right\}$$

DRL-based Action Selector

DRL training framework

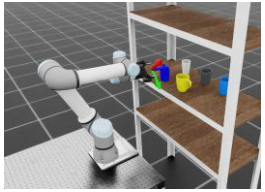


DRL-based Action Selector

DRL training framework

Isaac Sim

Visualization



Nucleus

- Object.usd
(Can, Cup, Mug, Bottle)
- Speedrack.usd
- UR5e.usd

Extensions

- Physics engine
 - NVIDIA PhysX
- Objects
 - Visual, Materials, Physics...
- Robots and Articulations
 - Joint state, Physics params...
- Sensors
 - Camera, IMU, Contact...
- Controllers
 - RMPflow, Dynamic control...

Isaac Sim

Visualization

NVIDIA Omniverse 기반으로 제작되었으며, Omniverse Kit에서 제공하는 기본 UI를 통해 USD 스테이지를 쉽게 편집 가능

Omniverse Nucleus

Omniverse Nucleus는 사용자가 자신의 클라우드를 구축할 수 있게 하며, 해당 클라우드에 저장된 파일과 데이터를 실시간 동기화 가능

Extensions

다양한 기능과 도구를 제공하며, 사용자는 로봇 시뮬레이션 환경을 설계 및 제어 가능

Learning Agent (RSL-RL)

RSL_RLCfg

- Proximal Policy Optimization (PPO)
 - MLP layer: [256, 128, 64]
 - Discount factor
 - Lambda
 - Learning rate

Train

- OnPolicyRunner

Play

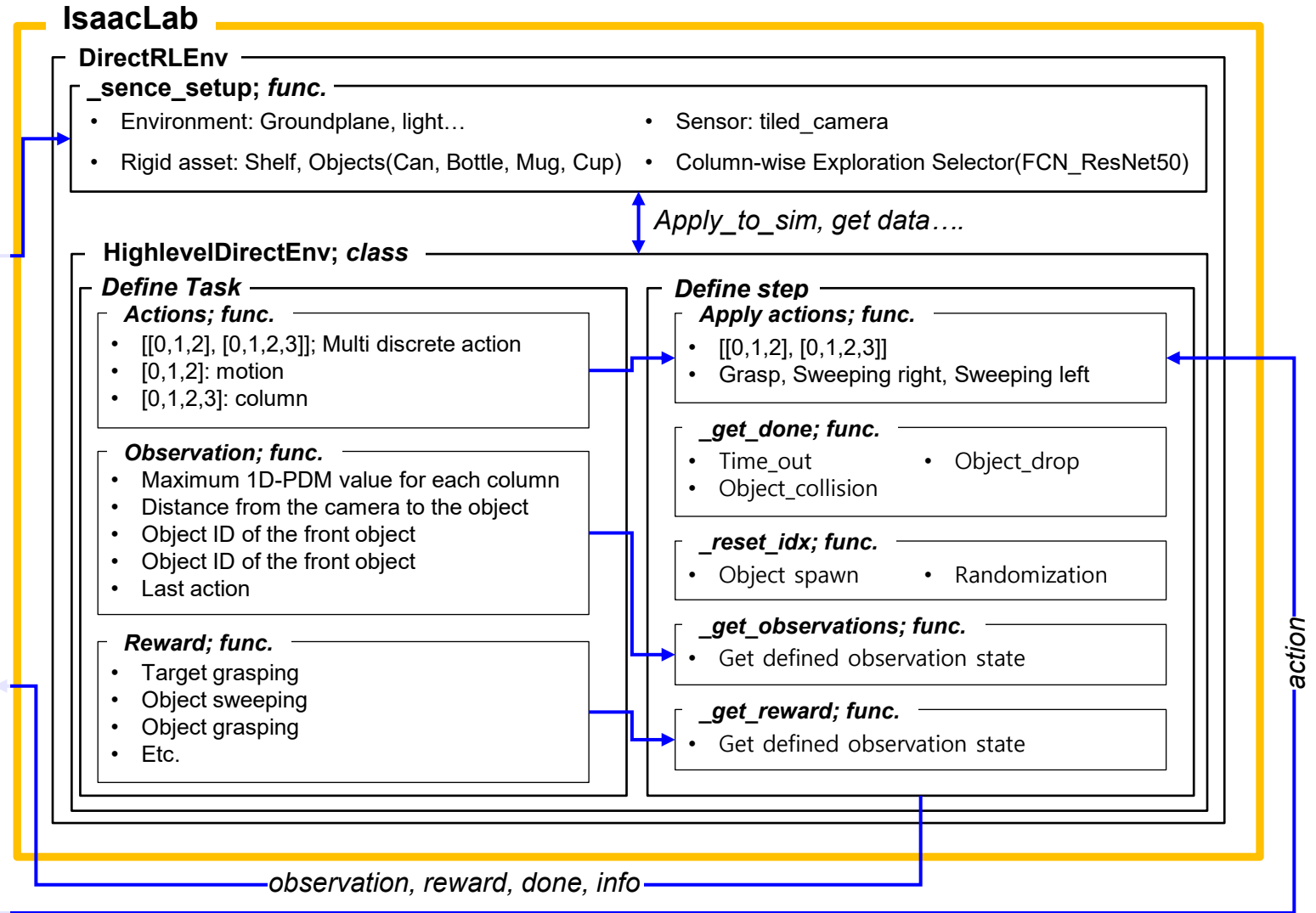
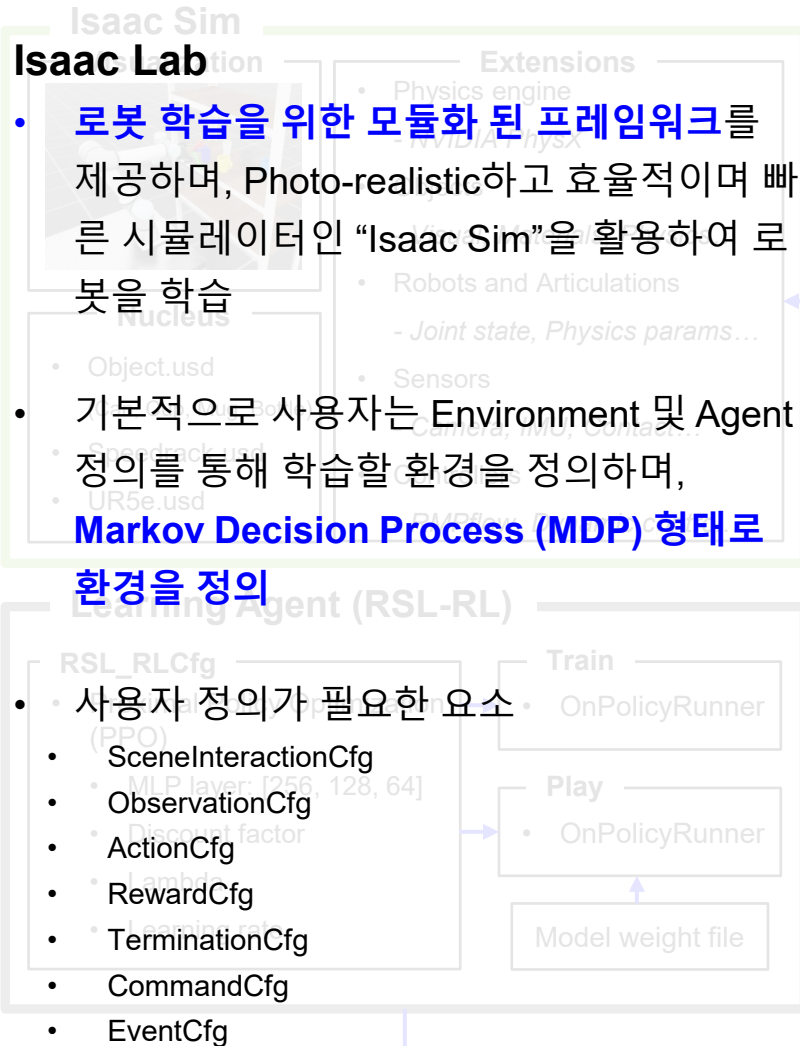
- OnPolicyRunner

Model weight file

Extension	Description
Physics engine	NVIDIA PhysX를 기반/ 중력, 마찰, 충돌 등 물리적 상호작용 모델링/ 시뮬레이션 파라미터 (시간 스텝, 충돌 감지..)
Object	객체(Box, Sphere, Cylinder, Corn...) 생성/ 사용자 정의 객체 생성/ 객체 속성(크기, 재질, 색상) 편집
Robot(Articulation)	로봇 가져오기(URDF, MJCF, USD 포맷 지원)/ 관절 제어(Position, Velocity, Torque)/ 로봇 시뮬레이션
Sensor	센서를 시뮬레이션에 추가하여 로봇의 환경 인식 지원 (Camera, LiDAR, IMU, Contact sensor, Force Sensor)
Controller	다양한 Physics object를 제어하는 api (Dynamic controller, ROS Control, Lula(FK, IK 기반 말단부 위치 및 자세 결정))

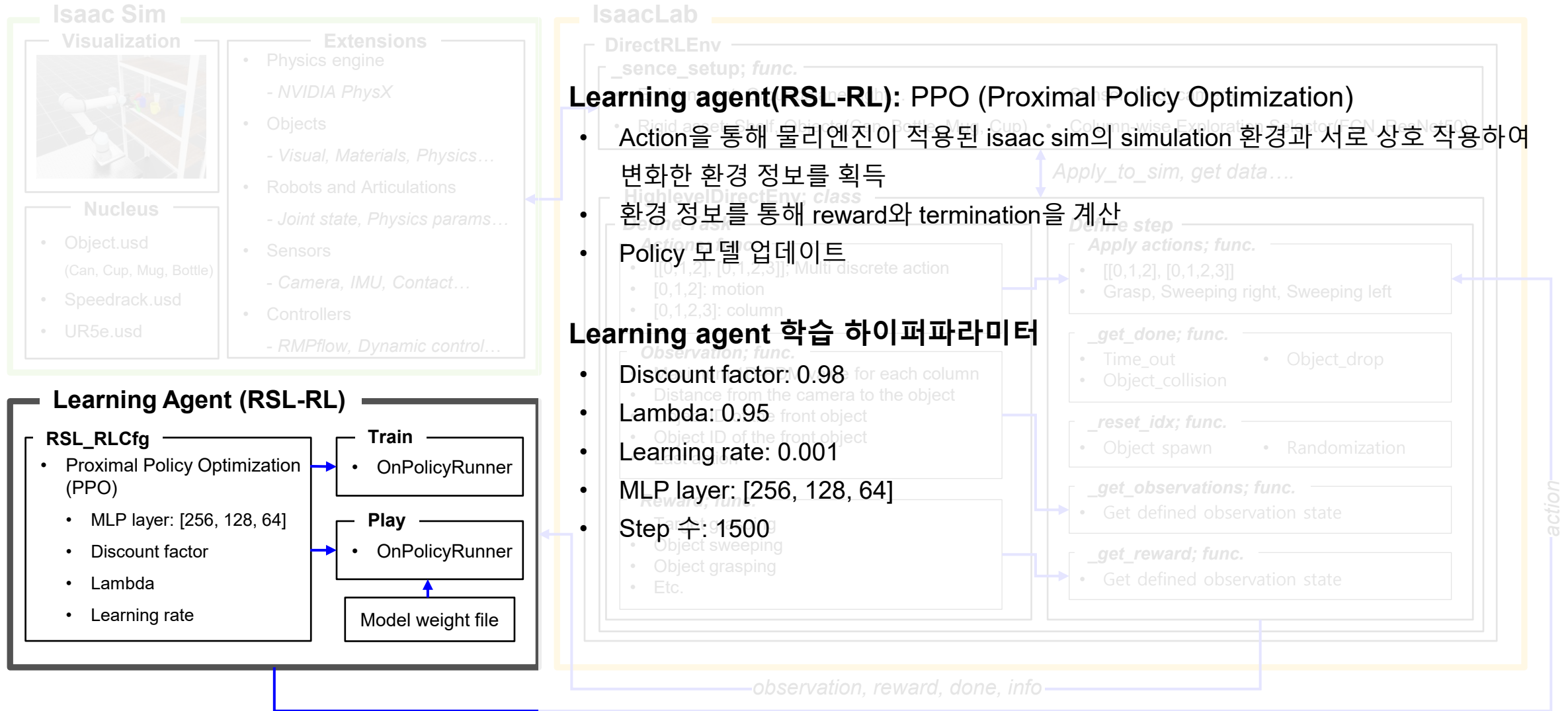
DRL-based Action Selector

DRL training framework



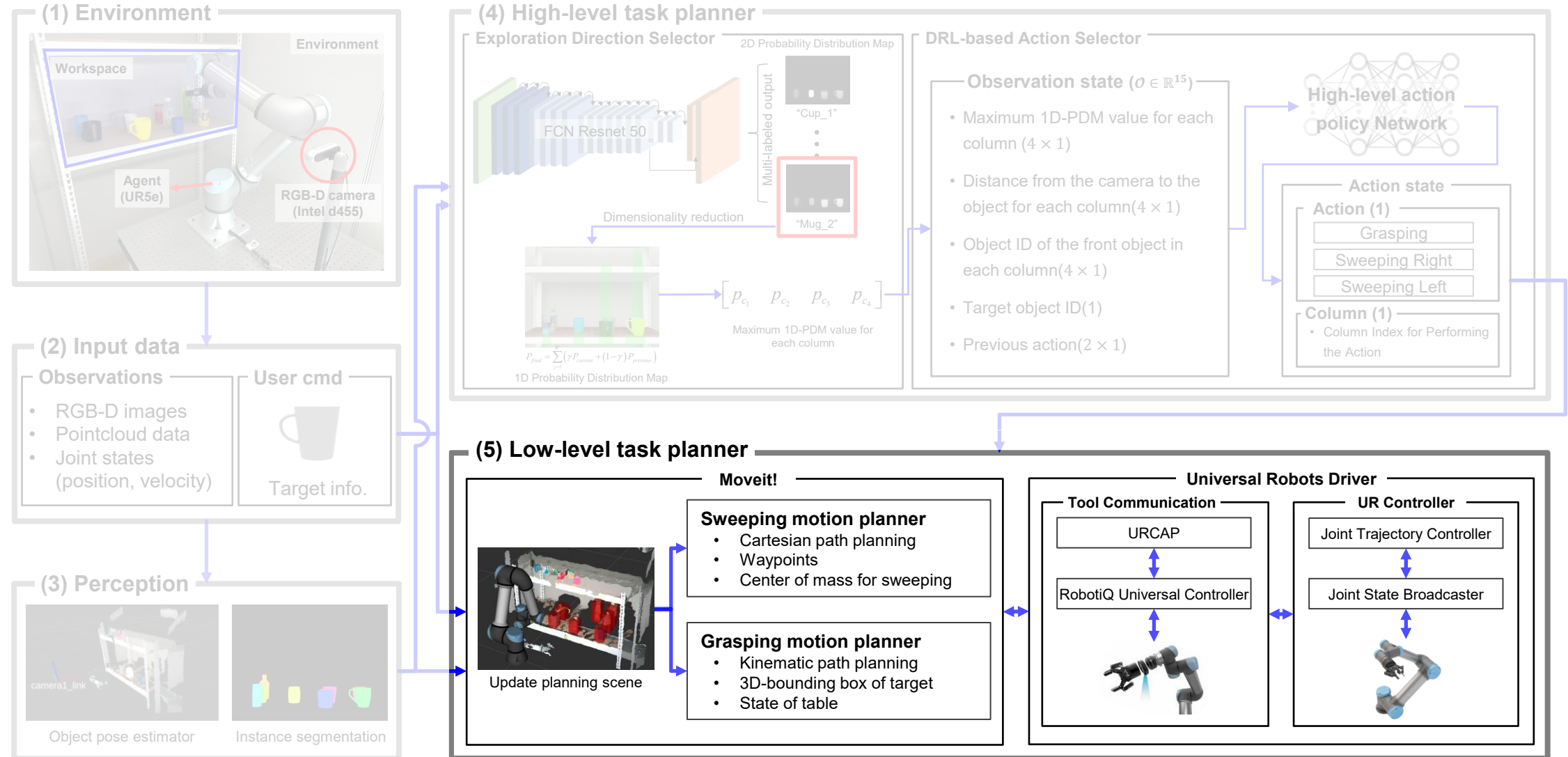
DRL-based Action Selector

DRL training framework



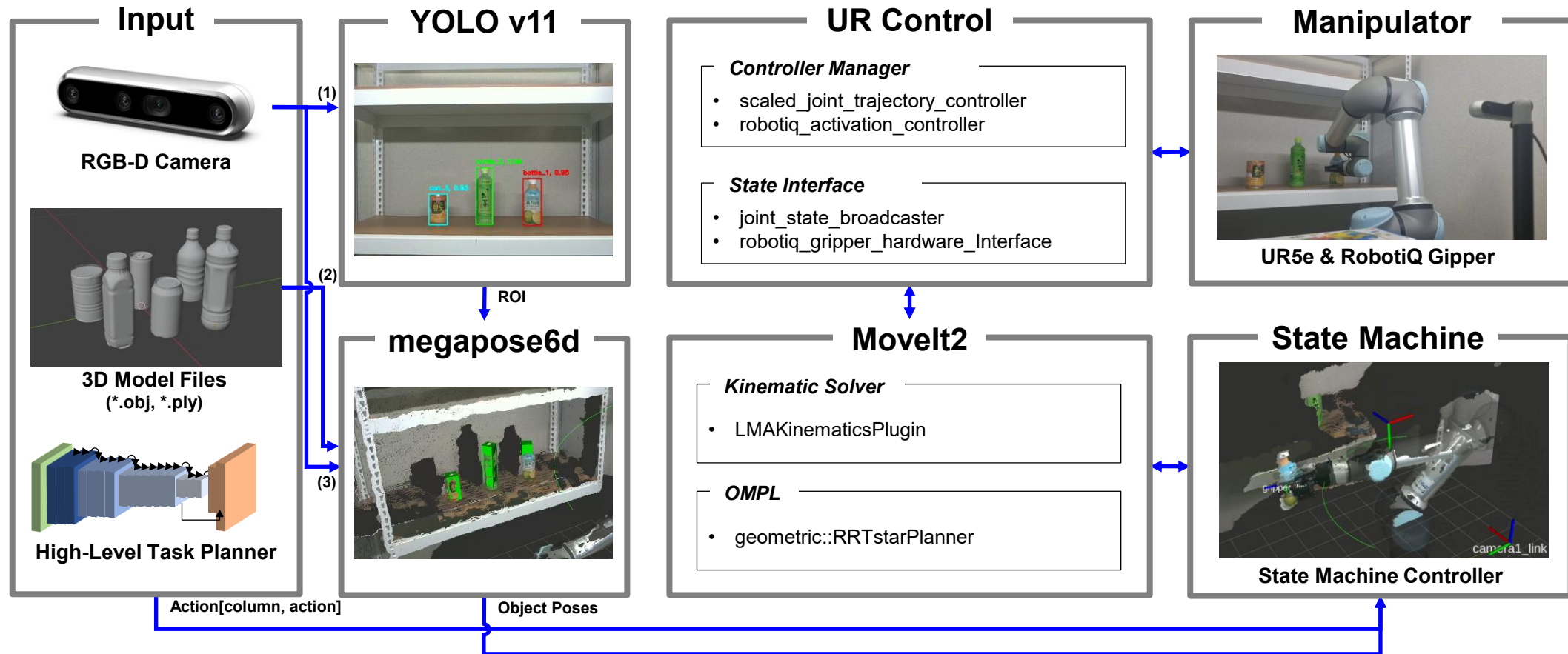
Method

Overall framework



Low-level task planner

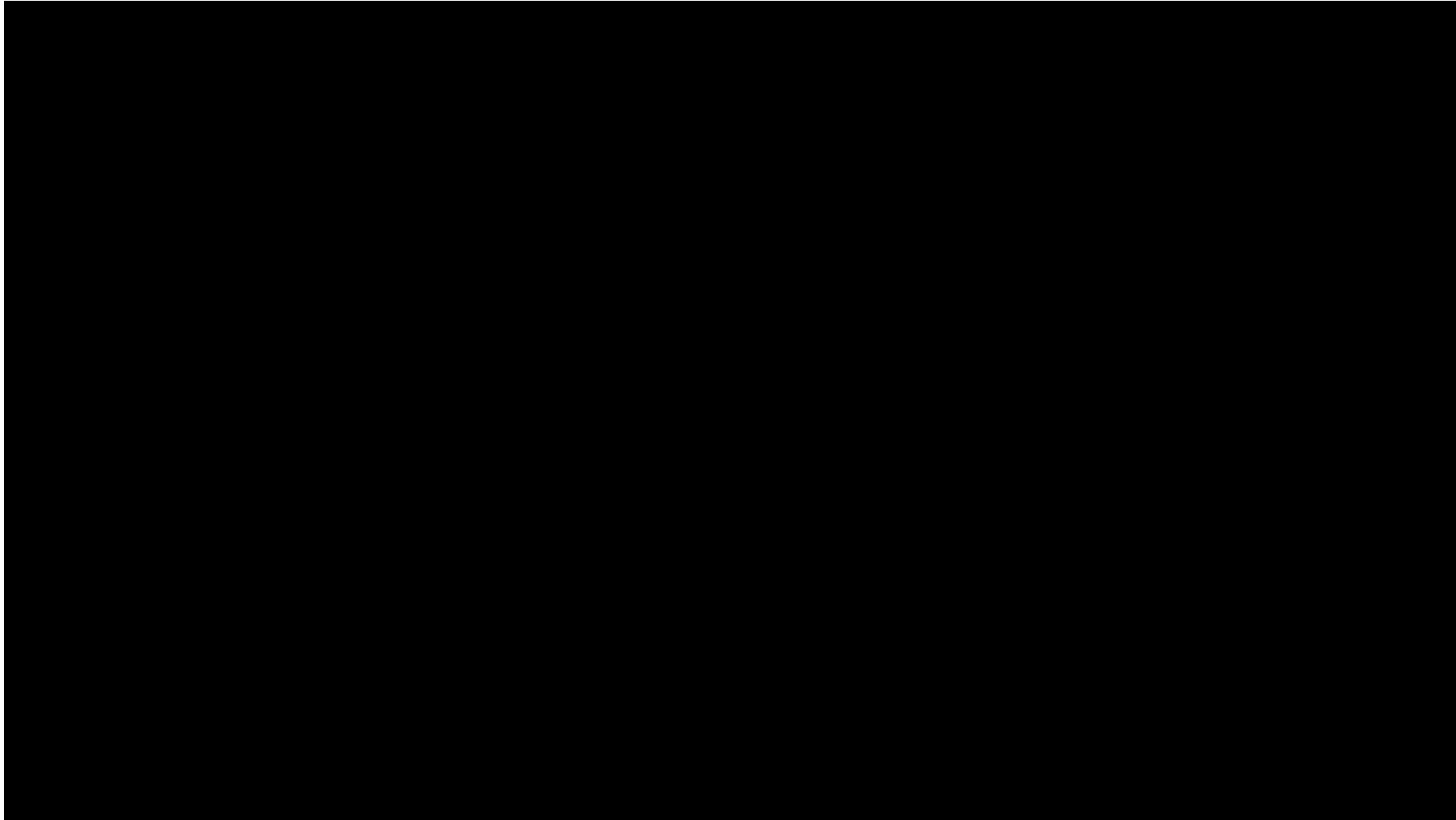
Overall framework of low-level task planner



- (1) RGB Image
- (2) CAD Model File
- (3) RGB Image, Camera Intrinsic Parameters, Depth Image(Optional)

Low-level task planner

Real-world experiment



Verification

Random Scenario



Scene #1



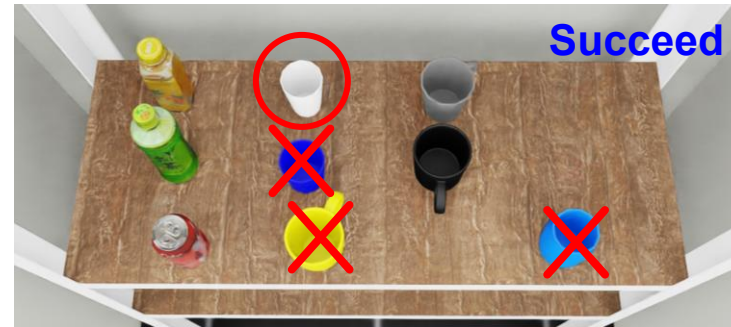
Scene #2

•
•
•



Scene #N

- **Task success rate:** 목표 물체까지 도달하여, 목표 물체 수거 여부



< Ours >

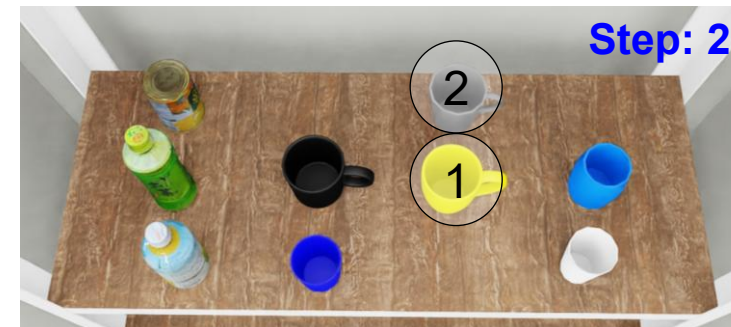


Target object



< Others >

- **Step count:** 몇번의 Searching step을 거쳐, 목표 물체를 수거 하는지 측정



< Ours >



Target object



< Others >

Experiment

Column-wise Exploration Selector

• Training data

- Training : 24,000
 - 각 class 별 2,000장
- Test : 1,200
 - 각 class 별 100장

• Hyperparameter

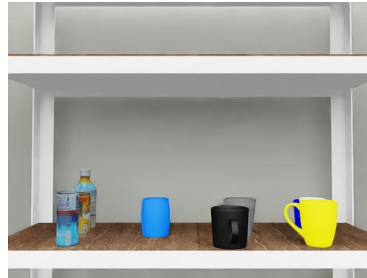
- Batch size: 16 (RTX 3090TI)
- Learning rate: 0.0001
- Epochs: 100
- Loss function: MSELoss

• Result

- Mean accuracy: 99.55%
- Mean IoU: 85.42%

• Predict result

• Simulation



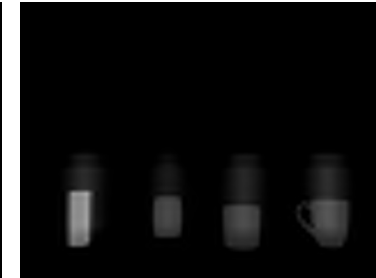
RGB



Can_2



Mug_2



Bottle_2



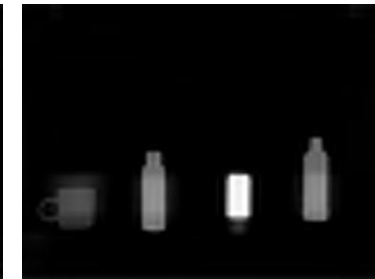
• Real-world



RGB



Mug_3



Can_1



Cup_1



Experiment

Column-wise Exploration Selector

- Shelf configuration



- Column index: [0, 1, 2, 3]
- Row index: [1st row: A, 2nd row: B, 3rd row: C]

- Target object

- Mug_2



Real-World FCN Network Experiment

Original Image



1D PDM

FCN Response

No command is given.

2D PDM

Experiment

Column-wise Exploration Selector

- Shelf configuration



- Column index: [0, 1, 2, 3]
- Row index: [1st row: A, 2nd row: B, 3rd row: C]

- Target object

- Can_3



Real-World FCN Network Experiment

Original Image



1D PDM

FCN Response

No command is given.

2D PDM

Experiment

DRL-based Action Selector

- Target object

- Mug_2



Step	1D-PDM	Action	Column
1	[6.94, 8.90, 39.15, 16.54]	grasping	3 rd (2)
2	[11.33, 15.31, 29.71, 34.50]	grasping	4 th (3)
3	[8.16, 11.35, 16.98, 36.27]	grasping	4 th (3)
4	[5.22, 7.44, 8.60, 31.09]	grasping	4 th (3)



Experiment

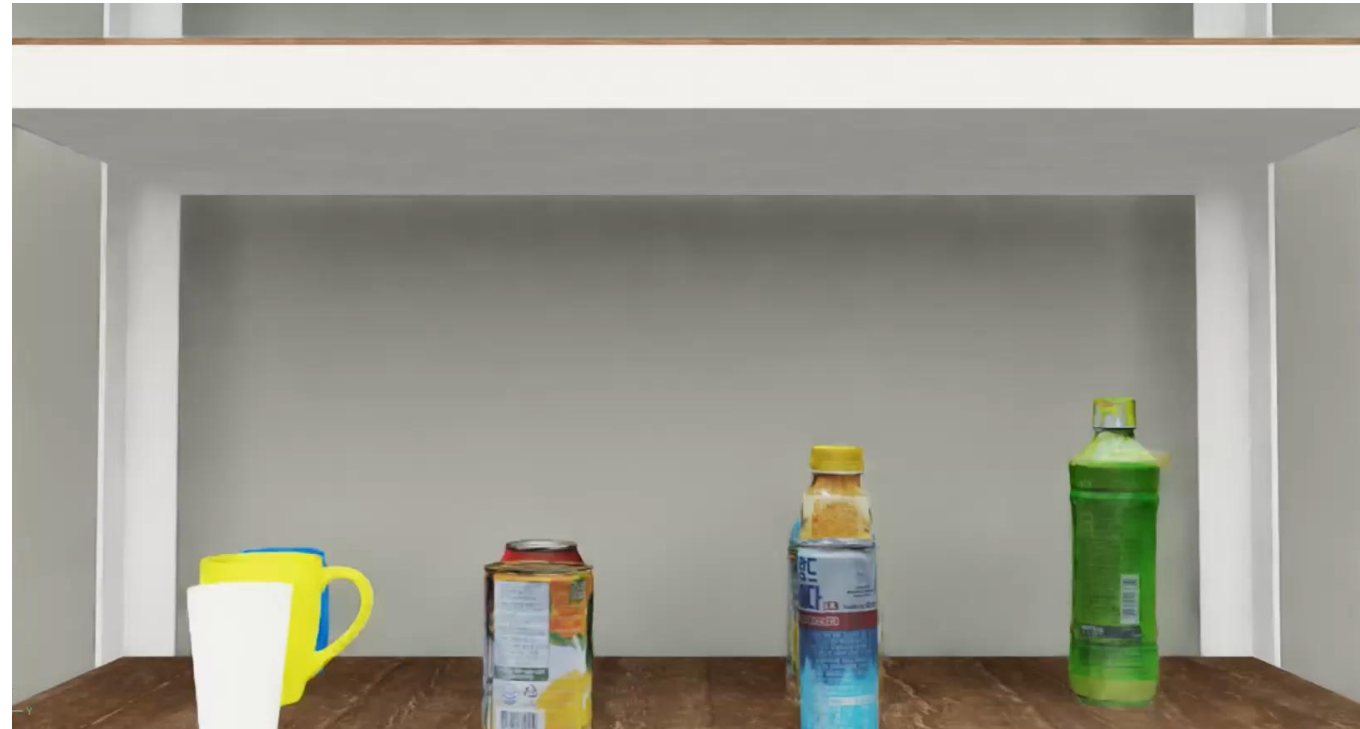
DRL-based Action Selector

- Target object

- Bottle_1



Step	1D-PDM	Action	Column
1	[15.54, 22.44, 24.29, 75.02]	grasping	4 th (3)
2	[13.31, 45.11, 54.37, 22.52]	grasping	3 rd (2)
3	[9.47, 34.04, 80.13, 6.75]	Sweeping_R	3 rd (2)
4	[5.97, 16.43, 63.88, 17.85]	grasping	3 rd (2)



Experiment

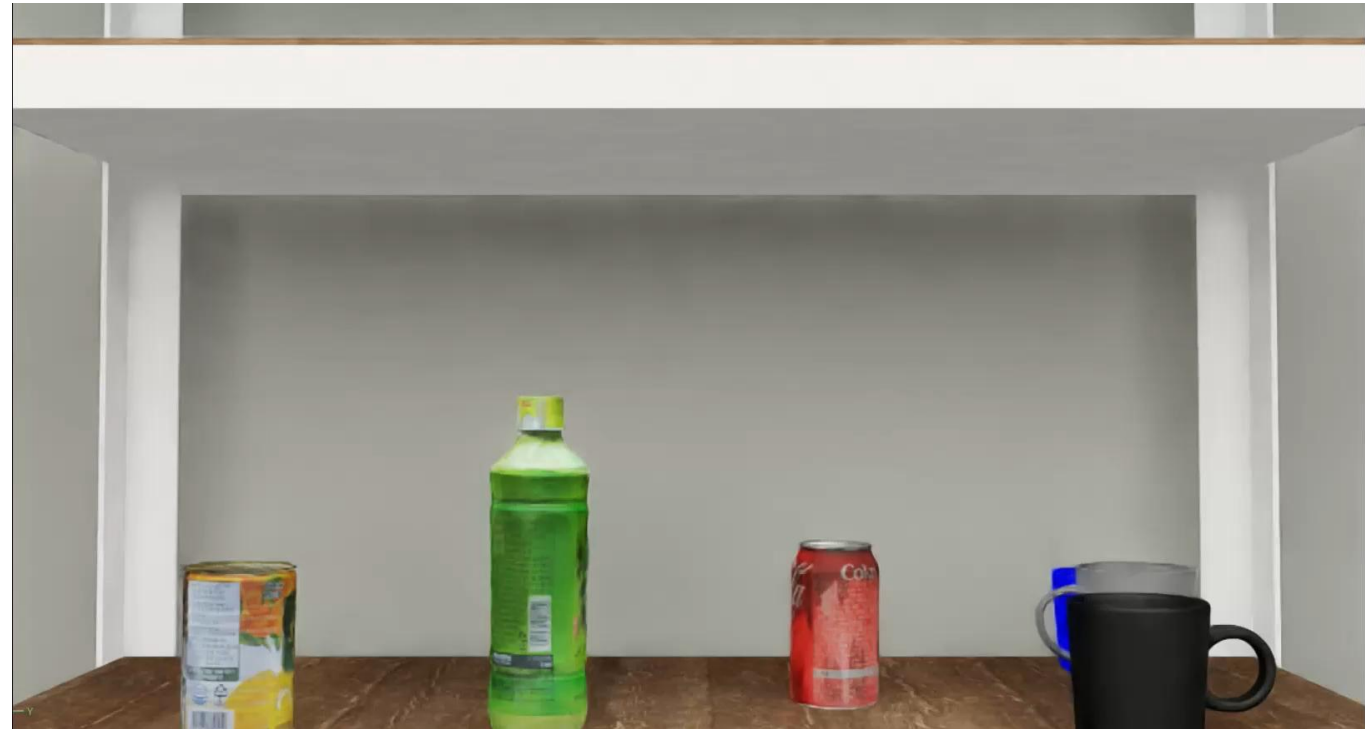
DRL-based Action Selector

- Target object

- Bottle_1



Step	1D-PDM	Action	Column
1	[28.85, 88.90, 21.20, 8.78]	grasping	1 st (0)
2	[6.36, 111.21, 24.29, 8.74]	Sweeping_L	2 nd (1)
3	[75.96, 84.45, 20.78, 8.36]	grasping	2 nd (1)
4	[43.08, 67.78, 11.50, 6.01]	grasping	2 nd (1)



Experiment

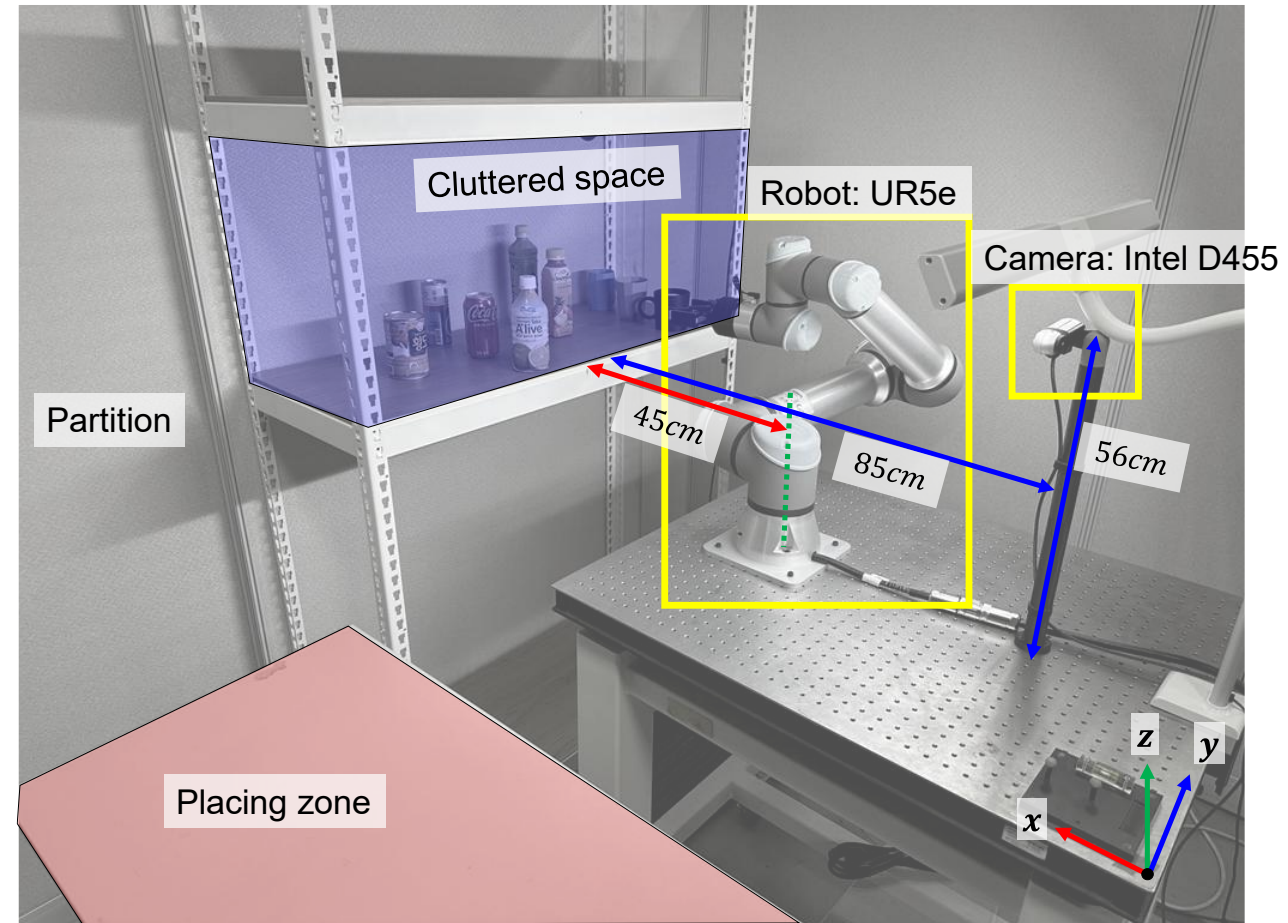
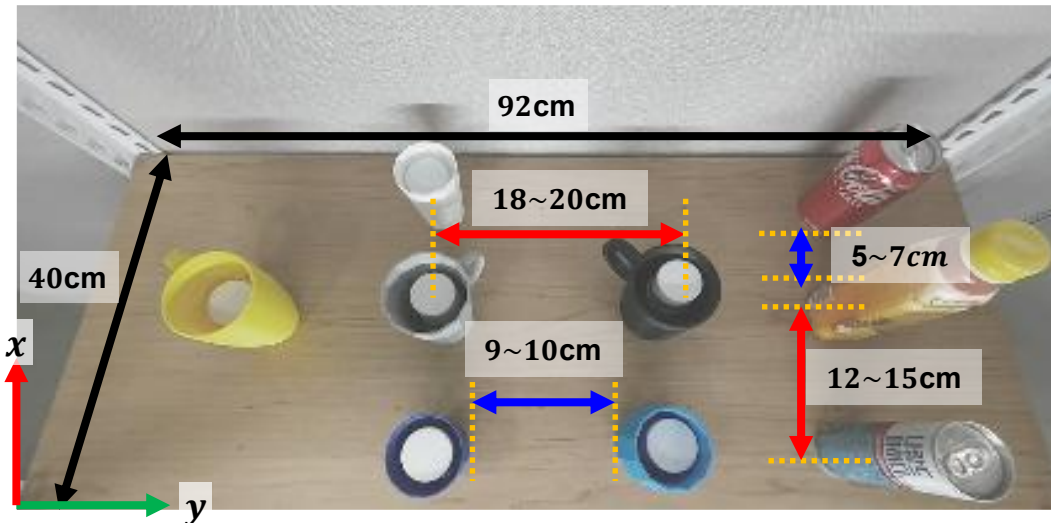
Test bench

• Robot

- Manipulator : UR5e
 - Workspace : 850mm
 - Distance from shelf
 - x : 45cm
- End-effector : 2f-85

• Sensor

- Camera : D455
 - Resolution : 640×480
 - Distance from shelf
 - x : 85cm
 - z : 56cm



Experiment

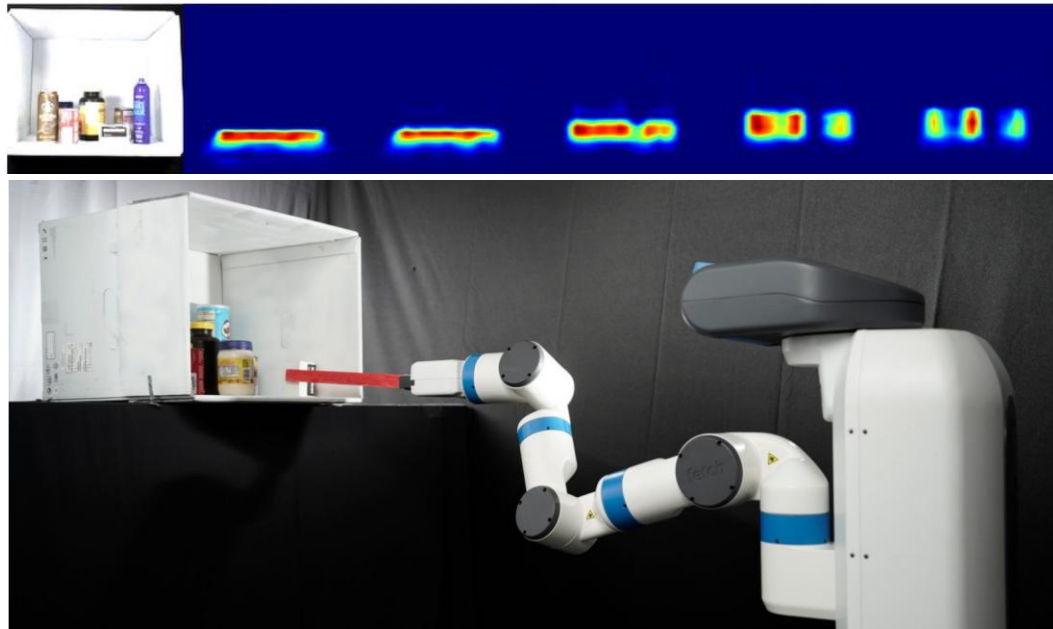
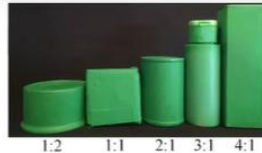
Comparison method

• X-ray(IROS 2022)

- Depth image와 FCN network를 통해 깊이 정보에 의해 목표 물체가 가려질 수 있는 영역의 확률 분포 추출
 - ➔ 선반에 물체가 가려질 수 있는 영역이 많은 곳(확률이 높은 곳)을 우선 탐색
- 최대 **9개의 물체들이 배치된 환경**에서 **6 step 만에 물체를 탐색** 하였음(평균 성공률 70~80%)

Contribution:

Extending of X-ray in lateral access



Experiment

Task scenario



Scenario 1



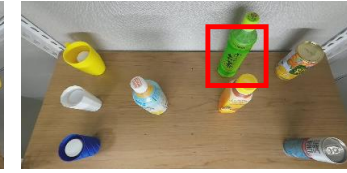
Scenario 2



Scenario 3



Scenario 4



Scenario 5



Scenario 6



Scenario 7



Scenario 8



Scenario 9



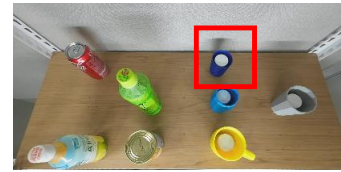
Scenario 10



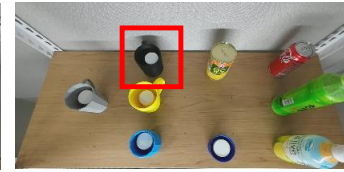
Scenario 11



Scenario 12



Scenario 13



Scenario 14



Scenario 15



Scenario 16



Scenario 17



Scenario 18



Scenario 19



Scenario 20



Scenario 21



Scenario 22



Scenario 23



Scenario 24



Scenario 25



Target

Experiment Result

No.	Scenario No.	Method	object num	Target	Step	Success
1	1	Ours	9	Can_1	4	O
2		X-ray			6	O
3		Random			7	X
4		Human			3	O
5	2	Ours	11	Cup_2	4	O
6		X-ray			5	O
7		Random			9	X
8		Human			3	O
9	3	Ours	9	Mug_1	2	O
10		X-ray			7	X
11		Random			6	O
12		Human			2	O
13	4	Ours	9	Cup_2	4	O
14		X-ray			5	O
15		Random			5	O
16		Human			3	O
17	5(12)	Ours	8	Bottle_2	2	O
18		X-ray			3	O
19		Random			5	O
20		Human			2	O
21	6(13)	Ours	9	Mug_3	3	O
22		X-ray			4	O
23		Random			8	X
24		Human			2	O
25	7(16)	Ours	9	Bottle_3	2	O
26		X-ray			3	O
27		Random			4	O
28		Human			2	O
29	8(17)	Ours	9	Can_1	2	O
30		X-ray			2	O
31		Random			4	O
32		Human			2	O
33	9(18)	Ours	11	Cup_3	4	O
34		X-ray			8	X
35		Random			5	O
36		Human			4	O
37	10(19)	Ours				
38		X-ray				
39		Random				
40		Human				
41	11(20)	Ours				
42		X-ray				
43		Random				
44		Human				
45	12(21)	Ours				
46		X-ray				
47		Random				
48		Human				
49	13(22)	Ours				
50		X-ray				
51		Random				
52		Human				
53	14(23)	Ours				
54		X-ray				
55		Random				
56		Human				
57	15(24)	Ours				
58		X-ray				
59		Random				
60		Human				
61	16(25)	Ours				
62		X-ray				
63		Random				
64		Human				
65	17(27)	Ours				
66		X-ray				
67		Random				
68		Human				
69	18(28)	Ours				
70		X-ray				
71		Random				
72		Human				
73	19(29)	Ours				
74		X-ray				
75		Random				
76		Human				
77	20(30)	Ours				
78		X-ray				
79		Random				
80		Human				
81	21(31)	Ours				
82		X-ray				
83		Random				
84		Human				
85	22(32)	Ours				
86		X-ray				
87		Random				
88		Human				
89	23(33)	Ours				
90		X-ray				
91		Random				
92		Human				
93	24(34)	Ours				
94		X-ray				
95		Random				
96		Human				
97	25(35)	Ours				
98		X-ray				
99		Random				
100		Human				

Experiment Result

Scenario 1

Target object



Can_2



Step	1D-PDM	Action	Column
1	[16.35, 17.86, 15.22, 3.95]	grasping	1 st (0)
2	[5.03, 22.67, 19.72, 5.16]	Sweeping_R	2 nd (1)
3	[1.64, 31.75, 23.55, 5.49]	Sweeping_L	2 nd (1)
4	[1.64, 31.75, 23.55, 5.49]	grasping	2 nd (1)

Experiment Result

Scenario 4

Target object



Cup_2



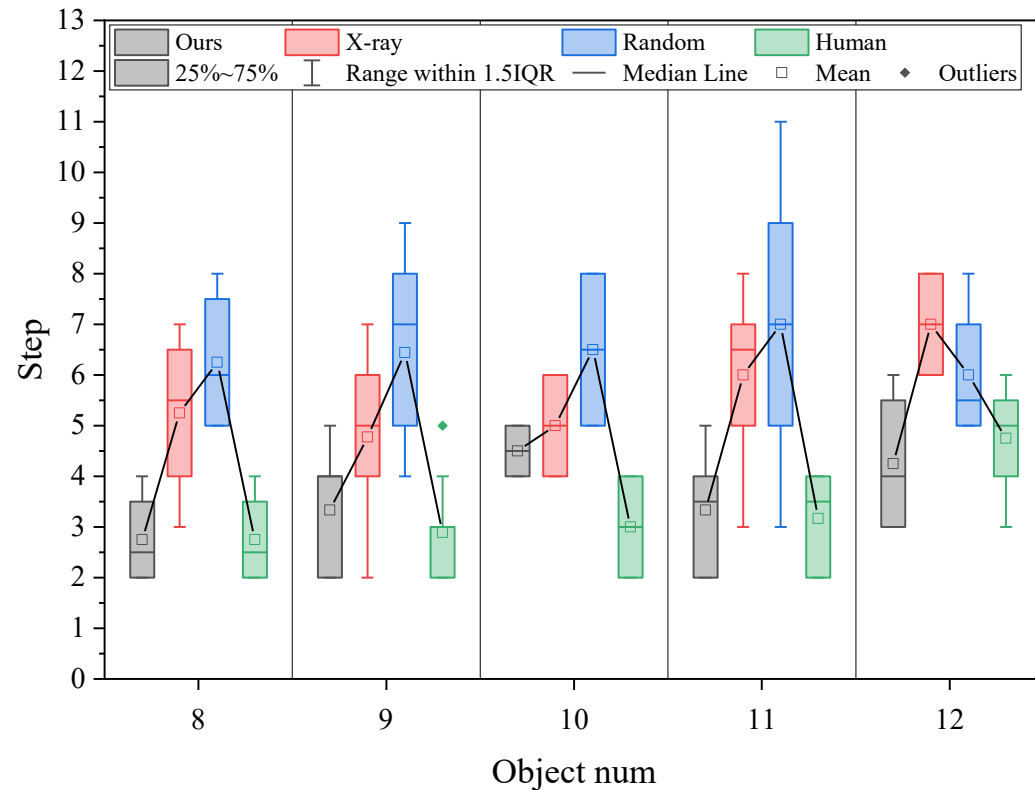
Step	1D-PDM	Action	Column
1	[8.25, 12.30, 26.10, 4.01]	grasping	3 rd (2)
2	[10.84, 15.67, 13.72, 5.85]	grasping	2 nd (1)
3	[10.45, 12.94, 11.19, 6.38]	grasping	2 nd (1)
4	[10.45, 12.94, 11.19, 6.38]	grasping	2 nd (1)

Experiment Result

Step count & success rate

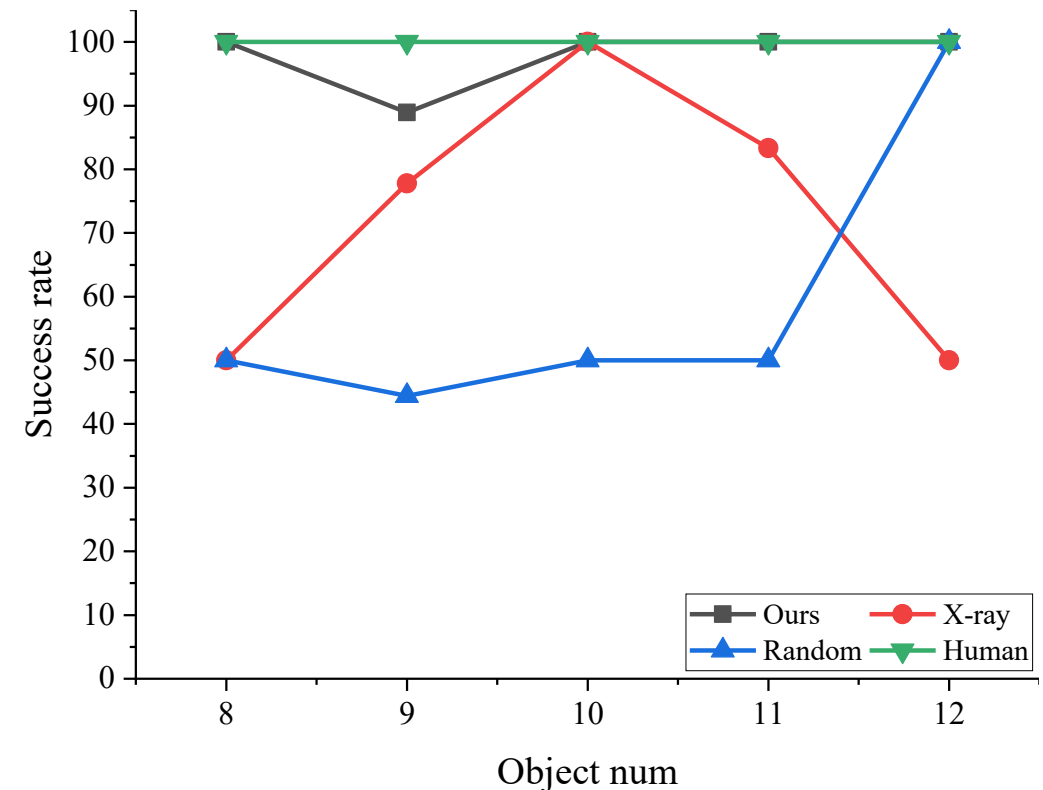
• Step count

- Human 대비 **7.4%** 낮은 성능
- X-ray 대비 약 **37%**, Random 대비 약 **47%** 향상



• Success rate

- Human 대비 **4%** 낮은 성능
- X-ray 대비 약 **33.3%**, Random 대비 약 **71.4%** 향상

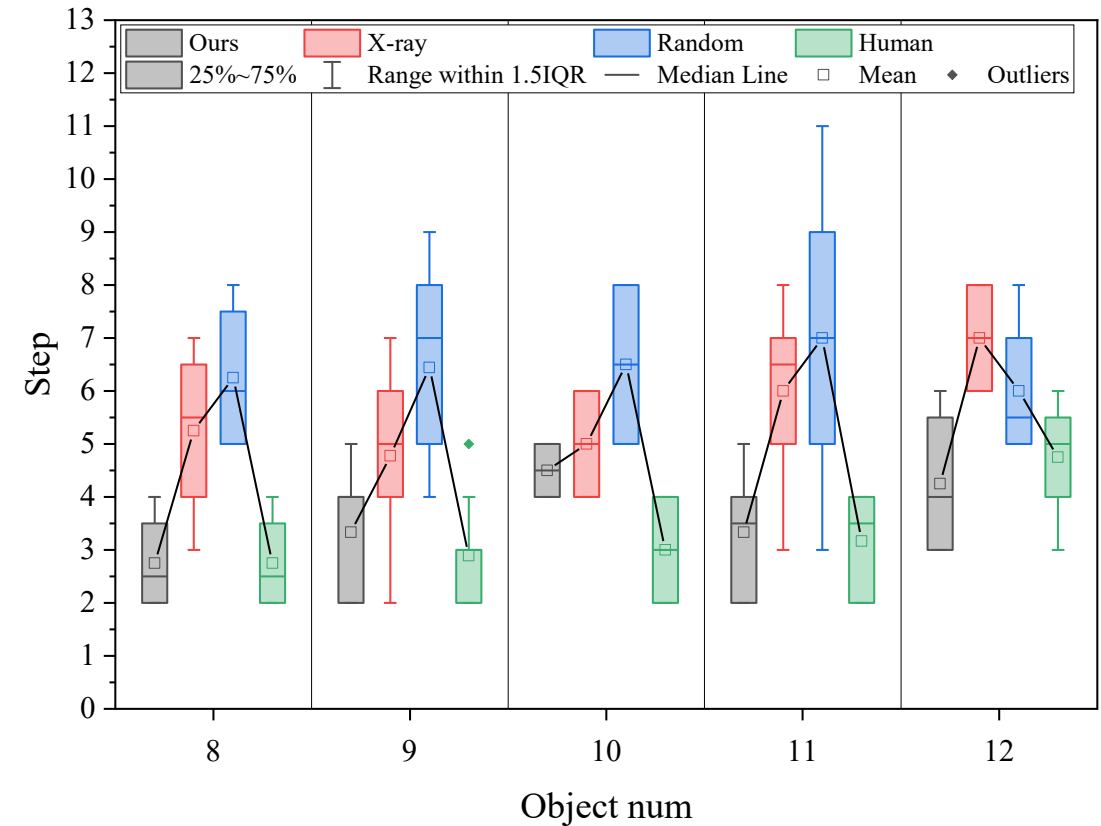


Experiment Result

Step count

Method	Mean step	Mean std
Ours	3.48	1.19
Human	3.24	1.20
X-ray	5.52	1.71
Random	6.48	1.92

Object num	Ours		Human		X-ray		Random	
	Mean step	Mean std	Mean step	Mean std	Mean step	Mean std	Mean step	Mean std
8	2.75	0.96	3.00	1.41	5.25	1.71	6.25	1.50
9	3.33	1.12	3.00	0.82	4.78	1.72	6.44	1.81
10	4.50	0.71	3.00	0.82	5.00	1.41	6.50	2.12
11	3.33	1.21	3.50	1.29	6.00	1.79	7.00	2.90
12	4.25	1.50	3.50	1.29	7.00	1.15	6.00	1.41



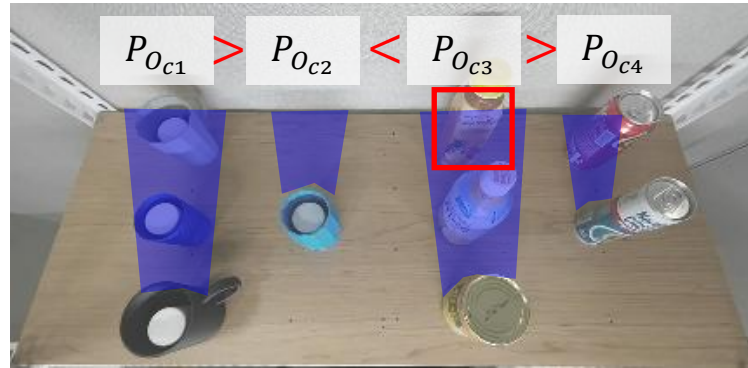
Human 대비 **7.4%** 낮지만, X-ray 대비 약 **37%**, Random 대비 약 **47%** 향상



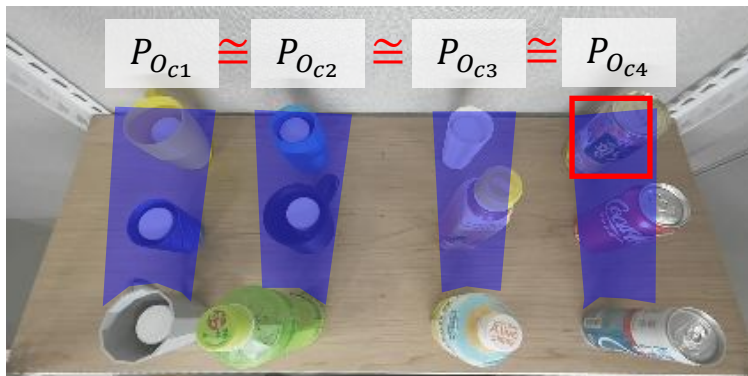
Experiment Result

Step count(X-ray)

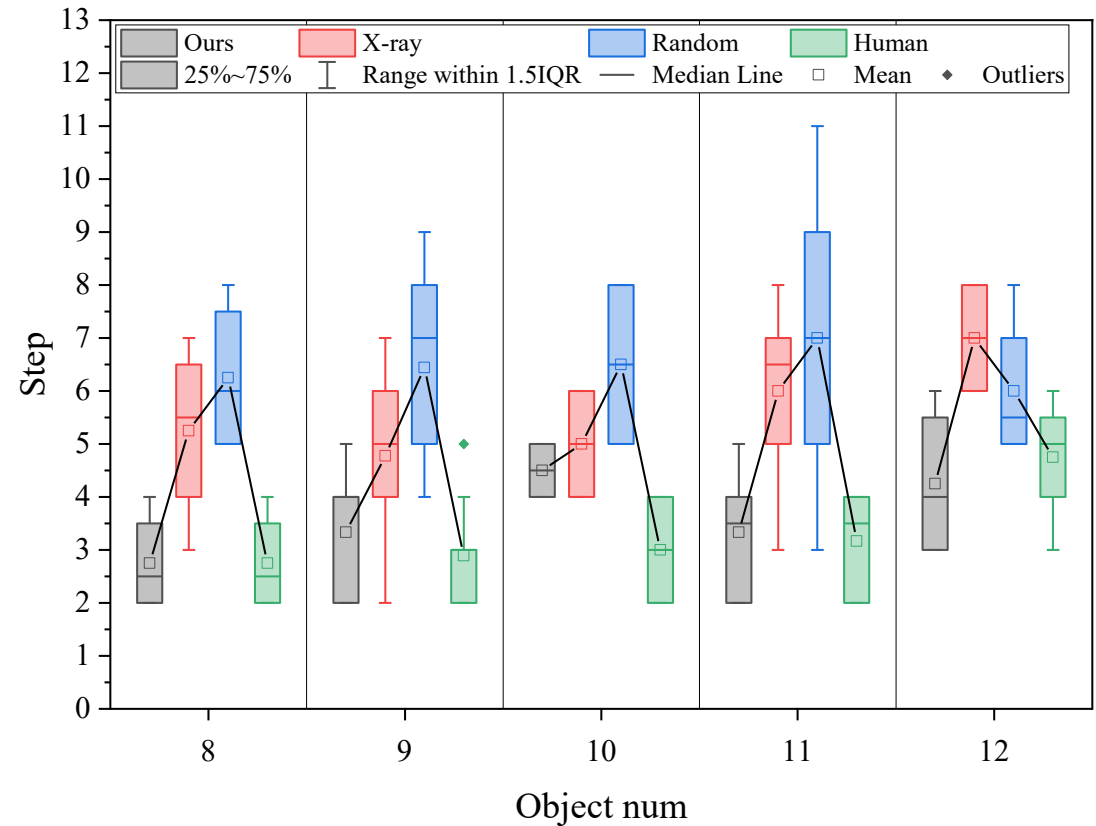
Occluded area
 Target



Object num : 9



Object num : 12



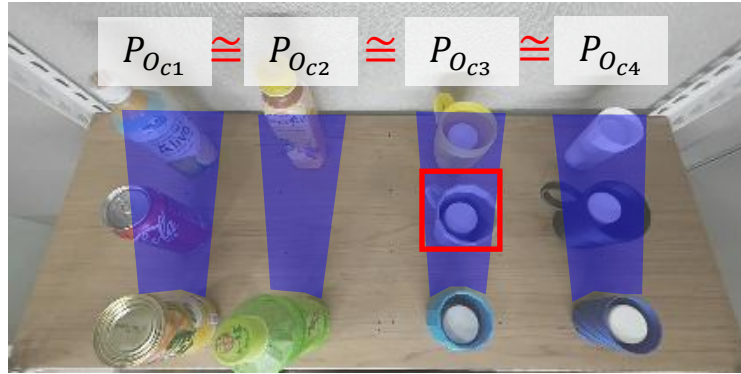
물체 개수가 많아질수록 **Random search와 유사하게 탐색하게 됨** → **step count 수 증가**

Experiment Result

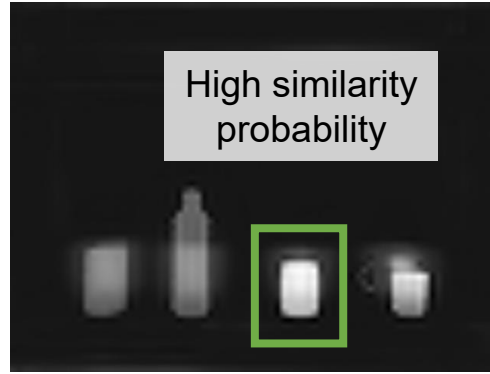
Step count(Ours)

 Occluded area

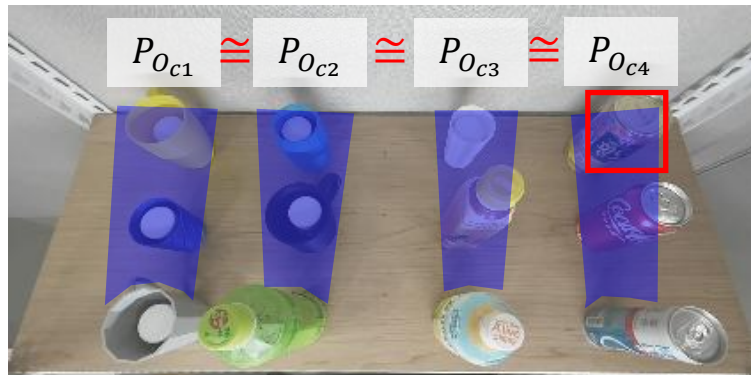
 Target



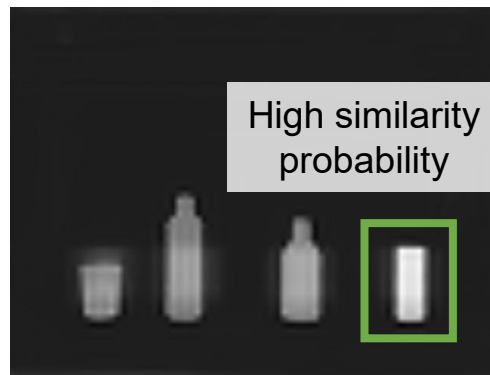
Object num : 11



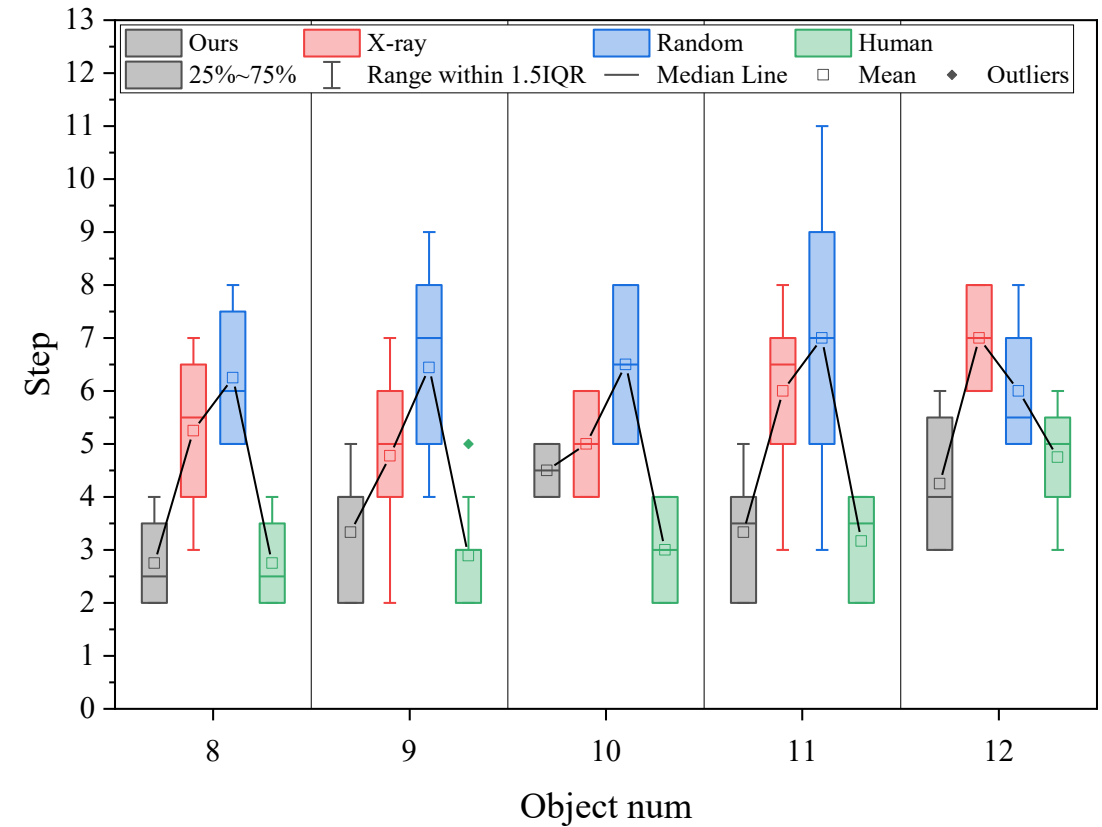
2D-PDM



Object num : 12



2D-PDM



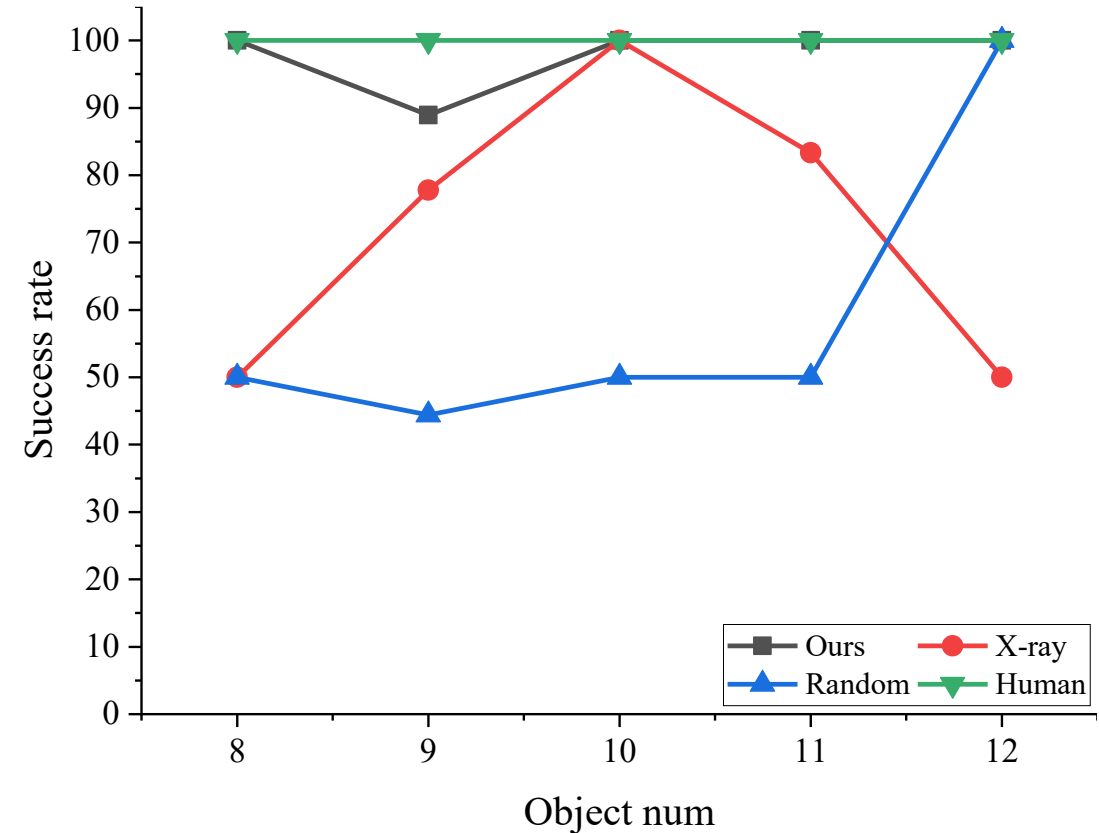
물체 더미들에 의해 가려질 확률이 비슷하면, 유사도에 의해 탐색을 수행

Experiment Result

Success rate

Method	Success rate (%)
Ours	96.0 %
Human	100.0 %
X-ray	72.0 %
Random	56.0 %

Object num	Ours	Human	X-ray	Random
8	100.0 %	100.0 %	50.0 %	50.0 %
9	88.9 %	100.0 %	77.8 %	44.4 %
10	100.0 %	100.0 %	100.0 %	50.0 %
11	100.0 %	100.0 %	83.3 %	50.0 %
12	100.0 %	100.0 %	50.0 %	100.0 %



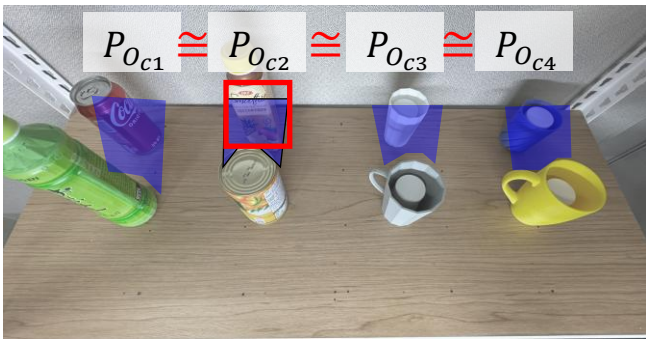
Human 대비 **4%** 낮지만, X-ray 대비 약 **33.3%**, Random 대비 약 **71.4%** 향상



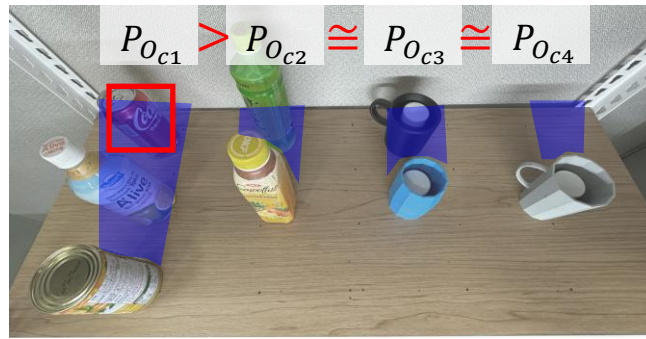
Experiment Result

Success rate(X-ray)

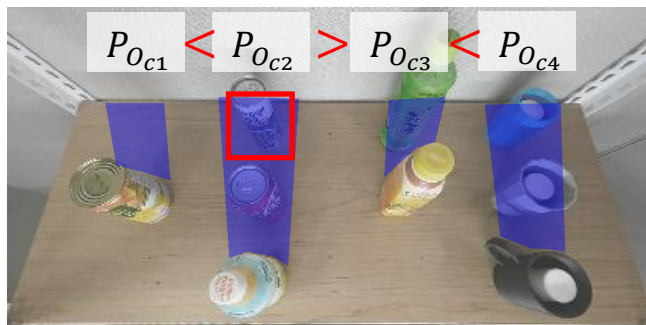
 Occluded area
  Target



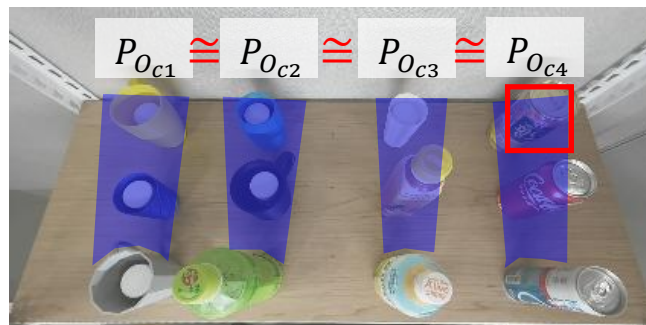
Object num : 8



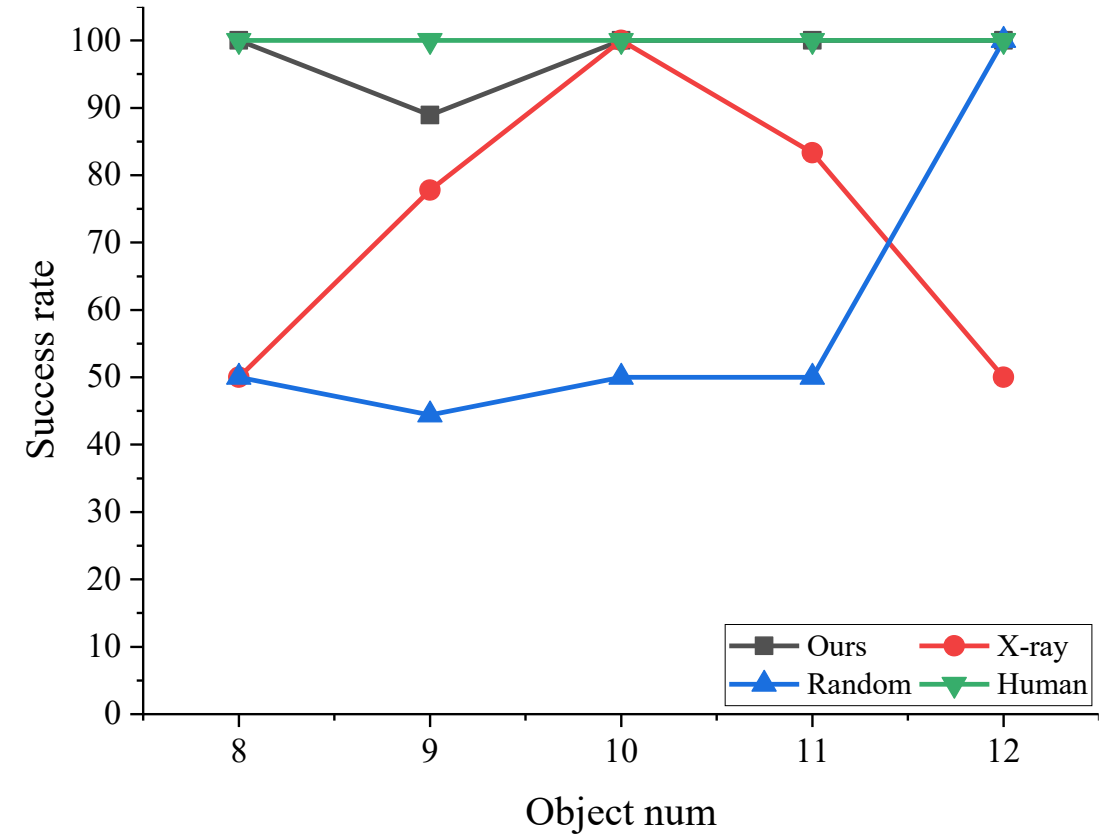
Object num : 8



Object num : 9



Object num : 12

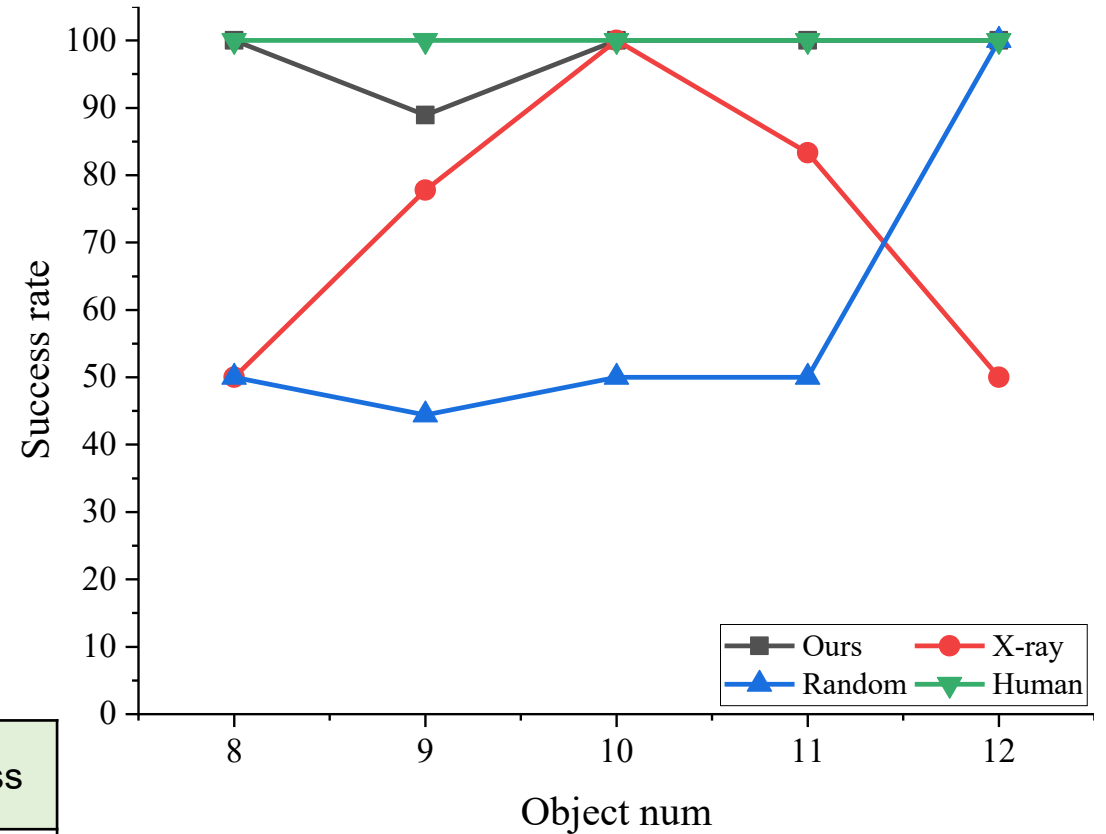


물체 개수가 적거나 많을수록 **Random search**와 유사하게 탐색하게 됨 → **success rate 감소**

Experiment Result

Success rate(ours)

- 비교 method와 달리 **sweeping 동작**이 존재
- 즉, 선반 안쪽 행에 위치한 물체를 sweeping 할 때 **로봇 팔의 두께(10cm)**로 인해 앞쪽 행의 물체를 넘어트림



Scenario No.	Method	Object num	Target	Step	Success
14	Ours	9	Mug_1	4	X
	X-ray			7	X
	Random			8	X

Experiment

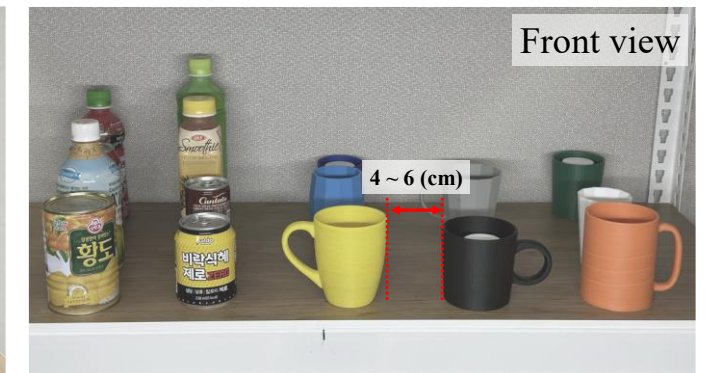
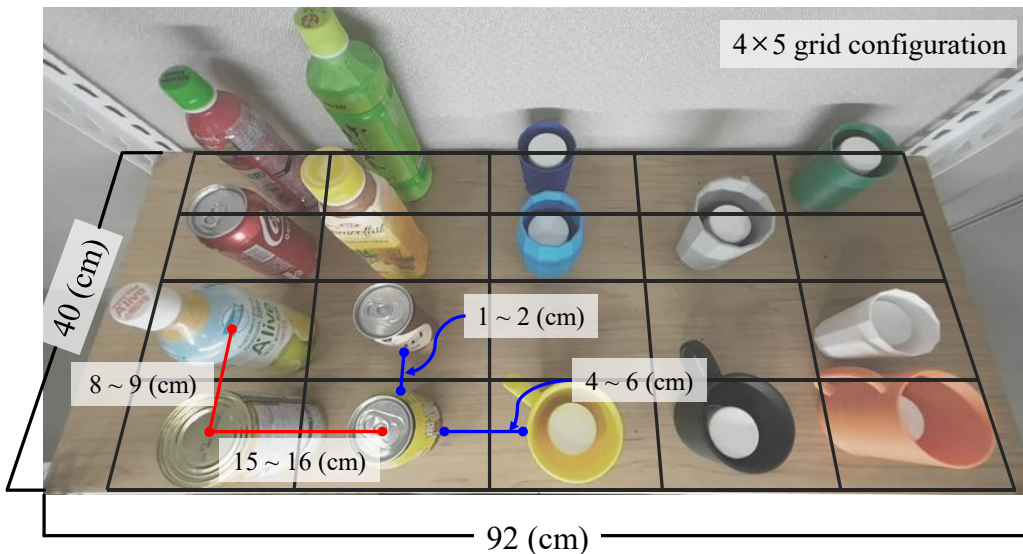
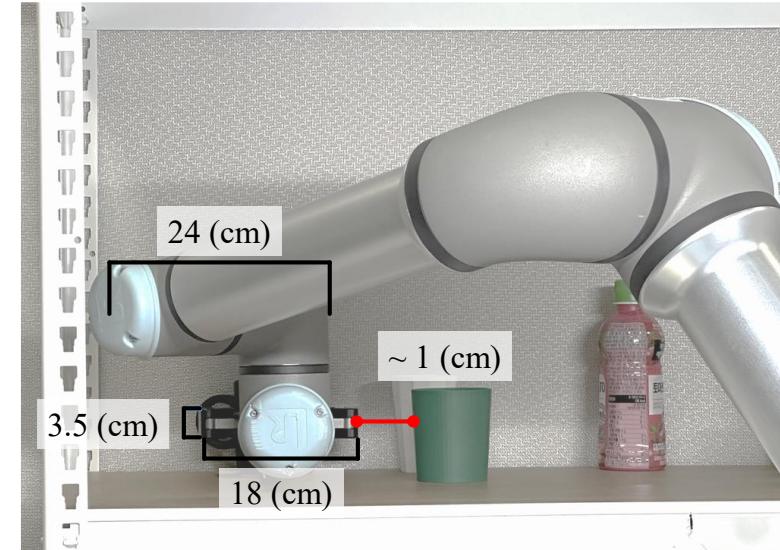
Evaluation of Generalization Performance under Extreme Environmental Conditions

• Object list

- 각 카테고리(Cup, Mug, Can, Bottle)별로 4개씩 총 16개

• Shelf Configuration

- 4 × 5배열
- x 축(깊이 방향) 물체 사이 공간: 1~2cm
- y 축(수평 방향) 물체 사이 공간: 4~6cm



Experiment

Task scenario: 4×5 configuration

 Target



Scenario 1



Scenario 2



Scenario 3



Scenario 4



Scenario 5



Scenario 6



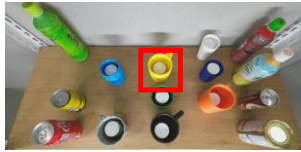
Scenario 7



Scenario 8



Scenario 9



Scenario 10



Scenario 11



Scenario 12



Scenario 13



Scenario 14



Scenario 15



Scenario 16



Scenario 17



Scenario 18



Scenario 19



Scenario 20



Scenario 21



Scenario 22



Scenario 23



Scenario 24



Scenario 25



Scenario 26



Scenario 27



Scenario 28



Scenario 29



Scenario 30



Scenario 31



Scenario 32



Scenario 33



Scenario 34



Scenario 35



Scenario 36



Scenario 37



Scenario 38



Scenario 39



Scenario 40

Experiment Result

No.	Scenario No.	Method	object num	Target	Step	Success
1	1	Ours	16	Mug_2	3	O
2		X-ray			2	O
3		Random			11	O
4		Human			2	O
5	2	Ours	16	Cup_3	3	O
6		X-ray			12	X
7		Random			6	O
8		Human			4	O
9	3	Ours	15	Can_4	3	O
10		X-ray			9	O
11		Random			6	O
12		Human			4	O
13	4	Ours	15	Bottle_4	3	O
14		X-ray			7	O
15		Random			5	O
16		Human			3	O
17	5	Ours	15	Can_2	5	O
18		X-ray			13	X
19		Random			6	O
20		Human			4	O
21	6	Ours	16	Mug_4	3	O
22		X-ray			9	O
23		Random			8	X
24		Human			5	O
25	7	Ours	16	Cup_4	3	O
26		X-ray			13	X
27		Random			15	X
28		Human			8	O
29	8	Ours	16	Mug_1	4	O
30		X-ray			9	O
31		Random			9	O
32		Human			7	O
33	9	Ours	15	Mug_2	5	O
34		X-ray			9	O
35		Random			7	O
36		Human			5	O
37	10	Ours	15	Mug_3	3	O
38		X-ray			8	O
39		Random			14	X
40		Human			5	O
41	11	Ours	14	Mug_4	5	O
42		X-ray			10	X
43		Random			6	O
44		Human			5	O
45	12	Ours	14	Cup_1	4	O
46		X-ray			12	X
47		Random			11	X
48		Human			6	O
49	13	Ours	14	Cup_2	3	O
50		X-ray			5	O
51		Random			10	X
52		Human			6	O

53	14	Ours	14	Cup_3	7	O
54		X-ray			11	X
55		Random			14	X
56		Human			4	O
57	15	Ours	14	Cup_4	4	O
58		X-ray			3	O
59		Random			5	O
60		Human			2	O
61	16	Ours	13	Can_1	4	O
62		X-ray			5	O
63		Random			12	X
64		Human			4	O
65	17	Ours	13	Can_2	3	O
66		X-ray			10	X
67		Random			7	O
68		Human			6	O
69	18	Ours	13	Can_3	5	O
70		X-ray			12	X
71		Random			8	O
72		Human			5	O
73	19	Ours	13	Can_4	4	O
74		X-ray			11	X
75		Random			13	X
76		Human			5	O
77	20	Ours	13	Bottle_1	5	O
78		X-ray			10	X
79		Random			5	O
80		Human			5	O
81	21	Ours	12	Bottle_2	2	O
82		X-ray			5	O
83		Random			5	O
84		Human			5	O
85	22	Ours	12	Bottle_3	6	O
86		X-ray			3	O
87		Random			11	X
88		Human			5	O
89	23	Ours	12	Bottle_4	3	O
90		X-ray			9	X
91		Random			3	X
92		Human			5	O
93	24	Ours	12	Mug_1	4	O
94		X-ray			10	X
95		Random			10	X
96		Human			3	O
97	25	Ours	12	Mug_2	5	O
98		X-ray			6	O
99		Random			10	X
100		Human			4	O
101	26	Ours	11	Mug_3	2	O
102		X-ray			5	O
103		Random			11	X
104		Human			5	O
105	27	Ours	11	Mug_4	4	O
106		X-ray			9	X
107		Random			4	O
108		Human			4	O

109	28	Ours	11	Cup_1	5	X
110		X-ray			9	X
111		Random			8	X
112		Human			4	O
113	29	Ours	11	Cup_2	4	O
114		X-ray			8	X
115		Random			11	X
116		Human			4	O
117	30	Ours	11	Cup_3	4	O
118		X-ray			8	X
119		Random			8	X
120		Human			6	O
121	31	Ours	11	Cup_4	5	O
122		X-ray			8	X
123		Random			11	X
124		Human			3	O
125	32	Ours	11	Can_1	3	O
126		X-ray			9	X
127		Random			4	O
128		Human			5	O
129	33	Ours	11	Can_2	3	O
130		X-ray			8	X
131		Random			11	X
132		Human			5	O
133	34	Ours	11	Can_3	4	O
134		X-ray			10	X
135		Random			10	X
136		Human			5	O
137	35	Ours	11	Can_4	5	O
138		X-ray			7	O
139		Random			10	X
140		Human			3	O
141	36	Ours	14	Cup_4	3	X
142		X-ray			10	X
143		Random			11	X
144		Human			6	O
145	37	Ours	10	Cup_4	3	O
146		X-ray			6	O
147		Random			5	O
148		Human			7	X
149	38	Ours	10	Mug_4	3	O
150		X-ray			10	X
151		Random			7	X
152		Human			5	O
153	39	Ours	10	Can_3	4	O
154		X-ray			8	X
155		Random			6	O
156		Human			3	O
157	40	Ours	10	Bottle_4	3	O
158		X-ray			6	O
159		Random			5	X
160		Human			3	O

Experiment Result

Scenario 1

Target object



Mug_2



Step	1D-PDM	Action	Column
1	[3.26, 1.77, 19.28, 35.36, 18.12]	grasping	3 rd (2)
2	[4.24, 2.28, 8.63, 48.44, 24.02]	Sweeping_L	4 th (3)
3	[4.24, 2.28, 8.63, 48.44, 24.02]	grasping	4 th (3)

Experiment Result

Scenario 35

Target object



Can_4



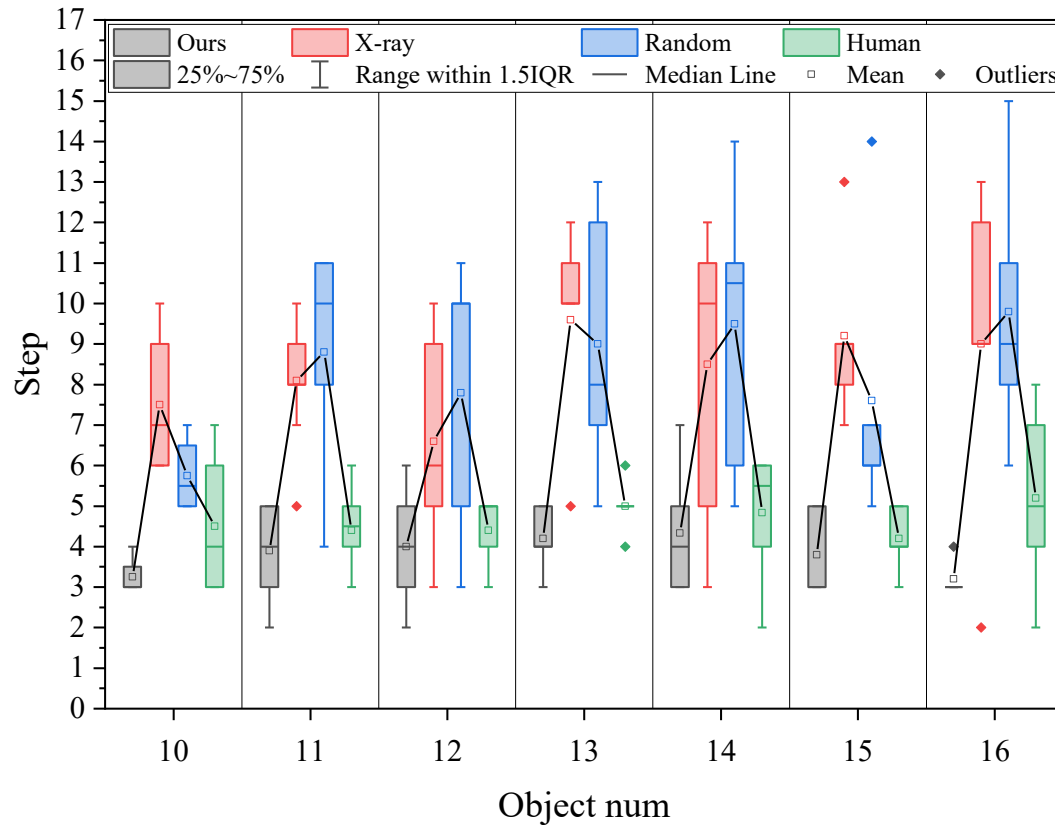
Step	1D-PDM	Action	Column
1	[2.84, 2.21, 14.05, 13.90, 13.54]	grasping	3 rd (2)
2	[3.70, 2.82, 29.08, 16.41, 16.88]	grasping	3 rd (2)
3	[3.87, 3.06, 8.88, 17.48, 16.62]	grasping	4 th (3)
4	[3.92, 3.00, 2.82, 26.21, 14.46]	Sweeping_L	4 th (3)
5	[3.92, 3.00, 2.82, 26.21, 14.46]	grasping	4 th (3)

Experiment Result

Step count & success rate

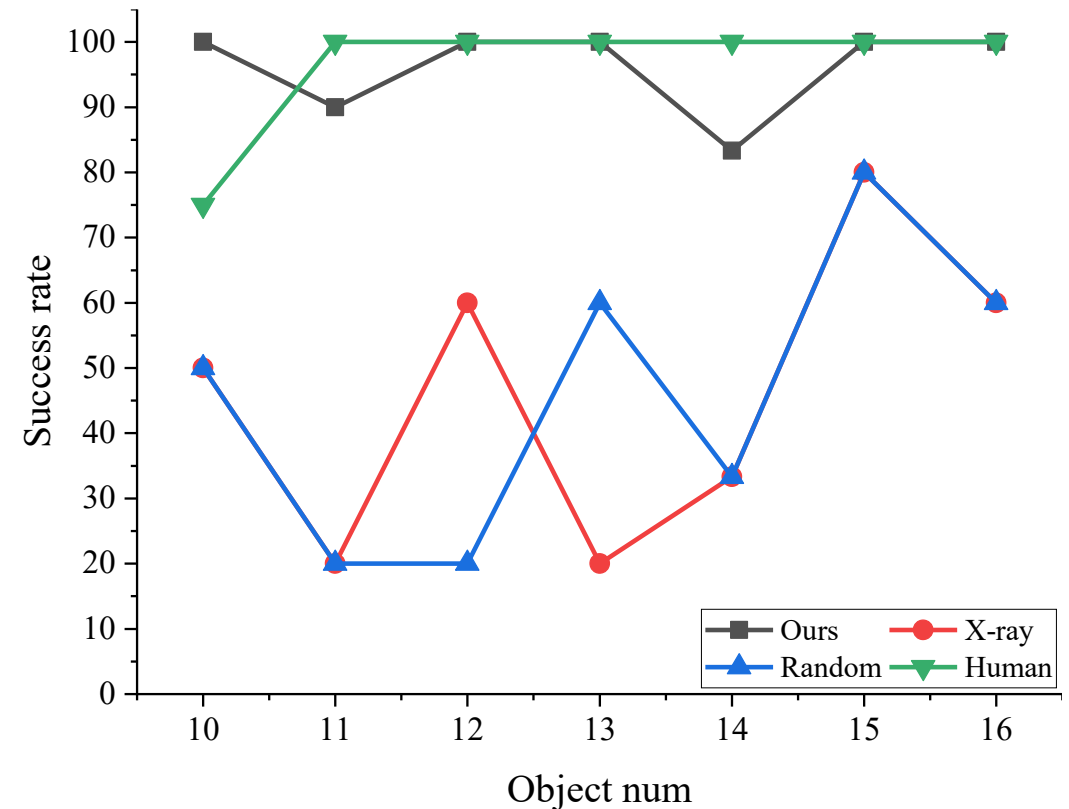
• Step count

- Human 대비 **16.6% 높은** 성능
- X-ray 대비 약 **53.9%**, Random 대비 약 **54.6%** 향상



• Success rate

- Human 대비 **2.5%** 낮은 성능
- X-ray, Random 대비 약 **123.5%** 향상

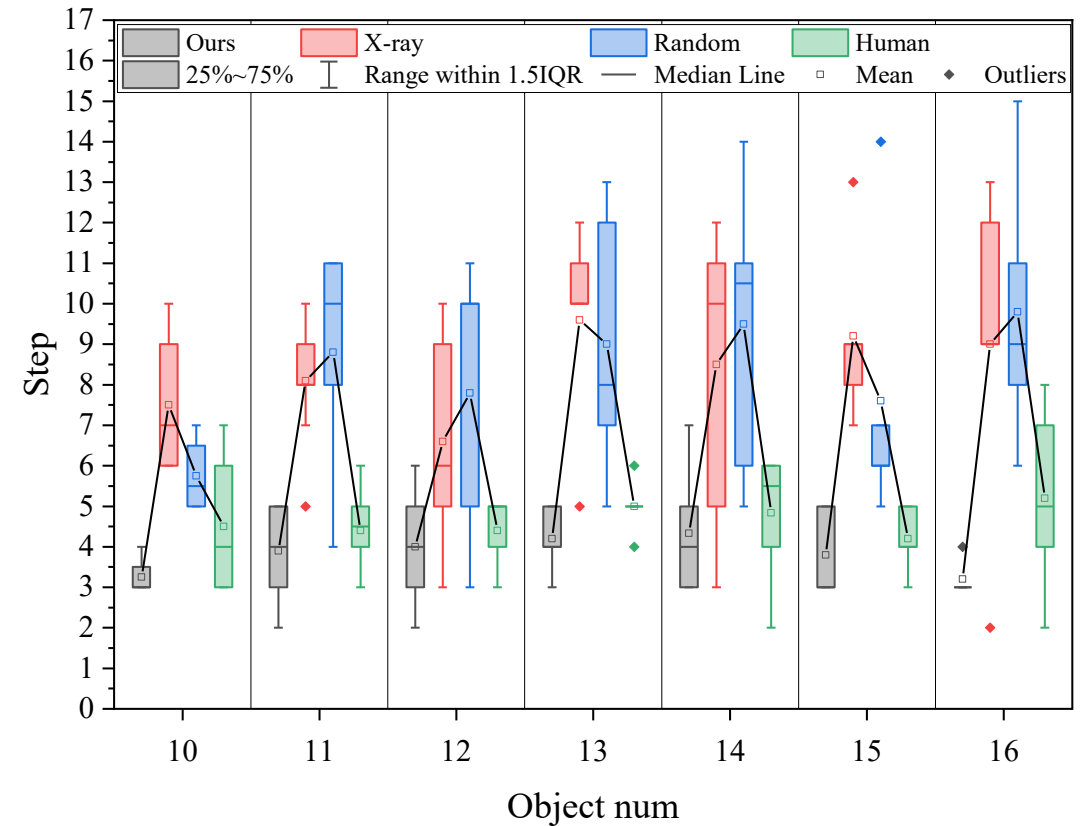


Experiment Result

Step count: 4×5 configuration

Method	Mean step	Mean std
Ours	3.85	1.08
Human	4.62	1.33
X-ray	8.35	2.69
Random	8.48	3.12

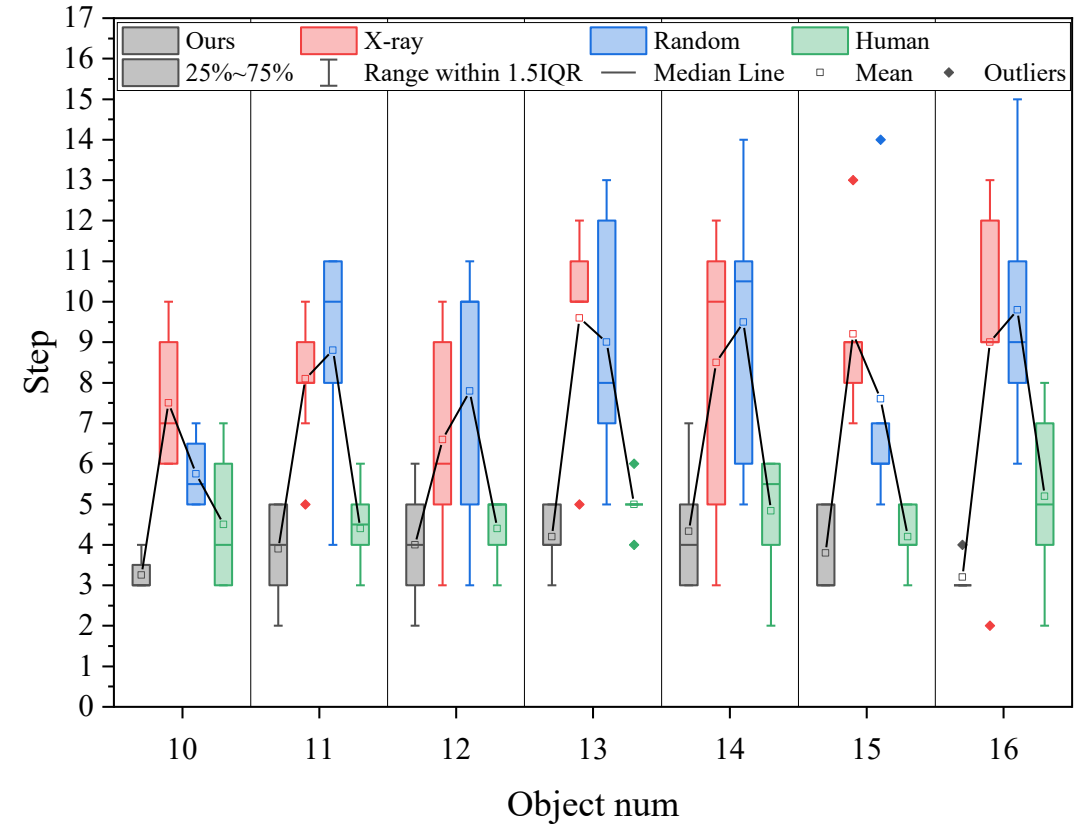
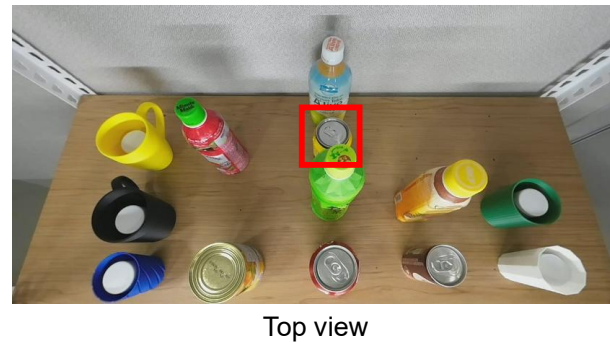
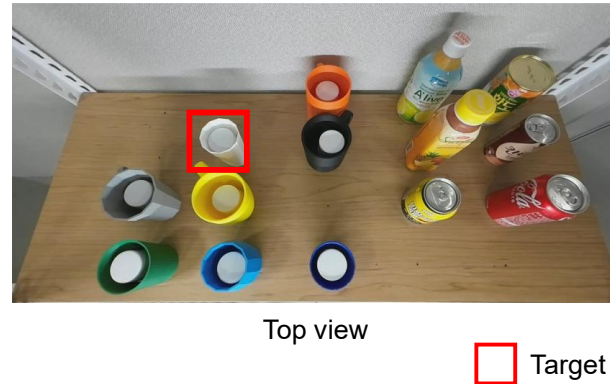
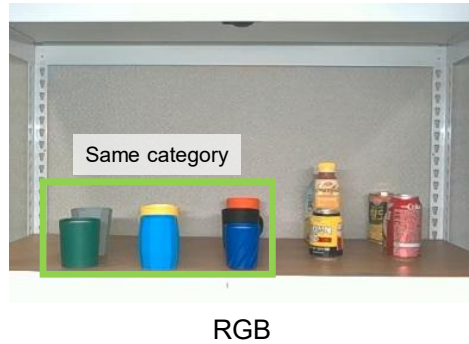
Object num	Ours		Human		X-ray		Random	
	Mean step	Mean std	Mean step	Mean std	Mean step	Mean std	Mean step	Mean std
10	3.25	0.50	4.50	1.91	7.50	1.91	5.75	0.96
11	3.90	0.99	4.40	0.97	8.10	1.37	8.80	2.78
12	4.00	1.58	4.25	0.96	6.60	2.88	7.80	3.56
13	4.20	0.84	5.00	0.71	9.60	2.70	9.00	3.39
14	4.33	1.51	4.83	1.60	8.50	3.62	9.50	3.39
15	3.80	1.10	4.20	0.84	9.20	2.28	7.60	3.65
16	3.20	0.45	5.20	2.39	9.00	4.30	9.80	3.42



Human 대비 약 **16.6%**, X-ray 대비 약 **53.9%**, Random 대비 약 **54.6%** 향상

Experiment Result

Step count(Human): 4×5 configuration



Target object와 같은 category 혹은 유사도가 높은 category가 여러 열에 배치된 경우 → Random search

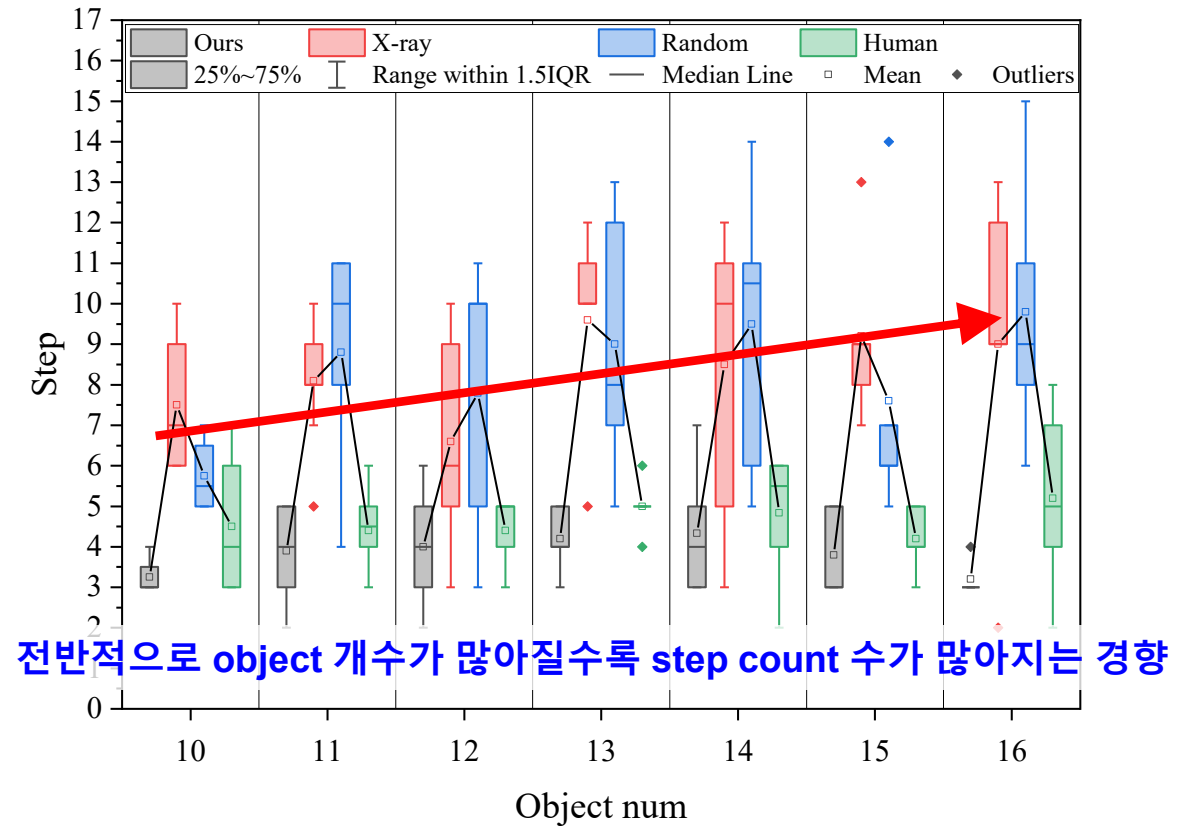


Experiment Result

Step count(X-ray): 4×5 configuration

- Human, 제안 연구
 - 물체를 인식 및 탐색 전략을 통해 일정한 step을 사용하여 목표 물체 탐색
- X-ray, Random
 - Depth 정보 및 random하게 탐색

Object num	Scenario num
10	4
11	10
12	5
13	5
14	6
15	5
16	5



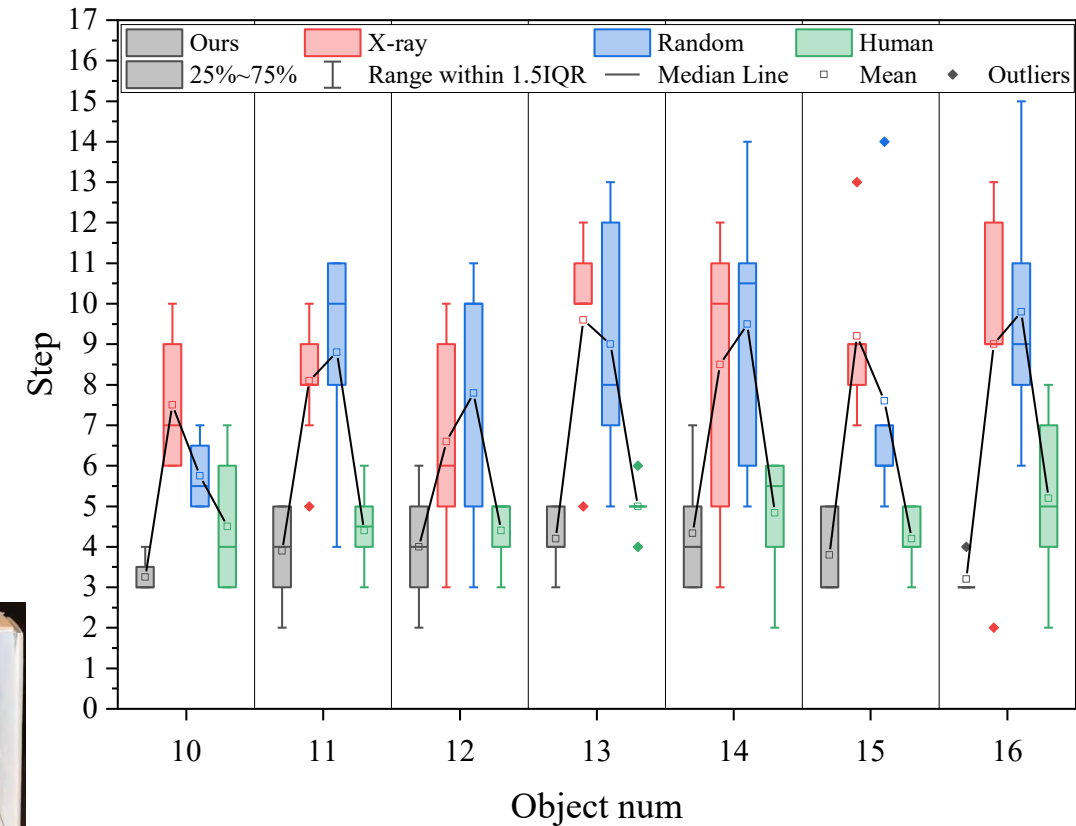
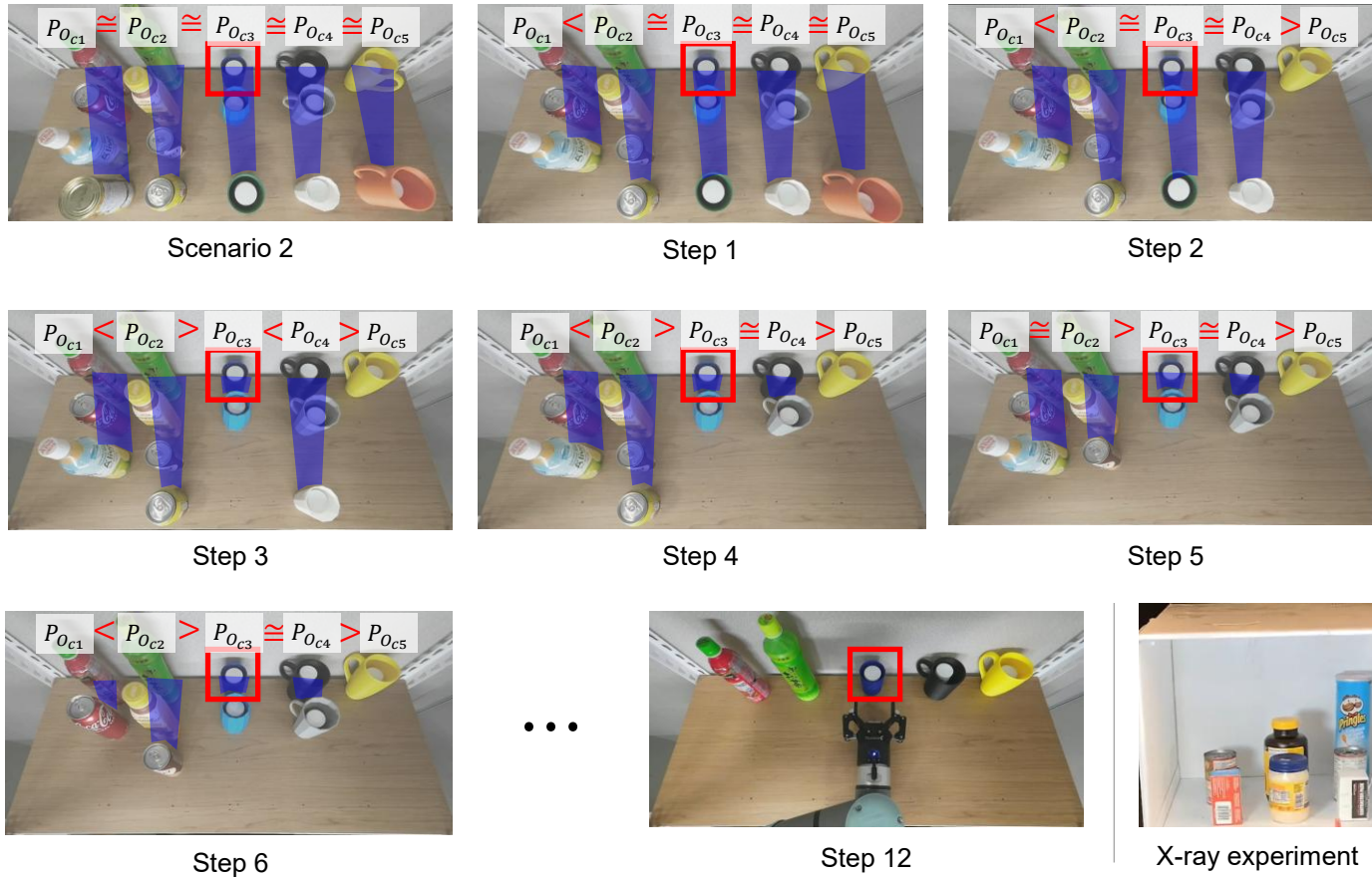
물체 개수별 시나리오 표본이 많으면 fluctuation이 줄어들 것으로 예상됨



Experiment Result

Step count(X-ray): 4×5 configuration

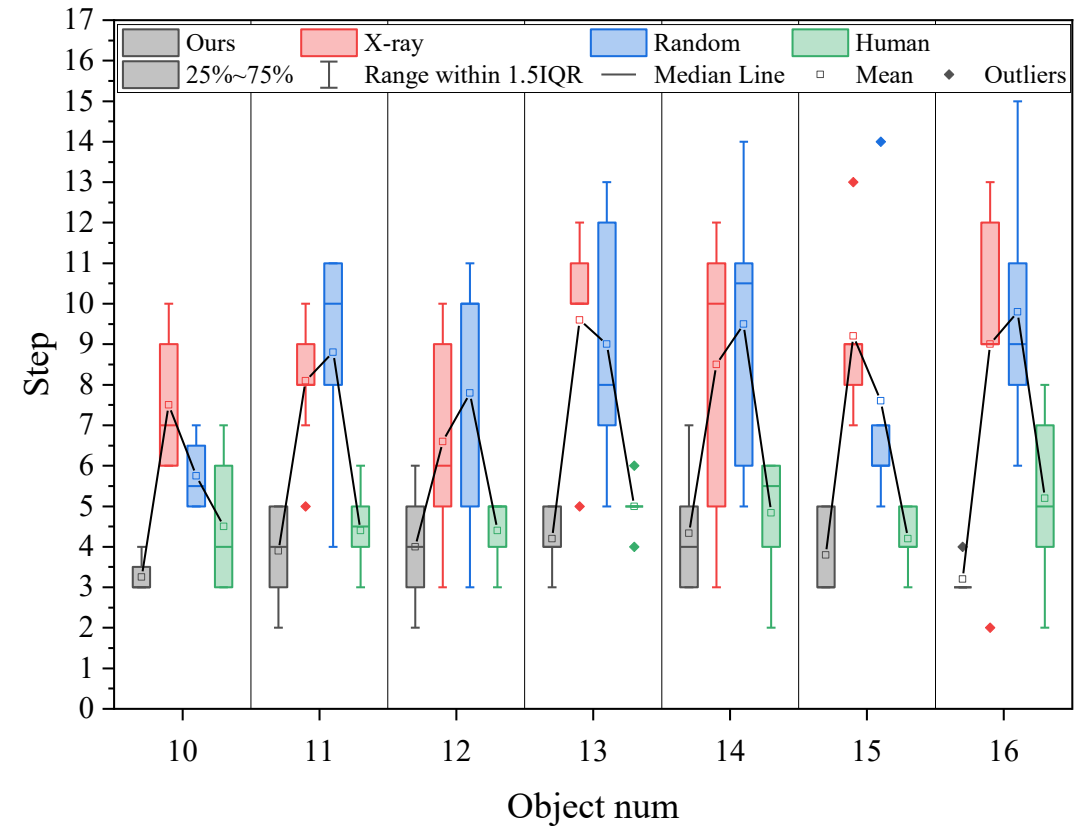
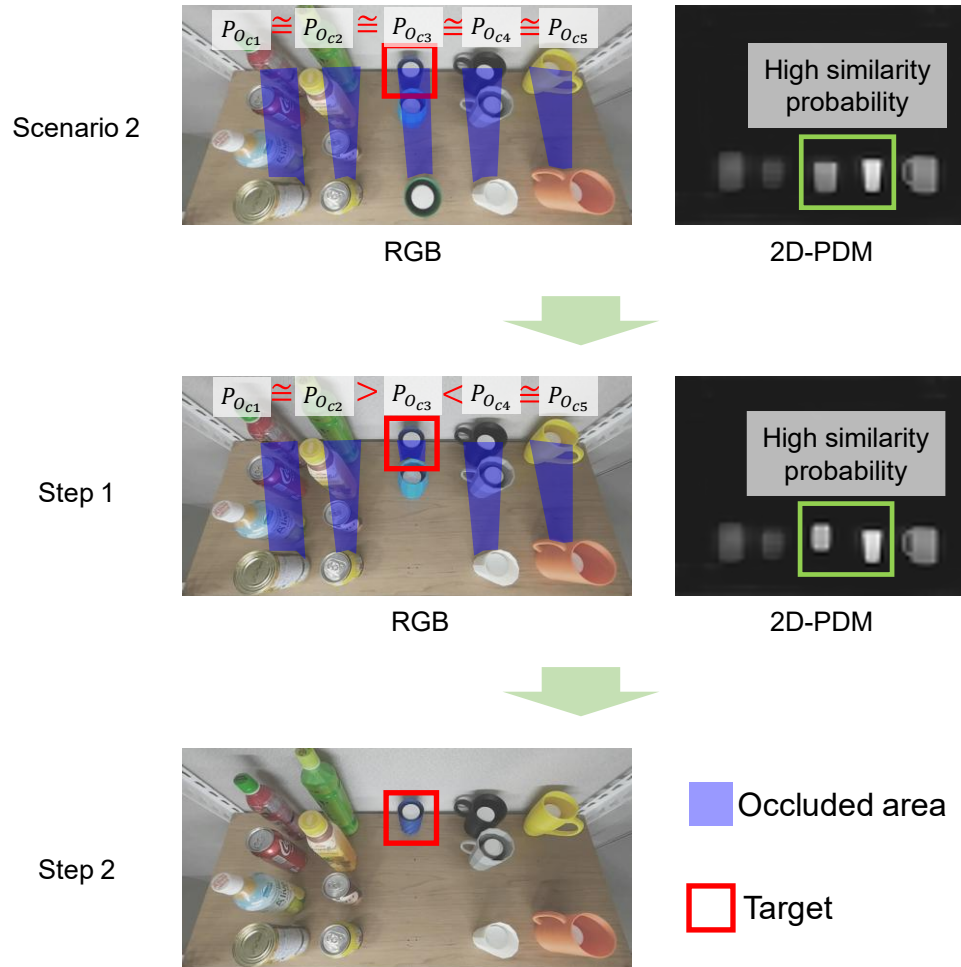
Occluded area Target



Depth에 의해 목표 물체 탐색을 수행: 수평 방향의 탐색만 하게 됨 → 깊이방향 탐색 불가능(일반화 X)

Experiment Result

Step count(ours): 4×5 configuration



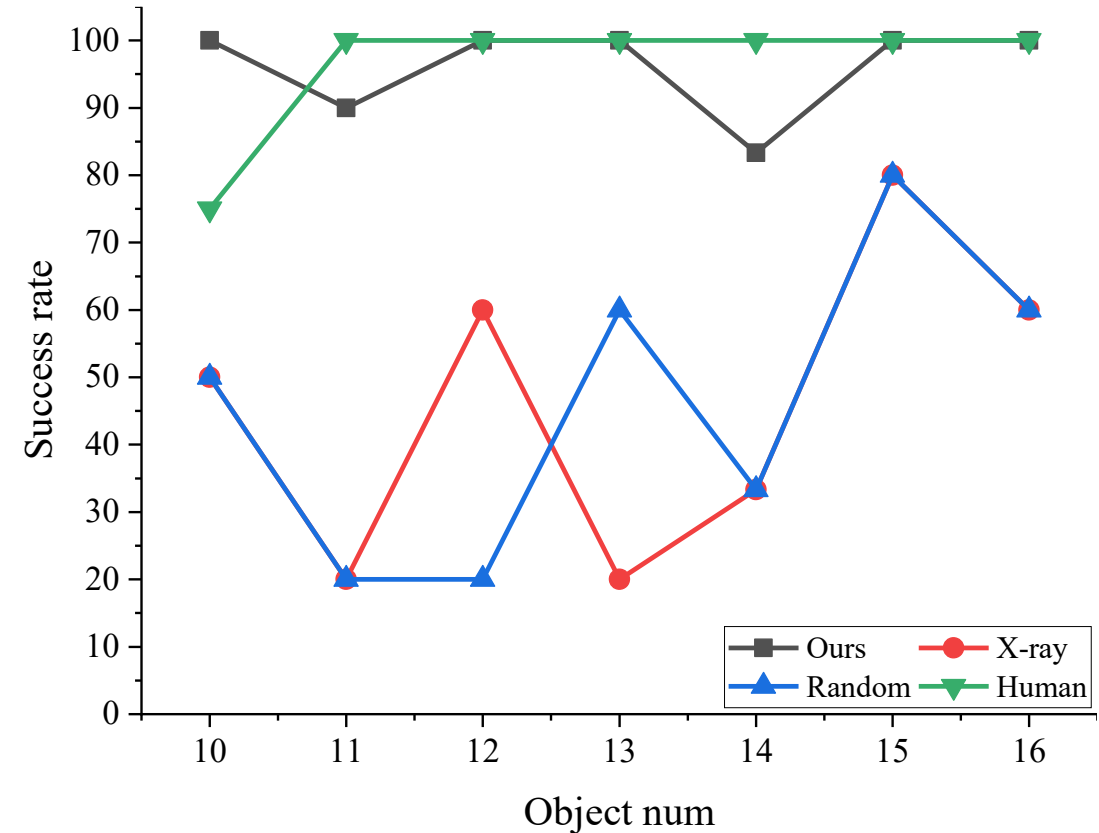
Depth정보에 의한 Occlusion value 보다 similarity value를 많이 반영($\beta: 0.4$) → 유사도에 의한 깊이방향 탐색 수행

Experiment Result

Success rate: 4×5 configuration

Method	Success rate (%)
Ours	95.0 %
Human	97.4 %
X-ray	42.5 %
Random	42.5 %

Object num	Ours	Human	X-ray	Random
10	100.0 %	75.0 %	50.0 %	50.0 %
11	90.0 %	100.0 %	20.0 %	20.0 %
12	100.0 %	100.0 %	60.0 %	20.0 %
13	100.0 %	100.0 %	20.0 %	60.0 %
14	83.3 %	100.0 %	33.3 %	33.3 %
15	100.0 %	100.0 %	80.0 %	80.0 %
16	100.0 %	100.0 %	60.0 %	60.0 %

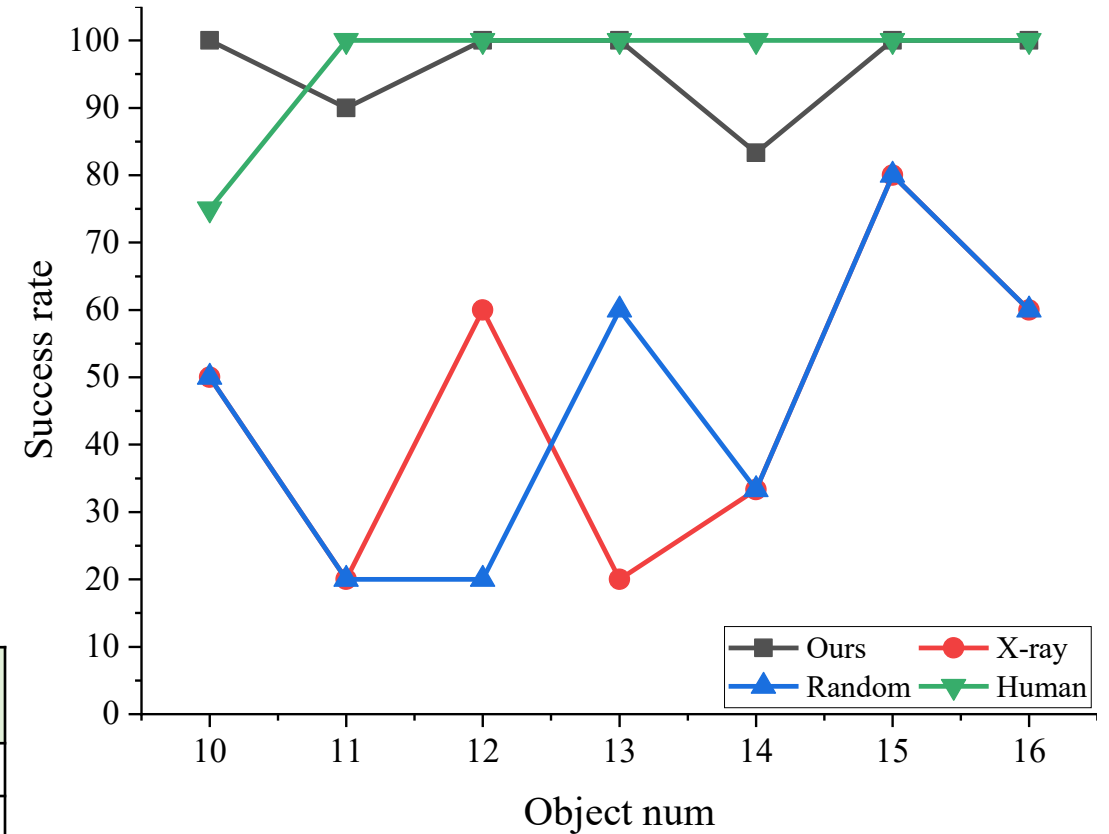


Human 대비 **2.5%** 낮지만, X-ray 대비 약 **123.5%**, Random 대비 약 **123.5%** 향상



Experiment Result

Success rate(ours): 4×5 configuration



Scenario No.	Method	Object num	Target	Step	Success
28	Ours	11	Cup_1	5	X
	X-ray			9	X
	Random			8	X
36	Ours	14	Cup_4	3	X
	X-ray			10	X
	Random			11	X

Discussion

- 모든 환경에서 Target object를 잘 탐색
 - 하지만, 우리가 원래 기대했던 행동은 아님
- 대부분의 sweeping이 가능한 환경에서 50~60% 정도는 sweeping보다 grasping을 하려는 경향이 있음



Target Object



Target Object



step	1D-PDM	Agent		Human	
		Action	Column	Action	Column
1	[3.84, 3.17, 13.07, 7.30]	grasping	2(3 rd)	Sweeping_R	2(3 rd)
2	[5.02, 4.11, 23.72, 9.49]	grasping	2(3 rd)	grasping	2(3 rd)
3	[5.02, 4.11, 23.72, 9.49]	grasping	2(3 rd)	grasping	2(3 rd)

step	1D-PDM	Agent		Human	
		Action	Column	Action	Column
1	[33.32, 32.98, 9.53, 4.70, 4.76]	grasping	1(2 nd)	Sweeping_L/R	1(2 nd)
2	[47.42, 40.76, 12.70, 6.08, 6.15]	grasping	1(2 nd)	grasping	1(2 nd)
3	[47.42, 40.76, 12.70, 6.08, 6.15]	grasping	1(2 nd)	grasping	1(2 nd)

Discussion

- 모든 환경에서 Target object를 잘 탐색
 - 하지만, 우리가 원래 기대했던 행동은 아님
- 대부분의 sweeping이 가능한 환경에서 50~60% 정도는 sweeping보다 grasping을 하려는 경향이 있음



- Sweeping 동작의 수행 및 reward를 받을 조건이 매우 sparse함(grasping 대비)
 - Environment, penalty
- 사람의 행동이 optimal policy가 아닐 수 있음
 - Environment, reward

Target Object

step	1D-PDM	Agent		Human	
		Action	Column	Action	Column
1	[3.84, 3.17, 13.07, 7.30]	grasping	2(3 rd)	Sweeping_R	2(3 rd)
2	[5.02, 4.11, 23.72, 9.49]	grasping	2(3 rd)	grasping	2(3 rd)
3	[5.02, 4.11, 23.72, 9.49]	grasping	2(3 rd)	grasping	2(3 rd)

step	1D-PDM	Agent		Human	
		Action	Column	Action	Column
1	[33.32, 32.98, 9.53, 4.70, 4.76]	grasping	1(2 nd)	Sweeping_L/R	1(2 nd)
2	[47.42, 40.76, 12.70, 6.08, 6.15]	grasping	1(2 nd)	grasping	1(2 nd)
3	[47.42, 40.76, 12.70, 6.08, 6.15]	grasping	1(2 nd)	grasping	1(2 nd)

Discussion

Conditions for executing the sweeping action and receiving a reward

• Reward

- r_{tg} : Target Grasping (60.0)
- r_{swr} : Sweeping Right Reward (35.0)
- r_{swl} : Sweeping Left Reward (35.0)
- r_{gr} : Non-target Grasping Reward (5.0)



Target Object

• Penalty

- p_{lpg} : Low Probability Grasping Penalty (-5.0)
- p_{lpswr} : Low Probability Sweeping Right Penalty (-5.0)
- p_{lpswl} : Low Probability Sweeping Left Penalty (-5.0)
- p_{ts} : Target Sweeping Penalty (-20.0)
- p_{ec} : Empty Column Action Penalty (-10.0)
- p_{lpg} : Grasping When Sweeping Possible Penalty (-20.0)
- p_{rsw} : Sweeping Same Object Again Penalty (-10.0)
- p_{rsg} : Grasping Previously Swept Object Penalty (-10.0)
- p_{ts} : Termination Penalty (-10.0)



step	1D-PDM	Agent				Human			
		Action	Column	Reward	Penalty	Action	Column	Reward	Penalty
1	[3.84, 3.17, 13.07, 7.30]	grasping	2(3 rd)	-	-20	Sweeping_R	2(3 rd)	35	-
2	[5.02, 4.11, 23.72, 9.49]	grasping	2(3 rd)	5	-	grasping	2(3 rd)	5	-
3	[5.02, 4.11, 23.72, 9.49]	grasping	2(3 rd)	5	-	grasping	2(3 rd)	5	-
Sum				-10				45	

Discussion

Conditions for executing the sweeping action and receiving a reward

• Reward

- r_{tg} : Target Grasping (60.0)
- r_{swr} : Sweeping Right Reward (35.0)
- r_{swl} : Sweeping Left Reward (35.0)
- r_{gr} : Non-target Grasping Reward (5.0)

• Penalty

- p_{lpg} : Low Probability Grasping Penalty (-5.0)
- p_{lpswr} : Low Probability Sweeping Right Penalty (-5.0)
- p_{lpawl} : Low Probability Sweeping Left Penalty (-5.0)
- p_{ts} : Target Sweeping Penalty (-20.0)
- p_{ec} : Empty Column Action Penalty (-10.0)
- p_{lpg} : Grasping When Sweeping Possible Penalty (-20.0)
- p_{rsw} : Sweeping Same Object Again Penalty (-10.0)
- p_{rsg} : Grasping Previously Swept Object Penalty (-10.0)
- p_{ts} : Termination Penalty (-10.0)



Target Object



step	1D-PDM	Agent				Human			
		Action	Column	Reward	Penalty	Action	Column	Reward	Penalty
1	[33.32, 32.98, 9.53, 4.70, 4.76]	grasping	1(2 nd)	-	-20	Sweeping_L/R	1(2 nd)	35	-
2	[47.42, 40.76, 12.70, 6.08, 6.15]	grasping	1(2 nd)	5	-	grasping	1(2 nd)	5	-
3	[47.42, 40.76, 12.70, 6.08, 6.15]	grasping	1(2 nd)	5	-	grasping	1(2 nd)	5	-
Sum				-10				45	

Discussion

Conditions for executing the sweeping action and receiving a reward



< Agent action >

- Action 선택
 - 3 (grasping, sweeping right, sweeping left)
- Column 선택
 - 4 (1st, 2nd, 3rd, 4th)
- 최종 경우의 수

$$4 \times 3 = 12$$

< Shelf environment >

(유사한 물체끼리 배치된다는 조건 제외 하고 단순 계산)

- 물체 선택
 - 12개 물체 중 k개 선택

$$C(12, k) = \frac{12!}{k!(12-k)!}$$

- 칸 선택
 - 12개 칸 중 k개 칸 선택

$$C(12, k) = \frac{12!}{k!(12-k)!}$$

- 물체를 칸에 배치
 - K개 물체를 k개 칸에 배치

$$k!$$

- 1개 이상 배치되는 경우의 수

$$S = \sum_{k=1}^{12} C(12, k) \times C(12, k) \times k! = 154,729,627,044$$

Discussion

Conditions for executing the sweeping action and receiving a reward

• Reward

- r_{tg} : Target Grasping (60.0)
- r_{swr} : Sweeping Right Reward (35.0)
- r_{swl} : Sweeping Left Reward (35.0)
- r_{gr} : Non-target Grasping Reward (5.0)

• Penalty

- p_{lpg} : Low Probability Grasping Penalty (-5.0)
- p_{lpshr} : Low Probability Sweeping Right Penalty (-5.0)
- p_{lpshl} : Low Probability Sweeping Left Penalty (-5.0)
- p_{ts} : Target Sweeping Penalty (-20.0)
- p_{ec} : Empty Column Action Penalty (-10.0)
- p_{lpg} : Grasping When Sweeping Possible Penalty (-20.0)
- p_{rsu} : Sweeping Same Object Again Penalty (-10.0)
- p_{rsg} : Grasping Previously Swept Object Penalty (-10.0)
- p_{ts} : Termination Penalty (-10.0)

— Grasping

— Sweeping



• Sweeping으로 reward를 받을 수 있는 확률

- 1st col을 sweeping right 했을 때만 가능

$$\frac{1}{12} = 0.083$$

→ 1D-PDM이 높은 2개의 column중 sweeping 해야 하기 때문

(1st step에서 우연히 sweeping을 했어도 2nd step에서 termination이 되어버리면 동일한 shelf configuration이 나올 확률 = 약 1/154,729,627,044)

Discussion

Optimal policy for human behavior

- **Reward**
 - Grasping
 - Sweeping
 - r_{tg} : Target Grasping (60.0)
 - r_{swr} : Sweeping Right Reward (35.0)
 - r_{swl} : Sweeping Left Reward (35.0)
 - r_{gr} : Non-target Grasping Reward (5.0)
- **Penalty**
 - p_{lpg} : Low Probability Grasping Penalty (-5.0)
 - p_{lpswr} : Low Probability Sweeping Right Penalty (-5.0)
 - p_{lpawl} : Low Probability Sweeping Left Penalty (-5.0)
 - p_{ts} : Target Sweeping Penalty (-20.0)
 - p_{ec} : Empty Column Action Penalty (-10.0)
 - p_{lpg} : Grasping When Sweeping Possible Penalty (-20.0)
 - p_{rsr} : Sweeping Same Object Again Penalty (-10.0)
 - p_{rsg} : Grasping Previously Swept Object Penalty (-10.0)
 - p_{ts} : Termination Penalty (-10.0)

Step	Action	Column	Reward	Penalty
1	Sweeping_R	1 st (0)	35	-
2	grasping	1 st (0)	-	-10(p_{ec})
2	Sweeping_L	2 nd (1)	-	-10(p_{swpso})
2	grasping	2 nd (1)	-	-10(p_{rsg})
2	grasping	3 rd (2) & 4 th (3)	-	-5(p_{lpg})

1st step



2nd step



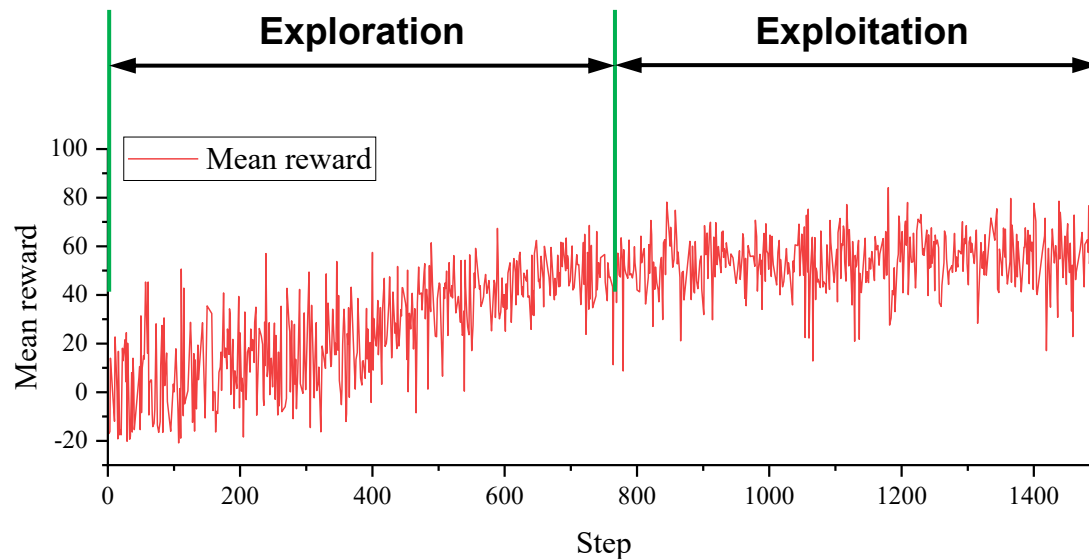
Target Object

- 의도했던 대로 action을 수행하면 다음 step에서는 전부 penalty
 - 인간이 생각한 policy는 특정 case에 대해서만 성립(Overfitted policy)
 - 인간처럼 행동하도록 하는 reward를 정의하기 애매모호함

→ RL이 학습한 policy(주로 grasping만 하는 경우)가 모든 shelf configuration에 대한 optimal policy일 수 있음

Discussion

- 앞선 이유들로 인해 초반 탐색 시 sweeping penalty가 너무 크게 작용해서 exploitation 단계에서 의도와는 다르게 grasping을 주로 하게 되는 경향이 발생



Exploration



• Future work

- 인간과 유사하게 agent 행동 학습을 위해 모방학습(Imitation Learning)의 일종인 Inverse Reinforcement Learning(IRL)을 통해 정의하기 애매모호한 reward를 추정 및 생성
- 이후, 추정 및 생성된 reward를 통해 RL 학습 진행

Conclusions and future works

• Conclusions

- 본 연구는 가려진 물체의 탐색을 위한 **심층강화학습 기반 탐색 방법을 제안**하였음
- Target object 탐색을 위해 탐색 방향을 알려주는 **column-wise exploration selector**를 설계 및 검증 완료
 - Simulation 및 real-world에서 검증 완료
- DRL 기반 target object 탐색을 위한 **action selector** 설계 및 검증 완료
 - Simulation 및 real-world에서 검증 완료
- Low level task planner와 결합하여 **최종 성능 평가 및 사전 연구와의 비교 검증 완료**
 - 3x4 shelf 환경
 - Step count: X-ray 대비 약 **37%**, Random 대비 약 **47% 향상**
 - Success rate: X-ray 대비 약 **33.3%**, Random 대비 약 **71.4% 향상**
 - 4x5 shelf 환경
 - Step count: X-ray 대비 약 **54.8%**, Random 대비 약 **54.6% 향상**
 - Success rate: X-ray 대비 약 **123.5%**, Random 대비 약 **123.5% 향상**

• Future works

- 실험 결과를 바탕으로 논문의 실험 및 결론 부분 보완 작성

Research achievements

• 논문

- ✓ IEEE RAL, "A New Data Augmentation Method for Time Series Wearable Sensor Data Using a Learning Mode Switching-Based DCGAN", 2021/08
- ✓ Sensors, "Fast Wearable Sensor-Based Foot-Ground Contact Phase Classification Using a Convolutional Neural Network with Sliding-Window Label Overlapping", 2020/09
- ✓ Mathematics, "Wearable Inertial Sensor-Based Hand-Guiding Gestures Recognition Method Robust to Significant Changes in the Body-Alignment of Subject", 2022/12
- ✓ Mathematics, "Explicit Identification of Pointwise Terrain Gradients for Speed Compensation of Four Driving Tracks in Passively Articulated Tracked Mobile Robot", 2023/02
- ✓ Mathematics, "Optimal Joint Path Planning of a New Virtual-Linkage-Based Redundant Finishing Stage for Additive-Finishing Integrated Manufacturing", 2023/12
- ✓ Sensors, "Bi-Directional Long Short-Term Memory-Based Gait Phase Recognition Method Robust to Directional Variations in Subject's Gait Progression Using Wearable Inertial Sensor", 2024/02
- ✓ KSPE, "Development of a Bowden-cable-based Exoskeleton Suit for Assisting Passive Stretching Exercises in Patients with Frozen Shoulder", 2024/07
- ✓ KSME, "Effect of the Muscle Activity of Each Quadriceps Femoris Muscle according to Walking on the Wearable Inertial Sensor Measurements", 2021/05
- ✓ KSME, "Development of a Test Bench for Performance Evaluation of Intelligent Social Robots", 2022/02
- ✓ Sensors, "A new multi-layered 3D NDT scan-matching method for robust localization in logistics warehouse environments", 2023/02
- ✓ Mathematics, "Hand-Guiding Gesture-Based Telemanipulation with the Gesture Mode Classification and State Estimation Using Wearable IMU Sensors", 2023/08
- ✓ Mathematics, "Upper Extremity Motion-Based Telemanipulation with Component-Wise Rescaling of Spatial Twist and Parameter-Invariant Skeletal Kinematics", 2024/01

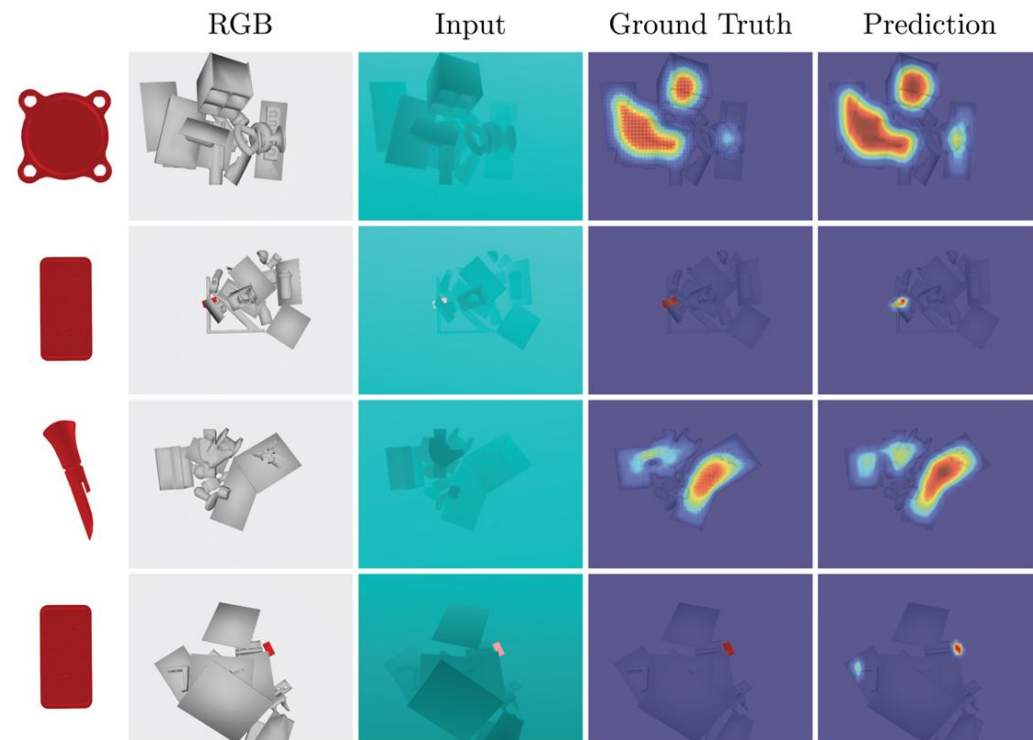
• 수상

- ✓ 대한기계학회, 2023년 춘계학술대회(동역학 및 제어), 우수논문상, "Bi-directional LSTM 기반 동적 Hand-guiding Gestures 인식 방법"
- ✓ 대한기계학회, 2023년 학술대회(생산 및 설계공학), 우수논문상, "6R Robotic-Arm 기반 Finishing 스테이지를 이용한 3D 조형물의 표면 마감: 실험적 조사"
- ✓ 대한기계학회, 2024년 춘계학술대회(생산 및 설계공학), 우수논문상, "Unstructured Environment 내에 위치한 Non-prehensile Object의 Grasp을 위한 계층적 Deep Reinforcement Learning 기반 Manipulation 방법"

감사합니다

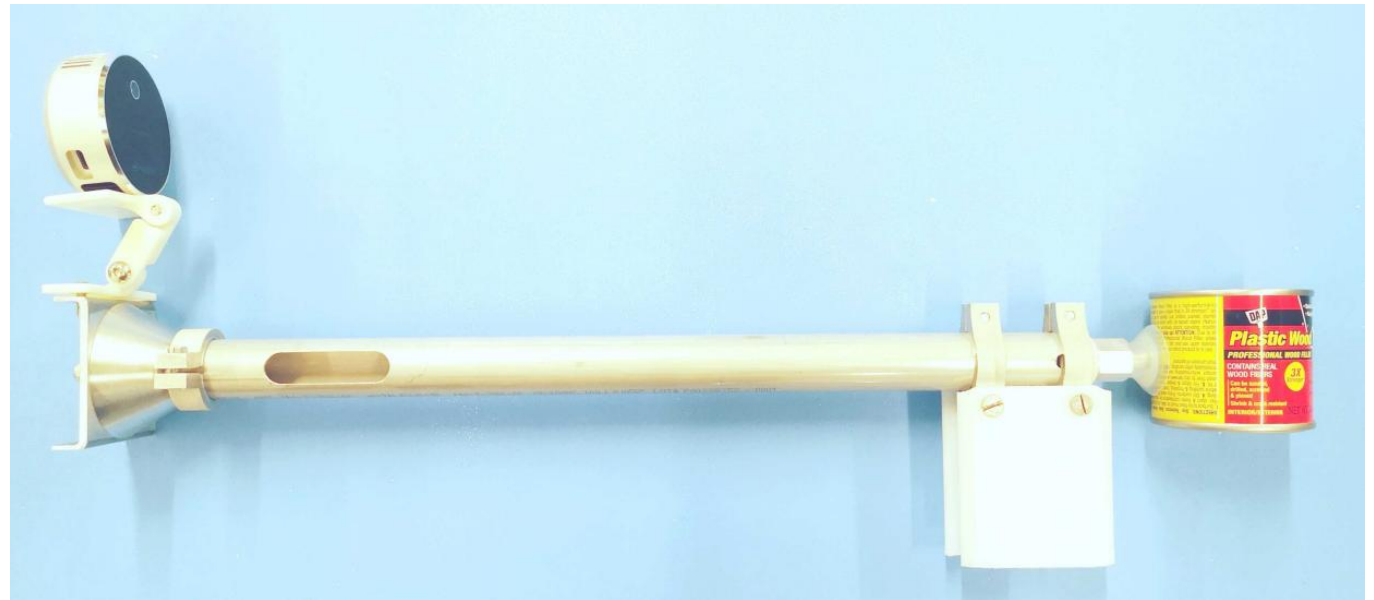
Reference

- Depth map만을 사용하여 학습
→ 더미 크기에 의존적인 탐색
- Target object가 obstacle과 대비되는 단색



Reference

- X-ray의 경우 일반 **상용 gripper는 크기로** 인해 선반 환경에서 **물체들과의 충돌로 인해 탐색에 어려움이 존재**
- 따라서 크기가 작고 길어 선반의 물체를 쉽게 탐색할 수 있는 “Bluction” tool을 제작하여 사용



Reference

• Cup



Cup_1



Cup_2

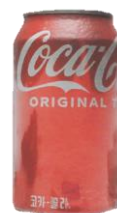


Cup_3



Cup_4

• Can



Can_1



Can_2



Can_3



Can_4

• Mug



Mug_1



Mug_2



Mug_3



Mug_4

• Bottle



Bottle_1



Bottle_2

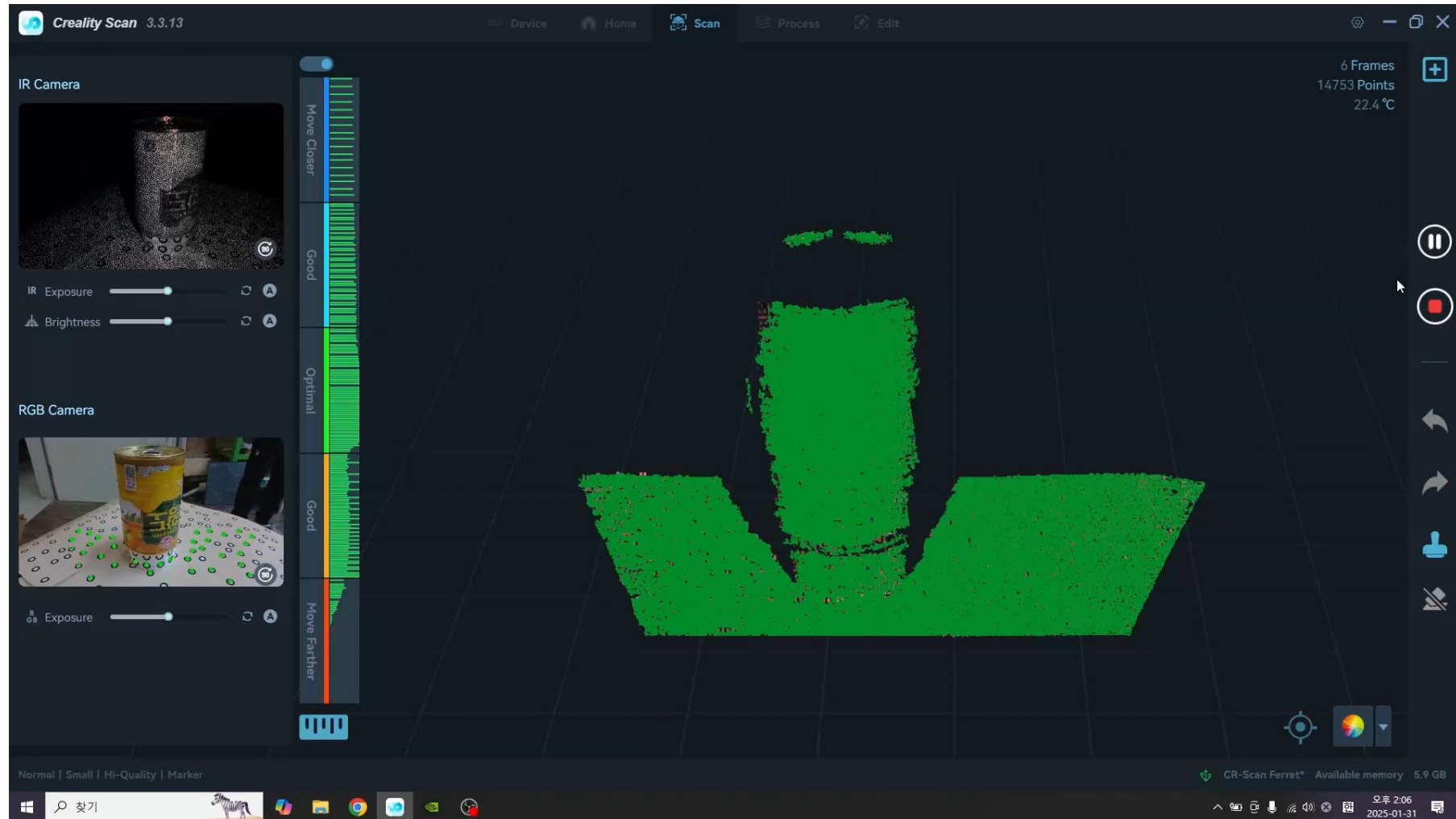


Bottle_3



Bottle_4

Reference



Reference

- Pytorch 에서 제공되는 segmentation backbone network

Weight	Mean IoU	pixelwise Acc	Params	GFLOPS	Recipe
DeepLabV3_ResNet101_Weights.COCO_WITH_VOC_LABELS_V1	67.4	92.4	61.0M	258.74	link
DeepLabV3_ResNet50_Weights.COCO_WITH_VOC_LABELS_V1	66.4	92.4	42.0M	178.72	link
FCN_ResNet101_Weights.COCO_WITH_VOC_LABELS_V1	63.7	91.9	54.3M	232.74	link
FCN_ResNet50_Weights.COCO_WITH_VOC_LABELS_V1	60.5	91.4	35.3M	152.72	link

- Resnet50**

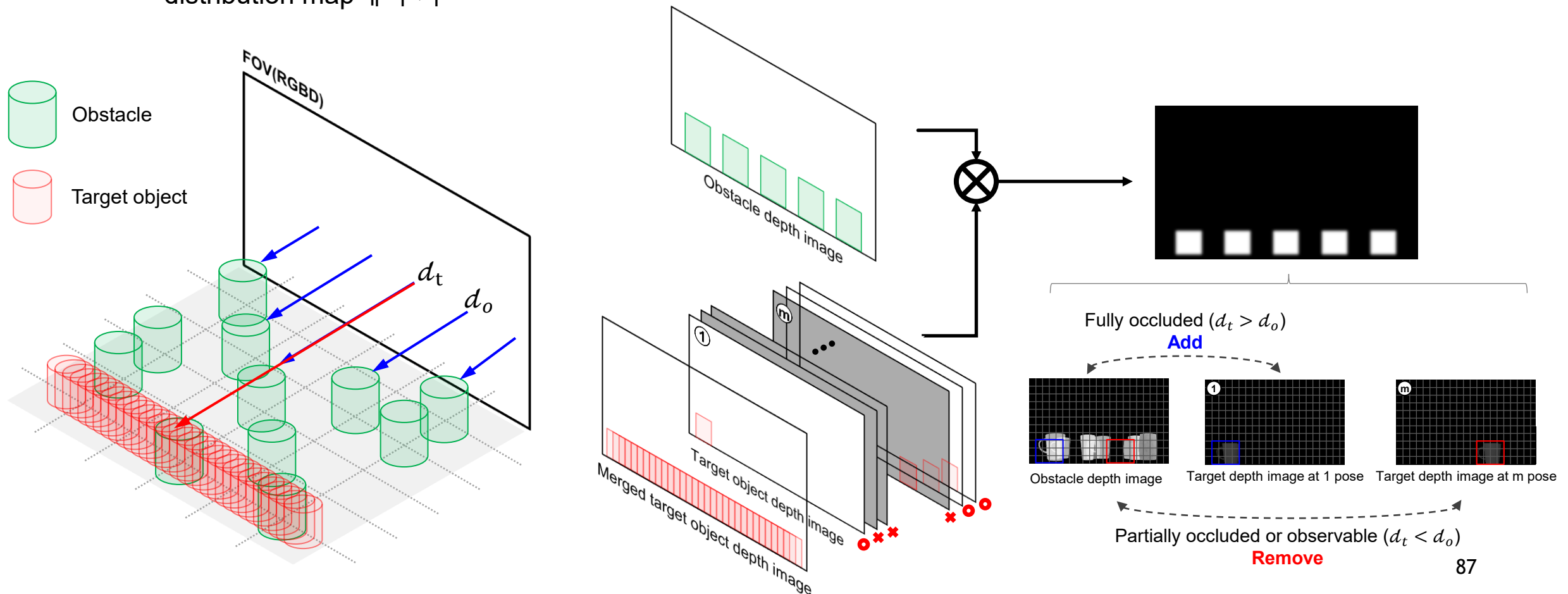
- 비슷한 성능
- 낮은 parameter 개수
- 낮은 GFLOPS 요구 사항

➔ 모델이 가벼우면서 성능은 비슷함

Reference

Data generation for training the FCN ResNet50 network

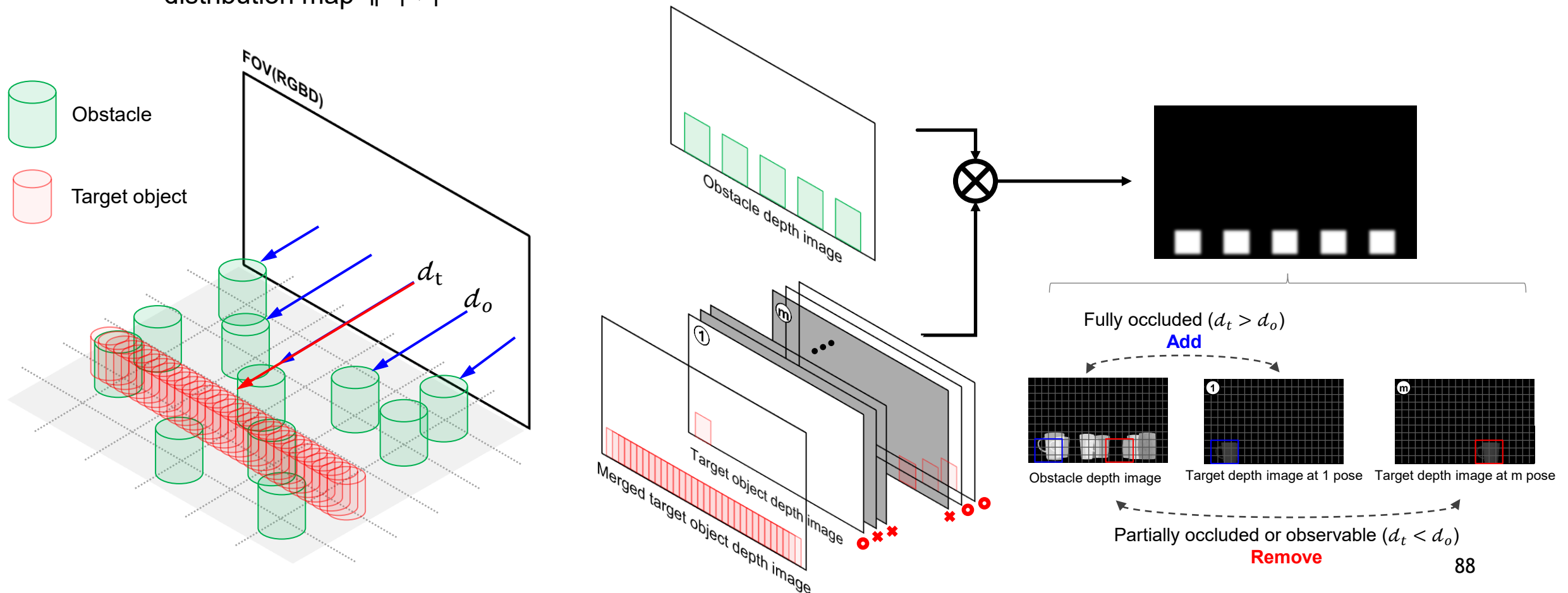
- (Occlusion distribution map) Heap(Dummy)에 의해 target object가 가려질 수 있는 영역에 대한 평가
 - Target object depth image와 Obstacle depth image를 서로 겹쳐서 target object가 obstacle에 가려지면 그 부분을 distribution map에 추가



Reference

Data generation for training the FCN ResNet50 network

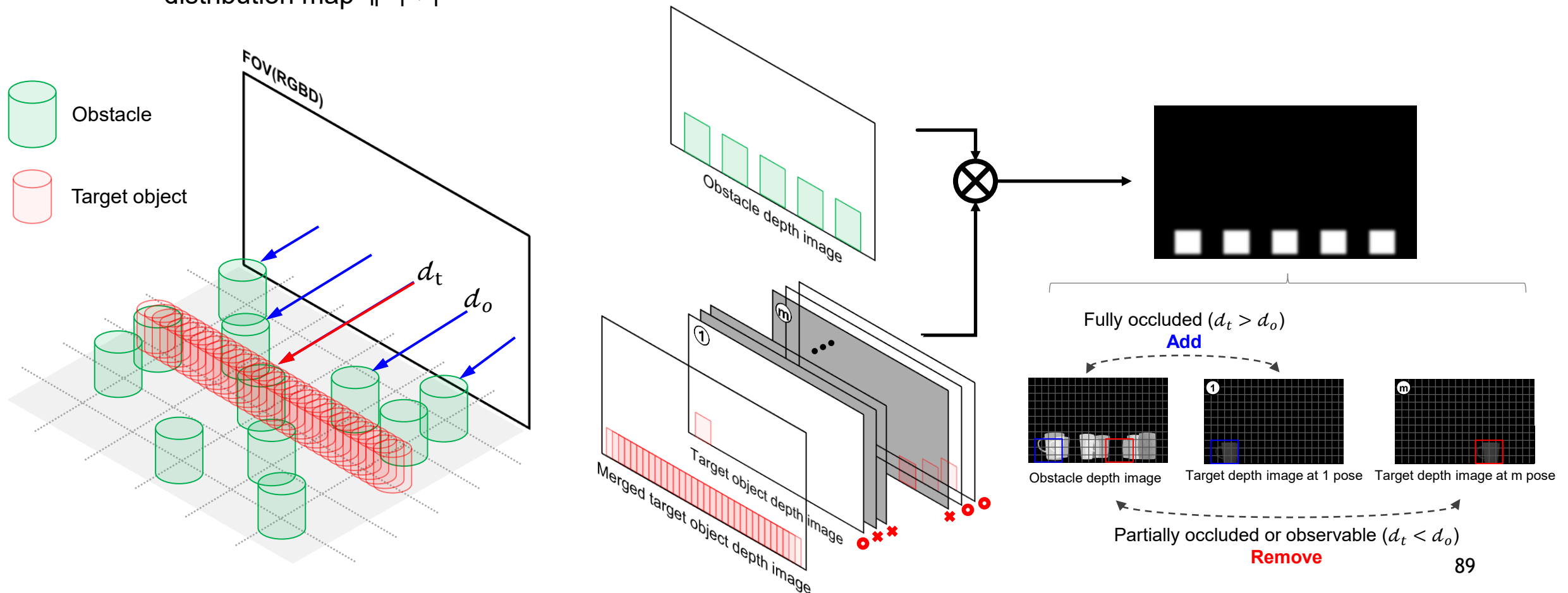
- (Occlusion distribution map) Heap(Dummy)에 의해 target object가 가려질 수 있는 영역에 대한 평가
 - Target object depth image와 Obstacle depth image를 서로 겹쳐서 target object가 obstacle에 가려지면 그 부분을 distribution map에 추가



Reference

Data generation for training the FCN ResNet50 network

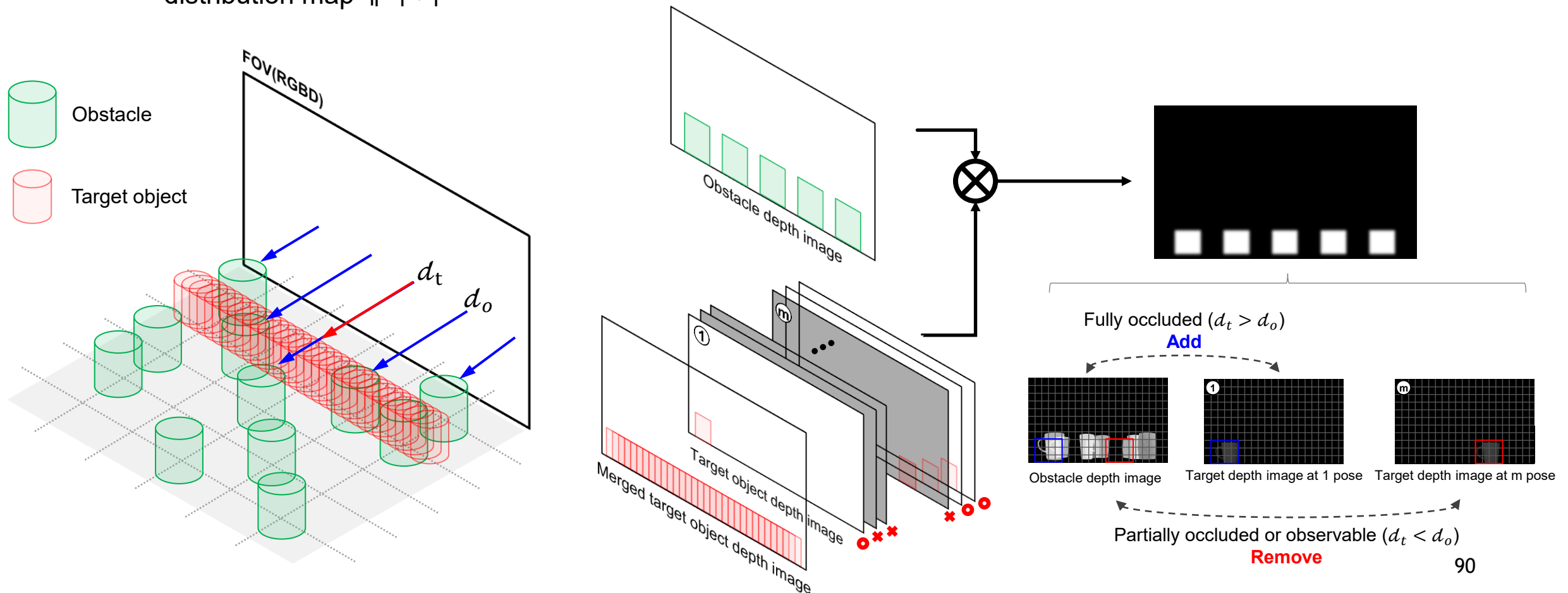
- (Occlusion distribution map) Heap(Dummy)에 의해 target object가 가려질 수 있는 영역에 대한 평가
 - Target object depth image와 Obstacle depth image를 서로 겹쳐서 target object가 obstacle에 가려지면 그 부분을 distribution map에 추가



Reference

Data generation for training the FCN ResNet50 network

- (Occlusion distribution map) Heap(Dummy)에 의해 target object가 가려질 수 있는 영역에 대한 평가
 - Target object depth image와 Obstacle depth image를 서로 겹쳐서 target object가 obstacle에 가려지면 그 부분을 distribution map에 추가

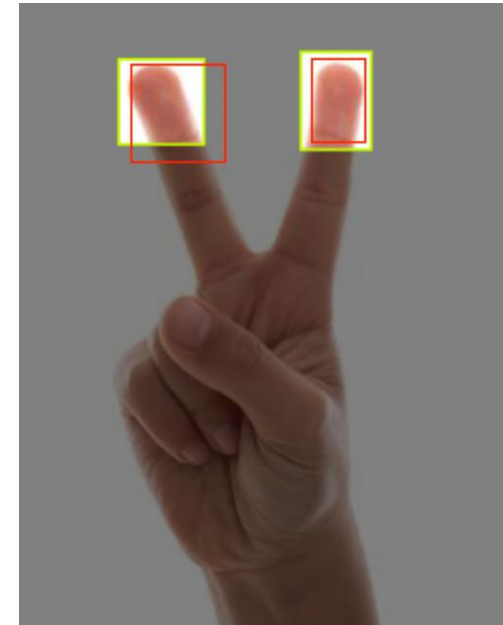


Reference

- **Intersection over Union(IoU)**

- Intersection(교집합)을 union(합집합)으로 나눈 비율
- 즉, IOU는 모델이 얼마나 object detection 혹은 segmentation을 정확히 했는지 정확도를 측정할 수 있는 지표

$$\text{IoU} = \frac{\text{Area of Overlap}}{\text{Area of Union}}$$

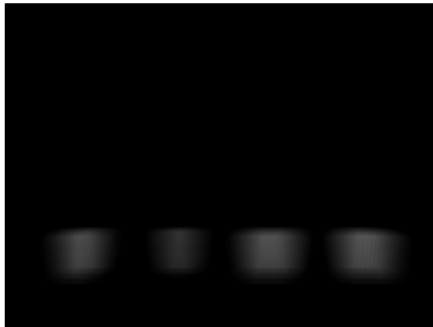
Reference

Occlusion distribution map

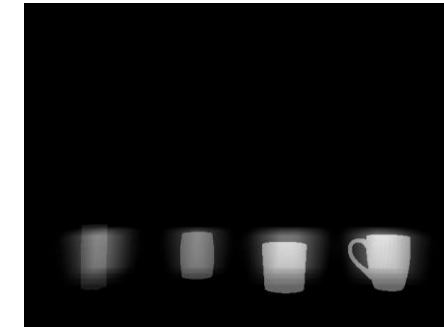
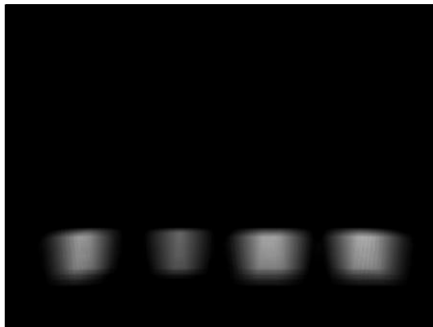
Similarity distribution map

Final distribution map

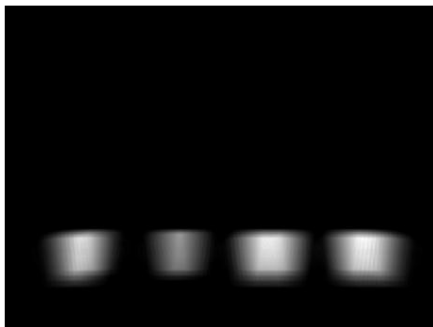
$\beta: 2$



$\beta: 5$



$\beta: 8$



Reference

- **Vectorized Training**

- RSL RL
- SKRL

- **학습 결과**

- SKRL : object들의 떨림 발생
 - 아직 해결 안됨(동일한 문제들로 인해 해결하고 있다고 함)

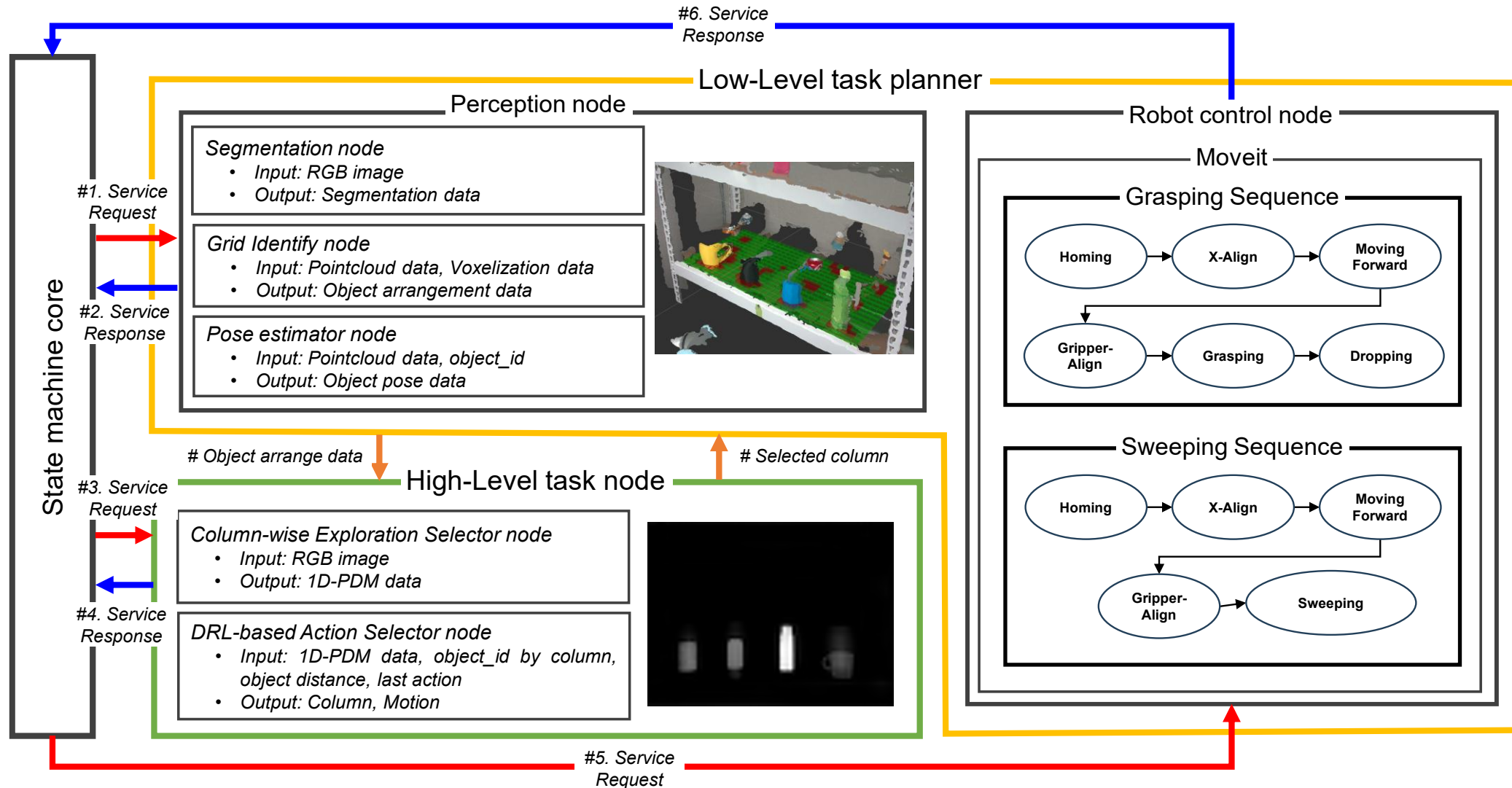
➔ **RSL-RL 사용**

Feature Comparison

Feature	RSL RL	SKRL	Stable Baselines3
Algorithms Included	PPO	Extensive List	Extensive List
Vectorized Training	Yes	Yes	No
Distributed Training	No	Yes	No
ML Frameworks Supported	PyTorch	PyTorch, JAX	PyTorch
Multi-Agent Support	PPO	PPO + Multi-Agent algorithms	External projects support
Documentation	Low	Comprehensive	Extensive
Community Support	Small Community	Small Community	Large Community
Available Examples in Isaac Lab	Large	Large	Small

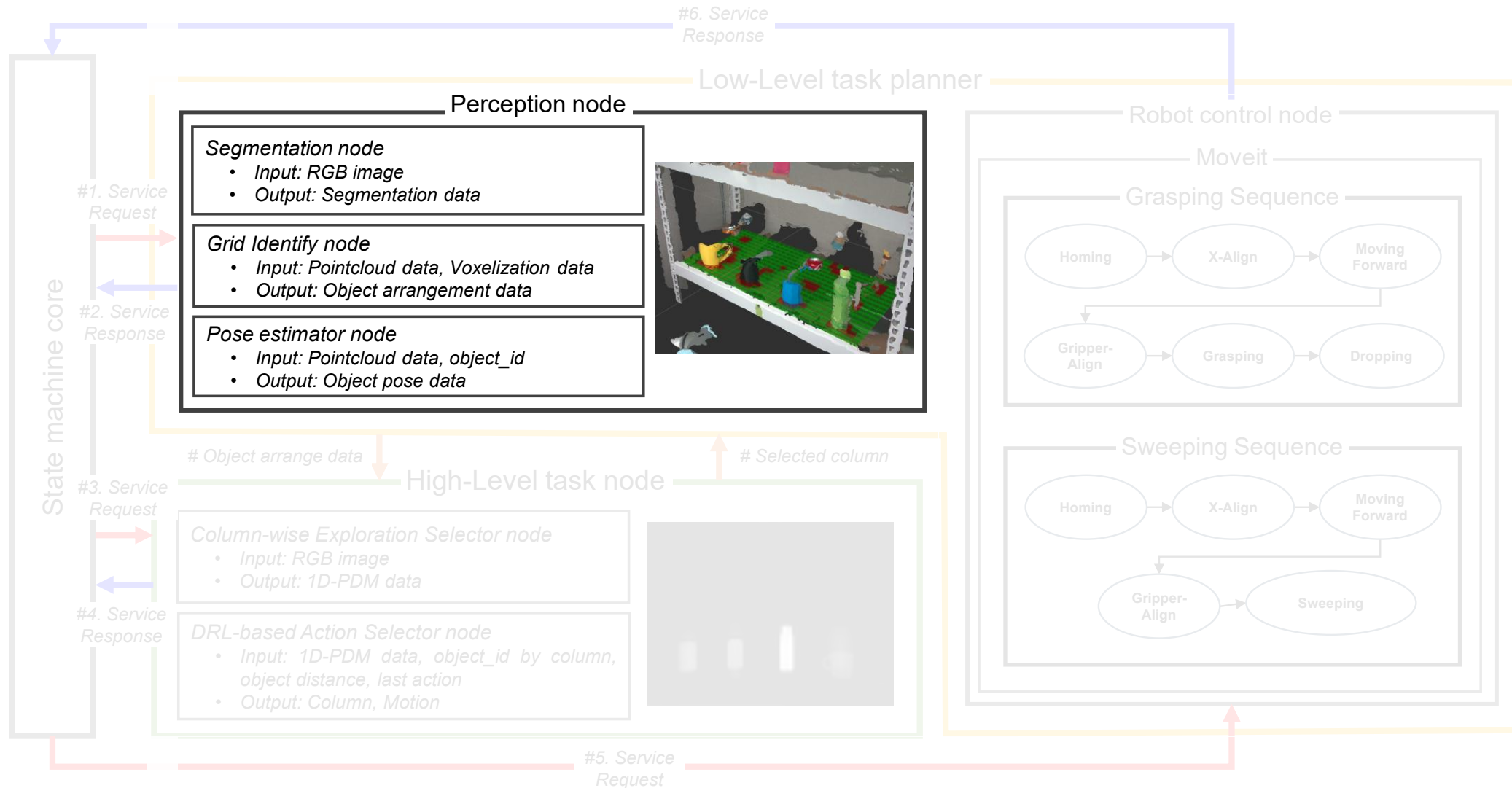
Comments

Computation cost comparison



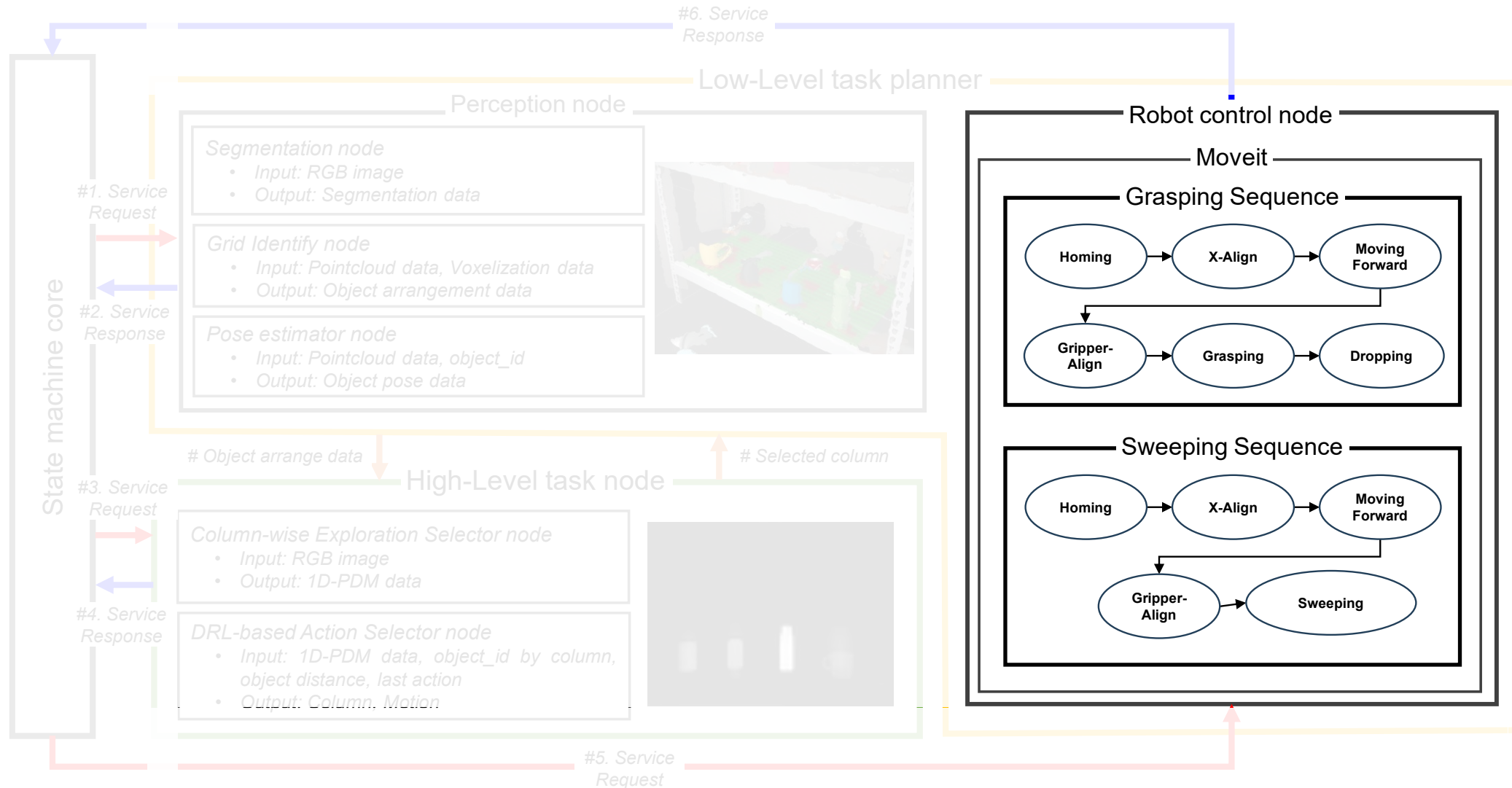
Comments

Computation cost comparison



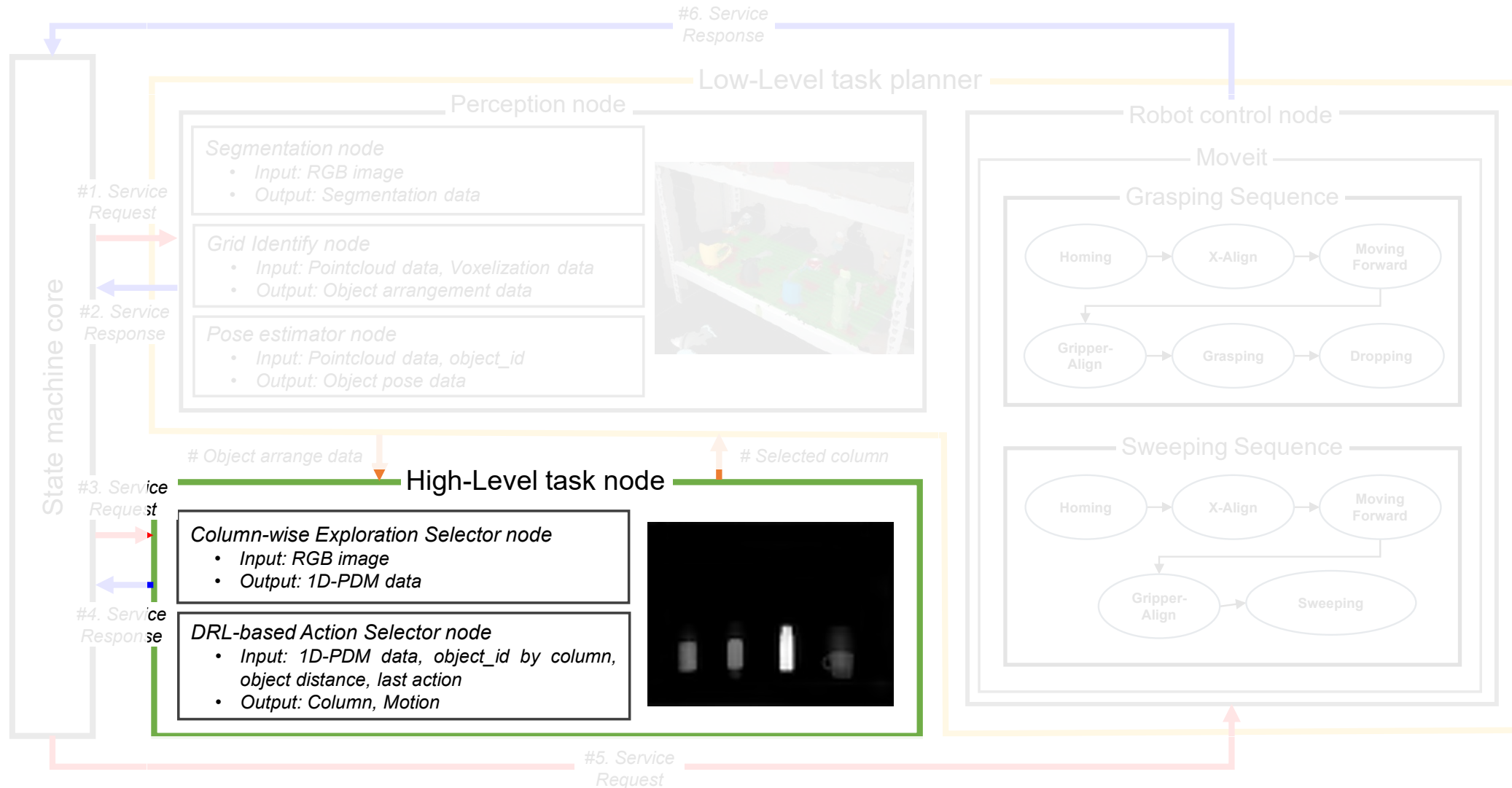
Comments

Computation cost comparison



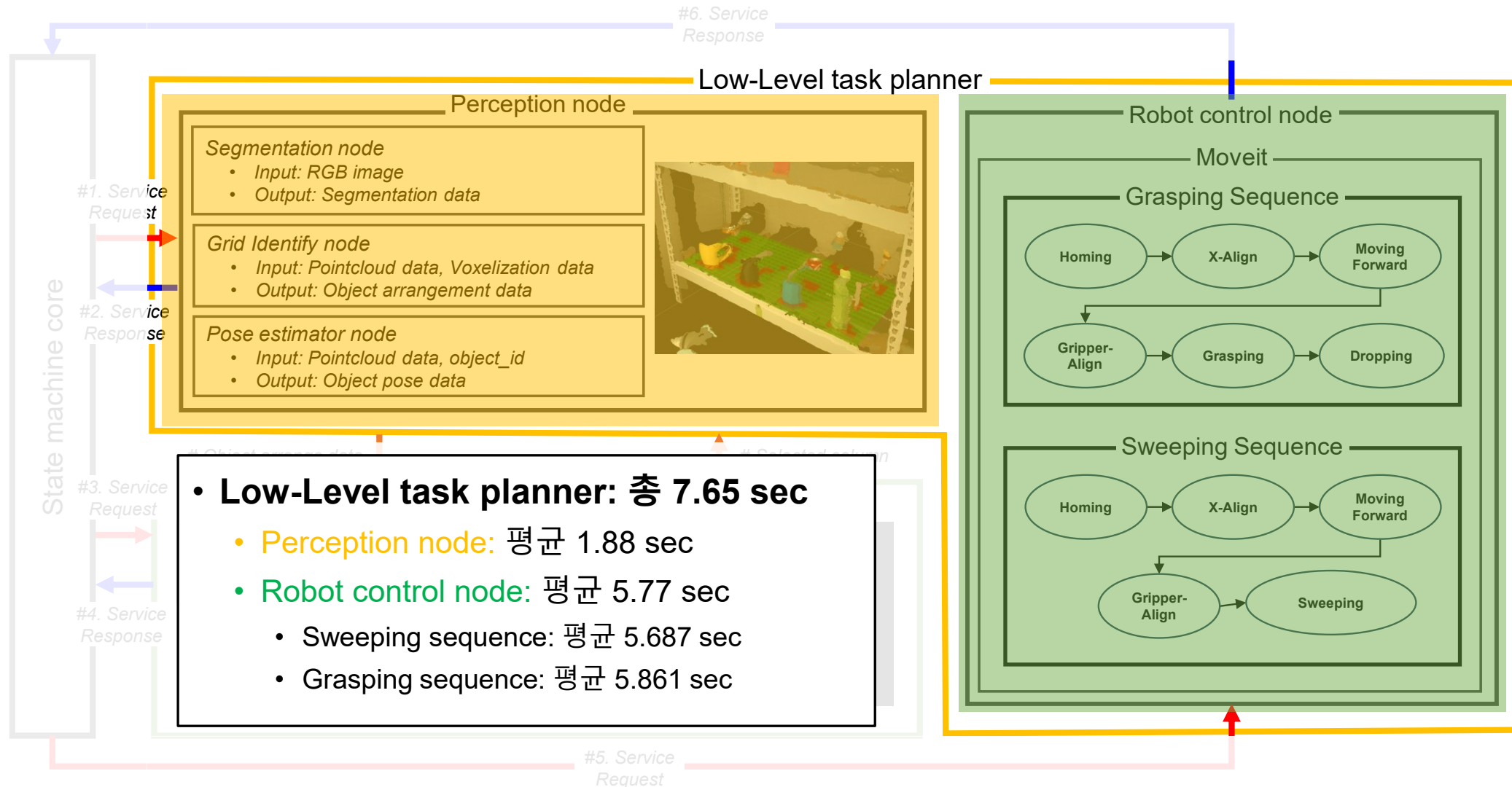
Comments

Computation cost comparison



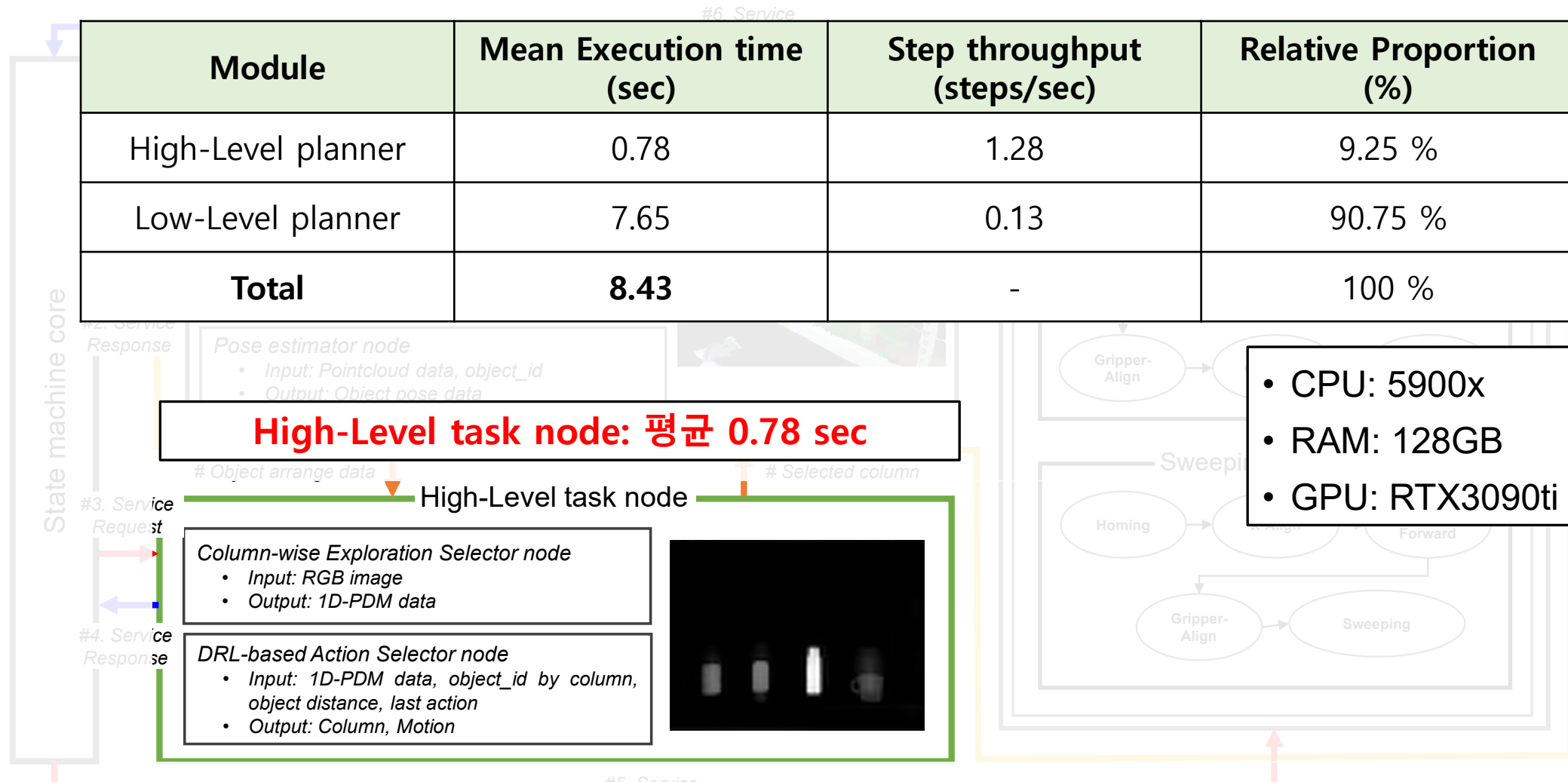
Comments

Computation cost comparison



Comments

Computation cost comparison



Low-level task planner가 High-level task planner 대비 약 880.8% 높은 computation cost를 가짐



Comments

Feasibility of replacing the high-level task planner with an end-to-end LLM-based approach

- Semantic Mechanical Search with Large Vision and Language Models – arxiv(2024)
 - 연구 목적: Occluded target object를 semantic prior 기반 mechanical search로 탐색하는 문제
 - Method: **VLM(ViLD/SAM) + LLM**
- Language-Grounded Dynamic Scene Graphs for Interactive Object Search with Mobile Manipulation – IEEE RAL(2024)
 - 연구 목적: 모바일 매니퓰레이터 기반 대규모 미탐색 공간에서의 언어기반 목표 물체를 탐색하는 문제
 - Method: **Hierarchical 3D Scene Graph + LLM(GPT-4)**
- Can an Embodied Agent Find Your “Cat-shaped Mug”? LLM-Based Zero-Shot Object Navigation – IEEE RAL(2024)
 - 연구 목적: 자연어로 기술된 목표 물체를 LLM을 활용한 zero-shot object navigation 문제
 - Method: **YOLO/BLIP + LLM(GPT-3) + GLIP**
- GMM-searcher: efficient object search in large-scale scenes using large language models – Nature, Scientific Reports(2025)
 - 연구 목적: LLM과 경험 기반 GMM을 활용한 대규모 환경에서의 목표 객체 탐색 효율화 문제
 - Method: **ARTG-GMM + LLM(GPT-4)**

LLM은 본질적으로 **언어 기반 모델**이기 때문에 **이미지나 센서 데이터를 텍스트 형식으로 변환**하는 추가적인 모듈이 필수적

Comments

DRL vs LLM

• DRL

• 장점

- 센서 raw 데이터 input 기반 학습 가능
(입력 데이터의 형식 다양함)
- 추론 시간이 매우 짧음
 - 평균 1초 미만(보통 ms 단위)
 - 제안 연구 0.78s
- Policy를 환경에 맞게 최적화 가능
- 정확한 목표 기반 reward shaping 가능

• 단점

- 일반화에 약함(LLM 대비 상대적)
 - 학습에 보지 못한 object 등
- 학습 자체의 샘플 비효율성과 긴 훈련 시간

• LLM

• 장점

- 고수준의 추론 가능
- 높은 일반화 성능
 - Zero-shot
 - 추론 기반 구조로 복잡한 구조/지시 변화에 유연
- 사전 학습된 LLM 사용시 학습 자체 비용 낮음

• 단점

- 긴 추론시간
 - 평균 11초 이상(최대 40초) – GPT4 모델 사용시 11초
- 본질적으로 언어 모델이기 때문에 추가적인 모듈과의 통합 필요
- 신뢰성 이슈
 - LLM의 잘못된 추론 및 답변으로 인한 로봇 오작동
- 프롬프트 설계에 따른 성능의 편차가 큼

감사합니다